

## SYLLABUS

### UNIT-I:- SOLUTIONS OF ALGEBRAIC AND TRANSCENDENTAL EQUATIONS

TOPICS:- Introduction, bisection method, method of false position, iteration method, Newton Raphson method (one variable)

### UNIT-II:- INTERPOLATION AND NUMERICAL INTEGRATION

TOPICS:- Introduction, errors in polynomial interpolation, finite differences, symbolic relations and separation of symbols, Forward differences, Backward differences, central differences, Difference of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lobachevsky's formulae, Trapezoidal rule, Simpson's  $\frac{1}{3}$  rule and  $\frac{3}{8}$  rule.

### UNIT-III:- NUMERICAL SOLUTIONS OF ORDINARY DIFFERENTIAL EQUATIONS

TOPICS:- Solution of ordinary differential equations by Taylor's series method, Picard method of successive approximations, Euler's method, Runge-Kutta method (2nd and 4th order)

### UNIT-IV:- PARTIAL DIFFERENTIATION

TOPICS:- Introduction, homogeneous function for single variable, Euler's theorem, total derivative, chain rule, generalised mean value theorem (without proof), Taylor's and Maclaurin's series, Expansion of function of two variables, Jacobian functional dependence.

Applications:-

maxima and minima of functions of two variables without constraints.

## UNIT-V:- PARTIAL DIFFERENTIAL EQUATIONS-I

TOPICS:- Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions.  
Solution of first order linear and non linear equations (standard type).

## UNIT-VI:- PARTIAL DIFFERENTIAL EQUATIONS-II

TOPICS:- Second and higher order linear equations, solutions to homogeneous and non homogeneous equations separation of variation method.



## UNIT-I

### SOLUTIONS OF ALGEBRAIC AND TRANSCENDENTAL EQUATIONS

#### polynomial Function:-

A function  $f(x)$  is said to be a polynomial function. If  $f(x)$  is a polynomial in  $x$ .

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n; a_0 \neq 0$$

And the coefficients  $a_0, a_1, a_2, a_n$  are real constants and  $n$  is a non negative integer is called a polynomial function of degree  $n$ .

#### Algebraic And Transcendental functions:-

A function which is sum or difference of two polynomials, is called the algebraic function otherwise the function is said to be transcendental function.

Ex 1:-  $f(x) = 5x^3 + 3x^2 + 9x + 5$

$g(x) = 6x^2 + 5x - 9$  is called algebraic functions

Ex 2:-  $f(x) = xe^x + \sin x$

$g(x) = 2\log x + 5x + 3$  is called transcendental functions

#### Root of an equation:-

A number  $\alpha$  is called a root or solution of an equation  $f(x) = 0$  if  $f(\alpha) = 0$

#### Intermediate value theorem:-

If  $f(x)$  is continuous in  $[a, b]$  and  $f(a), f(b)$  are opposite sign then there exists a point  $c$  in  $(a, b)$  such that  $f(c) = 0$

#### Bisection method:-

Suppose we know an equation of the form  $f(x) = 0$  has, exactly one root between two real numbers  $x_0$  and  $x_1$ , the number is chosen such that  $f(x_0)$  and  $f(x_1)$  will

have opposite signs.

let us bisect  $[x_0, x_1]$  into two half intervals and find the mid point.

$$x_2 = \frac{x_0 + x_1}{2}$$

case 'i':- If  $f(x_2)$  and  $f(x_0)$  have opposite signs then the root lies between  $x_0$  and  $x_2$ .

$$\text{let } x_3 = \frac{x_2 + x_0}{2}$$

case 'ii':- If  $f(x_2)$  and  $f(x_1)$  have opposite signs then the root lies between  $x_1$  and  $x_2$  then

$$x_3 = \frac{x_1 + x_2}{2}$$

Repeating the process of bisection we obtain successive sub intervals which are smaller. At each we get the mid point as better approximation of the root.

this process is terminated when the interval is smaller than the desired accuracy.

① Find an approximate value of the root of an equation  $x^3 - x - 1 = 0$  by bisection method.

Sol:- Given equation  $x^3 - x - 1 = 0$

$$\text{let } f(x) = x^3 - x - 1 = 0$$

$$x \quad 0 \quad 1 \quad 2$$

$$f(x) \quad -1 \quad -1 \quad 5$$

$$<0 \quad >0$$

$\therefore$  the root lies between 1 & 2.

$$\text{let } x_0 = 1 \text{ and } x_1 = 2$$

$$f(x_0) = -1 < 0$$

$$f(x_1) = 5 > 0$$

$\therefore$  the root lies between  $x_0$  &  $x_1$

by bisection method



$$x_2 = \frac{x_0 + x_1}{2} = \frac{(1+2)}{2} = 1.5$$

$$f(x_2) = f(1.5) = 0.875 > 0$$

$$f(x_2) > 0, f(x_0) < 0$$

∴ the root lies between  $x_2$  &  $x_0$

$$x_3 = \frac{x_2 + x_0}{2} = \frac{(1.5+1)}{2} = 1.25$$

$$f(x_3) = f(1.25) = -0.2968 < 0 \quad f(x_3) < 0, f(x_2) > 0$$

∴ the root lies between  $x_3$  &  $x_2$

$$x_4 = \frac{x_3 + x_2}{2} = \frac{(1.25+1.5)}{2} = 1.375$$

$$f(x_4) > 0, f(x_3) < 0$$

∴ the root lies between  $x_4$  &  $x_3$

$$x_5 = \frac{x_4 + x_3}{2} = \frac{(1.375+1.25)}{2} = 1.3125$$

$$f(x_5) = f(1.3125) = -0.0515 < 0$$

$$f(x_5) < 0, f(x_4) > 0$$

∴ the root lies between  $x_5$  and  $x_4$

$$x_6 = \frac{x_5 + x_4}{2} = \frac{(1.3125+1.375)}{2} = 1.3437$$

$$f(x_6) = f(1.3437) = 0.0826 > 0$$

$$f(x_6) > 0, f(x_5) < 0$$

∴ the root lies between  $x_6$  and  $x_5$

$$x_7 = \frac{x_6 + x_5}{2} = 1.3281$$

$$f(x_7) = f(1.3281) = 0.0144 > 0$$

$$f(x_7) > 0, f(x_5) < 0$$

∴ the root lies between  $x_7$  and  $x_5$

$$x_8 = \frac{x_7 + x_5}{2} = 1.3203$$

$$f(x_8) = -0.0187 < 0$$

$$f(x_8) < 0, f(x_7) > 0$$

∴ the root lies between  $x_8$  and  $x_7$

$$x_9 = \frac{x_8 + x_7}{2} = 1.3242$$

$$f(x_9) = f(1.3242) = -0.0022 < 0$$

$$f(x_9) < 0, f(x_7) > 0$$

∴ the root lies between  $x_9$  and  $x_7$

$$x_{10} = \frac{x_9 + x_7}{2} = 1.3261$$

$$f(x_{10}) = 0.0059 > 0$$

$$f(x_{10}) > 0, f(x_9) < 0$$

∴ the root lies between  $x_{10}$  and  $x_9$

$$x_{11} = \frac{x_{10} + x_9}{2} = 1.3251$$

$$f(x_{11}) = f(1.3251) = 0.0016 > 0$$

$$f(x_{11}) > 0, f(x_9) < 0$$

∴ the root lies between  $x_{11}$  and  $x_9$

$$x_{12} = \frac{x_{11} + x_9}{2} = 1.3246$$

$$f(x_{12}) = f(1.3246) = -0.0005 < 0$$

$$f(x_{12}) < 0, f(x_{11}) > 0$$

∴ the root lies between  $x_{12}$  and  $x_{11}$

$$x_{13} = \frac{x_{12} + x_{11}}{2} = 1.3248$$

∴ the approximate root of the equation 1.3248

② Find the approximate root of the equation  $x^3 + 5x + 1 = 0$  by using bisection method.

Sol:- Given equation  $x^3 + 5x + 1 = 0$

$$\text{let } f(x) = x^3 + 5x + 1 = 0, \quad f'(x) = \frac{2x^2 + 5}{1} = 2x^2 + 5$$

|        |       |       |       |
|--------|-------|-------|-------|
| $x$    | 0     | 1     | 2     |
| $f(x)$ | 1     | -3    | -1    |
|        | $> 0$ | $< 0$ | $< 0$ |

∴ the root lies between 0 and 1



Let  $x_0 = 0$  and  $x_1 = 1$

$$f(x_0) = 1 > 0$$

$$f(x_1) = -3 < 0$$

$\therefore$  the root lies between  $x_0$  and  $x_1$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{0 + 1}{2} = 0.5$$

$$f(x_2) = f(0.5) = -1.375 < 0$$

$$f(x_2) < 0, f(x_0) > 0$$

$\therefore$  the root lies between  $x_2$  and  $x_0$

$$x_3 = \frac{x_2 + x_0}{2} = \frac{0.5 + 0}{2} = 0.25$$

$$f(x_3) = f(0.25) = -0.234 < 0$$

$$f(x_3) < 0, f(x_0) > 0$$

$\therefore$  the root lies between  $x_3$  and  $x_0$

$$x_4 = \frac{x_3 + x_0}{2} = \frac{0.25 + 0}{2} = 0.125$$

$$f(x_4) = f(0.125) = 0.376 > 0$$

$$f(x_4) > 0, f(x_3) < 0$$

$\therefore$  the root lies between  $x_4$  and  $x_3$

$$x_5 = \frac{x_4 + x_3}{2} = \frac{0.25 + 0.125}{2} = 0.1875$$

$$f(x_5) = f(0.1875) = 0.0690 > 0$$

$$f(x_5) > 0, f(x_3) < 0$$

$\therefore$  the root lies between  $x_5$  and  $x_3$

$$x_6 = \frac{x_5 + x_3}{2} = \frac{0.1875 + 0.125}{2} = 0.15625$$

$$f(x_6) = f(0.15625) = -0.0830$$

$$\therefore f(x_6) < 0, f(x_5) > 0$$

$\therefore$  the root lies between  $x_6$  and  $x_5$

$$x_7 = \frac{x_6 + x_5}{2} = \frac{0.15625 + 0.1875}{2} = 0.171875$$

$$f(x_7) = f(0.171875) = -0.00712$$

$$f(x_7) < 0, f(x_5) > 0$$

∴ the root lies between  $x_7$  and  $x_8$

$$x_8 = \frac{x_7 + x_5}{2} = 0.1953$$

$$f(x_8) = 0.0309 > 0$$

$$f(x_8) > 0, f(x_7) < 0$$

∴ the root lies between  $x_8$  and  $x_7$

$$x_9 = \frac{x_8 + x_7}{2} = 0.1992$$

$$f(x_9) = f(0.1992) = 0.0119$$

$$f(x_9) > 0, f(x_7) < 0$$

∴ the root lies between  $x_9$  and  $x_7$

$$x_{10} = \frac{x_9 + x_7}{2} = 0.2011$$

$$f(x_{10}) = f(0.2011) = 0.0026 > 0$$

$$f(x_{10}) > 0, f(x_7) < 0$$

∴ the root lies between  $x_{10}$  and  $x_7$

$$x_{11} = \frac{x_{10} + x_7}{2} = 0.2021$$

$$f(x_{11}) = f(0.2021) = -0.00224 < 0$$

$$f(x_{11}) < 0, f(x_{10}) > 0$$

∴ the root lies between  $x_{11}$  and  $x_{10}$

$$x_{12} = \frac{x_{11} + x_{10}}{2} = 0.2016$$

$$f(x_{12}) = 0.00019 > 0$$

$$f(x_{12}) > 0, f(x_{11}) < 0$$

∴ the root lies between  $x_{12}$  and  $x_{11}$

$$x_{13} = \frac{x_{12} + x_{11}}{2} = 0.2018$$

∴ the approximate root of the equation 0.2018

③ Find the approximate root of the equation  $x - \cos x = 0$   
by using bisection method.

Sol:- Given equation  $x - \cos x = 0$



$$\text{Let } f(x) = x - \cos x$$

$$x \quad 0 \quad 1 \quad 2$$

$$f(x) = -1 \quad 0.45 \quad 0.41$$

$$< 0 > 0$$

the root lies between 0 and 1

$$\text{let } x_0 = 0 \text{ and } x_1 = 1$$

$$f(x_0) = -1 < 0$$

$$f(x_1) = 0.45 > 0$$

$\therefore$  the root lies between  $x_0$  and  $x_1$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{0 + 1}{2} = 0.5$$

$$f(0.5) = f(x_2) = -0.377 < 0$$

$$f(x_2) < 0, f(x_1) > 0$$

$\therefore$  the root lies between  $x_2$  and  $x_1$

$$x_3 = \frac{x_2 + x_1}{2} = \frac{0.5 + 1}{2} = 0.75$$

$$f(x_3) = f(0.75) = 0.0183 > 0$$

$$f(x_3) > 0, f(x_2) < 0$$

$\therefore$  the root lies between  $x_3$  and  $x_2$

$$x_4 = \frac{x_3 + x_2}{2} = \frac{0.5 + 0.75}{2} = 0.625$$

$$f(x_4) = f(0.625) = -0.185 < 0$$

$$f(x_4) < 0, f(x_3) > 0$$

$\therefore$  the root lies between  $x_4$  and  $x_3$

$$x_5 = \frac{x_4 + x_3}{2} = 0.6875$$

$$f(x_5) = f(0.6875) = -0.0853 < 0$$

$$f(x_5) < 0, f(x_3) > 0$$

$\therefore$  the root lies between  $x_5$  and  $x_3$

$$x_6 = \frac{x_5 + x_3}{2} = \frac{0.6875 + 0.75}{2} = 0.7187$$

$$f(x_6) = f(0.7187) = 0.0339 < 0$$

$$f(x_6) < 0, f(x_3) > 0$$

$\therefore$  the root lies between  $x_6$  and  $x_3$

$$x_7 = \frac{x_6 + x_3}{2} = 0.73435$$

$$f(x_7) = f(0.73435) = -0.0079 < 0$$

$$f(x_7) < 0, f(x_3) > 0$$

∴ the root lies between  $x_7$  and  $x_3$

$$x_8 = \frac{x_7 + x_3}{2} = 0.7421$$

$$f(x_8) = f(0.7421) = 0.0050 > 0$$

$$f(x_8) > 0, f(x_7) < 0$$

∴ the root lies between  $x_8$  and  $x_7$

$$x_9 = \frac{x_8 + x_7}{2} = 0.7382$$

$$f(x_9) = f(0.7382) = -0.0014 < 0$$

$$f(x_9) < 0, f(x_8) > 0$$

∴ the root lies between  $x_9$  and  $x_8$

$$x_{10} = \frac{x_9 + x_8}{2} = 0.74015$$

$$f(x_{10}) = f(0.74015) = 0.0016 > 0$$

$$f(x_{10}) > 0, f(x_9) < 0$$

∴ the root lies between  $x_{10}$  and  $x_9$

$$x_{11} = \frac{x_{10} + x_9}{2} = 0.73915$$

$$f(x_{11}) = f(0.73915) = 0.000024 > 0$$

$$f(x_{11}) > 0, f(x_9) < 0$$

∴ the root lies between  $x_{11}$  and  $x_9$

$$x_{12} = \frac{x_{11} + x_9}{2} = 0.7386$$

$$f(x_{12}) = f(0.7386) = -0.0008 < 0$$

∴ the root lies between  $x_{12}$  and  $x_{11}$

∴ the root lies between  $x_{11}$  and  $x_{12}$

$$x_{13} = \frac{x_{11} + x_{12}}{2} = 0.7388$$



$$f(x_{13}) = f(0.7388) = -0.0004 < 0$$

$$f(x_{12}) < 0, f(x_{11}) > 0$$

$\therefore$  the root lies between  $x_{11}$  and  $x_{13}$

$\therefore$  the approximate root of the equation is 0.7388

④ Find the approximate root of the equation  $\sin x = 1/x$ .

Sol:- Given equation  $\sin x = 1/x$

$$x \sin x - 1 = 0$$

$$\text{let } f(x) = x \sin x - 1 = 0$$

|        |    |       |      |
|--------|----|-------|------|
| $x$    | 0  | 1     | 2    |
| $f(x)$ | -1 | -0.15 | 0.81 |
|        | <0 | <0    | >0   |

$\therefore$  the root lies between 1 and 2

$$\text{let } x_0 = 1 \text{ and } x_1 = 2$$

$$f(x_0) = -0.15 < 0$$

$$f(x_1) = 0.81 > 0$$

$\therefore$  the root lies between  $x_0$  and  $x_1$

$$x_2 = \frac{x_0 + x_1}{2} = \frac{1+2}{2} = 1.5$$

$$f(x_2) = f(1.5) = 0.496 > 0$$

$$f(x_2) > 0, f(x_0) < 0$$

$\therefore$  the root lies between  $x_0$  and  $x_2$

$$x_3 = \frac{x_2 + x_0}{2} = \frac{1.5+1}{2} = 1.25$$

$$f(x_3) = f(1.25) = 0.186 > 0$$

$$f(x_3) > 0, f(x_0) < 0$$

$\therefore$  the root lies between  $x_3$  and  $x_0$

$$x_4 = \frac{x_3 + x_0}{2} = \frac{1.25+1}{2} = 1.125$$

$$f(x_4) = f(1.125) = 0.0150 > 0$$

$$f(x_4) > 0, f(x_0) < 0$$

$\therefore$  the root lies between  $x_4$  and  $x_0$

$$x_5 = \frac{x_4 + x_0}{2} = \frac{1.125+1}{2} = 1.0625$$

$$f(x_5) = f(1.0625) = -0.071 < 0$$

$$f(x_5) < 0, f(x_4) > 0$$

$\therefore$  the root lies between  $x_5$  and  $x_4$

$$x_6 = \frac{x_5 + x_4}{2} = \frac{1.0625 + 1.125}{2} = 1.0937$$

$$f(x_6) = f(1.0937) = -0.0284 < 0$$

$$f(x_6) < 0, f(x_4) > 0$$

$\therefore$  the root lies between  $x_6$  and  $x_4$

$$x_7 = \frac{x_6 + x_4}{2} = 1.1093$$

$$f(x_7) = f(1.1093) = -0.0067 < 0$$

$$f(x_7) < 0, f(x_4) > 0$$

$\therefore$  the root lies between  $x_7$  and  $x_4$

$$x_8 = \frac{x_7 + x_4}{2} = 1.11715$$

$$f(x_8) = f(1.1171) = 0.0040 > 0$$

$$f(x_8) > 0, f(x_7) < 0$$

$\therefore$  the root lies between  $x_8$  and 7

$$x_9 = \frac{x_8 + x_7}{2} = 1.1132$$

$$f(x_9) = f(1.1132) = -0.0013 < 0$$

$$f(x_9) < 0, f(x_8) > 0$$

$\therefore$  the root lies between  $x_9$  and  $x_8$

$$x_{10} = \frac{x_9 + x_8}{2} = 1.1151$$

$$f(x_{10}) = f(1.1151) = 0.0013 > 0$$

$$f(x_{10}) > 0, f(x_9) < 0$$

$\therefore$  the root lies between  $x_{10}$  and  $x_9$

$$x_{11} = \frac{x_{10} + x_9}{2} = \frac{1.1151 + 1.1132}{2} = 1.1141$$

$$f(x_{11}) = f(1.1141) = 0.00007 < 0$$

$$f(x_{11}) < 0, f(x_{10}) > 0$$

$\therefore$  the root lies between  $x_{11}$  and  $x_{10}$

$$x_{12} = \frac{x_{10} + x_{11}}{2} = 1.1146$$

$$f(x_{12}) = f(1.1146) = 0.0006 > 0$$

$\therefore$  the root lies between  $x_{11}$  and  $x_{12}$

$\therefore$  the approximate root of an

equation is 1.114 //

Regula-Falsi method :- (Method of false position)

To find the root of the equation  $f(x) = 0$  by this method consider two real numbers  $x_0, x_1$  such that  $f(x_0), f(x_1)$  have opposite signs. This implies the root lies between  $x_0$  and  $x_1$ .

$$f(x) = 0$$

$$x_0, x_1$$

$$f(x_0), f(x_1)$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$f(x_1) - f(x_0)$$

$$0 < 0.00007 = (1.1141 - 1.1132) f(1.1141) = (0.0007) f(1.1141)$$

$$x_2 = \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_0) - f(x_1)}$$

$$f(x_0) - f(x_1)$$

Case i:- If  $f(x_2)$  and  $f(x_0)$  have opposite signs then root is lie between  $x_2$  and  $x_0$ .

$$x_3 = \frac{x_2 f(x_0) - x_0 f(x_2)}{f(x_0) - f(x_2)}$$

$$f(x_0) - f(x_2)$$

Case ii:- If  $f(x_2)$  and  $f(x_1)$  have opposite signs then the root is lie between  $x_2$  and  $x_1$ .

$$x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$f(x_1) - f(x_2)$$



This process is repeated as many times as required to obtain the desired accuracy.

① Find a root of the equation  $x \log x_{10} = 1.2$  By using regula-falsi method.

Sol:- Given equation  $x \log x_{10} = 1.2$

$$x \log x - 1.2 = 0$$

$$\text{Let } f(x) = x \log x - 1.2 = 0$$

|        |      |         |        |
|--------|------|---------|--------|
| $x$    | 1    | 2       | 3      |
| $f(x)$ | -1.2 | -0.5979 | 0.2313 |
|        |      | <0      | >0     |

Root lies between 2 and 3

$$\text{Let } x_0 = 2 ; x_1 = 3$$

$$f(x_0) = -0.5979 ; f(x_1) = 0.2313$$

By Regula-falsi method,

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = 2.7210$$

$$f(x_2) = f(2.7210) = -0.0171 < 0$$

$$f(x_2) < 0, f(x_1) > 0$$

$\therefore$  the root lies between  $x_2$  and  $x_1$

$$x_1 = 3 ; x_2 = 2.7210$$

$$f(x_1) = 0.2313 ; f(x_2) = -0.0171$$

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

$$x_3 = 2.7402$$

$$f(x_3) = f(2.7402) = -0.00038 < 0$$

$$f(x_3) < 0, f(x_1) > 0$$

$\therefore$  the root lies between  $x_3$  &  $x_1$ .

$$x_3 = 2.7402 ; x_1 = 3$$

$$f(x_3) = -0.0003 ; f(x_1) = 0.2313$$

$$x_4 = \frac{x_1 f(x_3) - x_3 f(x_1)}{f(x_3) - f(x_1)}$$

$$x_4 = 2.7405$$

$$f(x_4) = f(2.7405) = -0.00012 < 0$$

$\therefore$  the approximate root of an equation  $\boxed{2.7405}$

② Find the approximate root of the equation  $e^x \sin x = 1$ .

Sol:- Given equation  $e^x \sin x = 1$   
 $e^x \sin x - 1 = 0$

$$\text{Let } f(x) = e^x \sin x - 1 = 0$$

$$f(0) = 0 ; f(2) = 2.718 - 1 = 1.718$$

$$f(x) = (1) - 1 = 0 ; f(2) = 2.718 - 1 = 1.718$$

$$<0 \quad >0$$

Root lies between 0 and 1

$$\text{Let } x_0 = 0 ; x_1 = 1$$

$$f(x_0) = -1 ; f(x_1) = 1.2873$$

By Regula-falsi method

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = 0.4371$$

$$f(x_2) = f(0.4371) = -0.3446$$

$$f(x_2) < 0, f(x_1) > 0$$

$\therefore$  the root lies between  $x_2$  and  $x_1$ .



$$x_2 = 0.4371$$

$$x_1 = 1$$

$$f(x_2) = -0.3446$$

$$f(x_1) = 1.2873$$

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)}$$

$$x_3 = 0.5559$$

$$f(x_3) = f(0.5559) = -0.0799 < 0$$

$$f(x_3) < 0, f(x_1) > 0$$

∴ the root lies between  $x_3$  &  $x_1$

$$x_3 = 0.5559$$

$$x_1 = 1$$

$$f(x_3) = -0.0799$$

$$f(x_1) = 1.2873$$

$$x_4 = \frac{x_1 f(x_3) - x_3 f(x_1)}{f(x_3) - f(x_1)}$$

$$x_4 = 0.5818$$

$$f(x_4) = f(0.5818) = -0.0167 < 0$$

$$f(x_4) < 0, f(x_1) > 0$$

∴ the root lies between  $x_4$  and  $x_1$

$$x_4 = 0.5818$$

$$x_1 = 1$$

$$f(x_4) = -0.0167$$

$$f(x_1) = 1.2873$$

$$x_5 = \frac{x_1 f(x_4) - x_4 f(x_1)}{f(x_4) - f(x_1)} = 0.5871$$

$$f(x_5) = f(0.5871) = -0.9815$$

$$f(x_5) < 0, f(x_1) > 0$$

∴ the root lies between  $x_5$  and  $x_1$

$$x_6 = \frac{x_1 f(x_5) - x_5 f(x_1)}{f(x_5) - f(x_1)}$$

$$x_6 = 0.5859$$

$$f(x_6) = -0.0065 < 0$$

$$f(x_6) < 0; f(x_1) > 0$$

∴ the roots lie between  $x_6$  and  $x_1$

$$x_7 = \frac{x_1 f(x_6) - x_6 f(x_1)}{f(x_6) - f(x_1)}$$

$$x_7 = 0.5879$$

$$f(x_7) = -0.00158 < 0$$

$$f(x_7) < 0, f(x_1) > 0$$

∴ the root lies between  $x_7$  &  $x_1$

$$x_8 = \frac{x_1 f(x_7) - x_7 f(x_1)}{f(x_7) - f(x_1)}$$

$$x_8 = 0.5874$$

∴ the approximate roots of the equation is 0.587

③ By using Regula falsi method find the approximate root of the equation  $2x^7 + x^5 + 1$  correct upto two decimal places near  $a = -1$  and  $b = 1$ .

Sol Given equ  $2x^7 + x^5 + 1$

$$\text{let } f(x) = 2x^7 + x^5 + 1$$

$$f(x) = 2x^7 + x^5 + 1$$

the root lies between  $x_0$  &  $x_1$

$$f(x_0) = -2; f(x_1) = 4$$

∴ the root lies between  $x_0$  &  $x_1$

By Regula falsi method

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = -0.333$$



$$f(x_2) = f(-0.333) = 0.9949 > 0$$

$$f(x_2) > 0, f(x_0) < 0$$

∴ the root lies b/w  $x_2$  and  $x_0$

$$x_3 = \frac{x_0 f(x_2) - x_2 f(x_0)}{f(x_2) - f(x_0)}$$

$$x_3 = -0.5547$$

$$f(x_3) = f(-0.5547) = 0.9151 > 0$$

$$f(x_3) > 0, f(x_0) < 0$$

∴ the root lies b/w  $x_3$  and  $x_0$

$$x_3 = -0.5547 \quad x_0 = -1$$

$$f(x_3) = 0.9151 \quad f(x_0) = -2$$

$$x_4 = \frac{x_0 f(x_3) - x_3 f(x_0)}{f(x_3) - f(x_0)}$$

$$x_4 = -0.6944$$

$$f(x_4) = 0.6828 > 0$$

$$f(x_4) > 0, f(x_0) < 0$$

∴ the root lies b/w  $x_4$  and  $x_0$

$$x_4 = -0.6944 \quad x_0 = -1$$

$$f(x_4) = 0.6828 \quad f(x_0) = -2$$

$$x_5 = \frac{x_0 f(x_4) - x_4 f(x_0)}{f(x_4) - f(x_0)}$$

$$x_5 = -0.7721$$

$$f(x_5) = f(-0.7721) = 0.3984 > 0$$

$$f(x_5) > 0, f(x_0) < 0$$

∴ the root lies between  $x_5$  and  $x_0$

$$x_5 = -0.7721 \quad x_0 = -1$$

$$f(x_5) = 0.3984 \quad f(x_0) = -2$$

$$x_6 = \frac{x_0 f(x_5) - x_5 f(x_0)}{f(x_5) - f(x_0)}$$

$$x_6 = -0.8099$$

$$f(x_6) = f(-0.8099) = 0.1943 > 0$$

$$f(x_6) > 0, f(x_0) < 0$$

∴ the root lies between  $x_6$  and  $x_0$

$$x_6 = -0.8099 \quad x_0 = -1$$

$$f(x_6) = 0.1943 \quad f(x_0) = -2$$

$$x_7 = \frac{x_0 f(x_6) - x_6 f(x_0)}{f(x_6) - f(x_0)}$$

$$x_7 = -0.8267$$

$$f(x_7) = f(-0.8267) = 0.0860 > 0$$

$$f(x_7) > 0, f(x_0) < 0$$

∴ the root lies b/w  $x_7$  and  $x_0$

$$x_7 = -0.8267 \quad x_0 = -1$$

$$f(x_7) = 0.0860 \quad f(x_0) = -2$$

$$x_8 = \frac{x_0 f(x_7) - x_7 f(x_0)}{f(x_7) - f(x_0)}$$

$$x_8 = -0.8338$$

$$f(x_8) = f(-0.8338) = 0.0366 > 0$$

$$f(x_8) > 0, f(x_0) < 0$$

∴ the root lies b/w  $x_8$  and  $x_0$

$$x_8 = -0.8338 \quad x_0 = -1$$

$$f(x_8) = 0.0366 \quad f(x_0) = -2$$

$$x_9 = \frac{x_0 f(x_8) - x_8 f(x_0)}{f(x_8) - f(x_0)}$$

$$x_9 = -0.8367$$

$$f(x_9) = 0.0157$$

∴ the approximate root of the equation is  $-0.8367$

Iteration method: (method of successive approximation)

Given equation  $f(x)=0$ ; which can take in the form

$$x = \phi(x) \rightarrow (1)$$

let  $[a, b]$  be an interval containing root  $\alpha, \sqrt{\alpha}$  of equation (1)

if  $|\phi'(x)| < 1$  for all  $x \in (a, b)$ ; then  $x_n = \phi(x_{n-1})$ ;  $n \geq 1$ .

To find the approximate value of the root we start initial approximation  $x_0$

$$x_1 = \phi(x_0)$$

$$x_2 = \phi(x_1)$$

$$x_3 = \phi(x_2)$$

$$x_n = \phi(x_{n-1})$$

and  $x_n$  is an approximate root.

① Find root of the equation  $x^3 - 2x - 5$  which lies near  $x=2$  by using iteration method.

Sol: - Given  $x^3 - 2x - 5 = 0$

$$x^3 = 2x + 5$$

$$x = (2x + 5)^{1/3}$$

$$\text{let } \phi(x) = (2x + 5)^{1/3}$$

$$\phi'(x) = \frac{1}{3}(2x + 5)^{-2/3} \times 2$$

$$= \frac{2}{3 \times (2x + 5)^{2/3}}$$

At  $x=2$

$$|\phi'(x)| = \left| \frac{2}{3 \times (2 \times 2 + 5)^{2/3}} \right| = 0.154 < 1$$

$$\therefore \phi(x) = (2x + 5)^{1/3}$$

$$x_0 = 2$$

$$x_n = \phi(x_{n-1})$$

$$x_1 = \phi(x_0) = 2.08$$

$$x_2 = \phi(x_1) = 2.0923$$

$$x_3 = \phi(x_2) = 2.0942$$

$$x_4 = \phi(x_3) = 2.0944$$

$\therefore$  The approximate root of the equation is 2.0944

$$1677.0 = 2x$$

$$0 < 1687.0 = (1677.0 - 2x) = (2x)$$

$$0 > (0x)7, 0 < (2x)7$$

$$1 = 0x$$

$$1677.0 = 2x$$

$$5 = (0x)7$$

$$1687.0 = (2x)7$$



② Find the root of the equation

$$f(x) = x^3 - 3x + 1 \text{ correct upto three}$$

decimal places starting with  $x=1$   $\therefore$  the approximate root of the equation is 0.383.  
by using successive approximation method.

Sol:- Given that  $x^3 - 3x + 1$   
 $x^3 = 3x - 1$   
 $x = (3x - 1)^{1/3}$

Let  $\phi(x) = (3x - 1)^{1/3}$   
 $\phi'(x) = \frac{1}{3}(3x - 1)^{-2/3} \times 3$   
 $\phi'(x) = \frac{3}{2(3x - 1)^{1/2}}$

At  $x=1$

$$|\phi'(x)| = \left| \frac{3}{2(3(1) - 1)^{1/2}} \right| = 1.06 > 1$$

$$\therefore \phi(x) \neq (3x - 1)^{1/3}$$

$$3x = x^2 + 1$$

$$x = \frac{x^2 + 1}{3}$$

$$\phi(x) = \frac{x^2 + 1}{3}$$

$$\phi'(x) = \frac{2x}{3}$$

At  $x=1$

$$|\phi'(x)| = \left| \frac{2(1)}{3} \right| = \frac{2}{3} = 0.666$$

$$\phi(x) = \frac{x^2 + 1}{3}$$

$$x_0 = 1, x_n = \phi(x_{n-1})$$

$$x_1 = \phi(x_0) = 0.666$$

$$x_2 = \phi(x_1) = 0.481$$

$$x_3 = \phi(x_2) = 0.410$$

$$x_4 = \phi(x_3) = 0.389$$

$$x_5 = \phi(x_4) = 0.383$$

③ Find the root of the equation

$$x^4 - x - 10 = 0$$

A Given equation  $x^4 - x - 10 = 0$

$$\text{Let } f(x) = x^4 - x - 10 = 0$$

|        |     |     |     |
|--------|-----|-----|-----|
| $x$    | 0   | 1   | 2   |
| $f(x)$ | -10 | -10 | -4  |
|        |     | < 0 | > 0 |

$\therefore$  the root lies between 1 and 2

Given  $x^4 - x - 10 = 0$

$$x^4 = x + 10$$

$$x = (x + 10)^{1/4}$$

$$\text{Let } \phi(x) = (x + 10)^{1/4}$$

$$\phi'(x) = \frac{1}{4}(x + 10)^{-3/4} \times 1$$

$$\phi'(x) = \frac{1}{4(x + 10)^{3/4}}$$

At  $x=0$

$$|\phi'(x)| = \left| \frac{1}{4(0 + 10)^{3/4}} \right| = 0.044 < 1$$

$$\therefore \phi(x) = (x + 10)^{1/4}$$

$$x_0 = 1.5, x_n = \phi(x_{n-1})$$

$$x_1 = \phi(x_0) = 1.841$$

$$x_2 = \phi(x_1) = 1.855$$

$$x_3 = \phi(x_2) = 1.855$$

$\therefore$  the approximate value of equation is 1.855 //

Q. Using iteration method find a root of the equation.

$$3x = \cos x + 1 \quad \text{in } (0, 1)$$

Sol:- Given that  $3x = \cos x + 1$

$$x = \frac{\cos x + 1}{3}$$

$$\text{Let } \phi(x) = \frac{\cos x + 1}{3}$$

$$\phi'(x) = \frac{-\sin x}{3}$$

at  $x=0$

$$|\phi'(x)| = \left| \frac{-\sin x}{3} \right| = 0 < 1$$

$$\therefore \phi(x) = \frac{\cos x + 1}{3}$$

$$x_0 = 0.5, \quad x_n = \phi(x_{n-1})$$

$$x_1 = \phi(x_0) = 0.625$$

$$x_2 = \phi(x_1) = 0.603$$

$$x_3 = \phi(x_2) = 0.607$$

$$x_4 = \phi(x_3) = 0.607$$

$\therefore$  the approximate root of the equation 0.607 //

Newton-Raphson Method:-

Newton Raphson method is a powerfull and elegant method to find the root of an equation. This method is generally used to improve the results obtained by the previous method.

Let  $x_0$  be an approximate root of the equation  $f(x)=0$  then

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}; \quad n \geq 1$$

For  $n=1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



for  $n=2$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$\therefore$  The approximate root is given by  $x_n$ .

① By Newton Raphson method, correct upto three decimal places find a root of an equation.

$x^6 - x^4 + x^3 + 1$  near 1.5 correct upto sol:- Given equation three decimal spaces.

Sol:- Given equation

$$x^6 - x^4 + x^3 + 1$$

$$x^6 - x^4 + x^3 - 1 = 0$$

$$\text{let } f(x) = x^6 - x^4 + x^3 - 1$$

$$f'(x) = 6x^5 - 4x^3 + 3x^2$$

$$\text{let } x_0 = 1.5$$

By Newton Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1.4228$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 1.4045$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 1.4036$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$x_4 = 1.4036$$

$\therefore$  The approximate root of the equation 1.4036

② using Newton Raphson method find the root of the equation

$$x \sin x + \cos x = 0 \text{ near } x = \pi$$

$$x \sin x + \cos x = 0$$

$$f(x) = x \sin x + \cos x$$

$$f'(x) = x \cos x + \sin x - \sin x$$

$$f'(x) = x \cos x$$

$$\text{let } x_0 = \pi$$

By Newton Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 2.823$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 2.798$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 2.798$$

$\therefore$  The approximate root of the equation is 2.798

③ Find a real root of the equation

$$x e^x - \cos x = 0$$

Sol:- Given equation

$$x e^x - \cos x$$

$$f(x) = x e^x - \cos x$$

$$f'(x) = x e^x + e^x + \sin x$$

$$f(x) = xe^x + \sin x$$

$$x \quad 0 \quad 1 \quad 2$$

$$f(x) \quad -1 \quad 2.17$$

$$26 > 0$$

∴ the root lies between 0 and 1

$$\text{let } x_0 = 0.5$$

By Newton Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 0.518$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 0.517$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 0.517$$

∴ the approximate root of the given equation is 0.517

④ Find the approximate root of the equation  $x \log_{10} x = 1.2$  by using Newton Raphson method.

Sol: - Given equation  $x \log_{10} x = 1.2$

$$f(x) = x \log_{10} x = 1.2$$

$$x \log_{10} x - 1.2 = 0$$

$$f(x) \quad 0 \quad 1 \quad 2 \quad 3$$

$$f(x) \quad 0 \quad -1.2 \quad -0.59 \quad 0.23$$

$$< 0 \quad > 0$$

∴ the root lies between 2 and 3

$$f(x) = x \log_{10} x - 1.2$$

$$f(x) = x \log_{10} x \log_{10} e - 1.2$$

$$= x \log_{10} e (0.4343) - 1.2$$

$$f(x) = (0.4343) (x \cdot \frac{1}{x} \log x)$$

$$= 0.4343 (1 + \log x)$$

$$f(x) = 0.4343 + 0.4343 \log x$$

$$\text{let } x_0 = 2.5$$

By Newton Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 2.7465$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 2.7406$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 2.7406$$

∴ the approximate root of the equation 2.7406

⑤ using Newton Raphson find the square root of the and hence find  $\sqrt{24}$ .

Sol: - let  $x = \sqrt{N}$ ; where N is a number whose square is formed

$$x^2 = N$$

$$x^2 - N = 0$$

$$\text{let } f(x) = x^2 - N$$

$$f'(x) = 2x$$

By Newton Raphson



$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}; i=0,1,2, \dots$$

$$N=29$$

$$x_0 = 3.0123$$

$$= x_i - \frac{(x_i^3 - N)}{3x_i^2}$$

$$= \frac{2x_i^3 - x_i^3 + N}{3x_i^2}$$

$$x_{i+1} = \frac{2x_i^3 + 29}{3x_i^2} = 3.0123$$

$$x_1 = \frac{2x_0^3 + 29}{3x_0^2} = 3.0123$$

$$x_{i+1} = \frac{x_i^2 + N}{2x_i}; i=0,1,2, \dots$$

$$x_2 = \frac{2x_1^3 + 29}{3x_1^2} = 3.0123$$

$$\text{let } N=24$$

$$x_0 = 4.8989$$

$$x_{i+1} = \frac{x_i^2 + 24}{2x_i}$$

$$x_1 = \frac{x_0^2 + 24}{2x_0} = 4.8989$$

$$x_2 = \frac{x_1^2 + 24}{2x_1} = 4.8989$$

The root of the member 3.0123,

⑦ By using Newton Raphson method. find a reciprocal of a member and hence find reciprocal of 18.

Sol:- Let  $x = \frac{1}{N}$ , where N is a number whose reciprocal to formed.

The approximate root of eqn?

$$x = \frac{1}{N}$$

$$Nx = 1$$

$$Nx - 1 = 0$$

⑧ By using Newton Raphson method

find a cube root of a member

$$\text{let } f(x) = Nx - 1$$

$$f'(x) = N$$

and hence find  $\sqrt[3]{29}$

Let  $x = \sqrt[3]{N}$ ; where N is a number

By Newton Raphson method

whose cube to formed

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}; i=0,1,2, \dots$$

$$x^3 = N$$

$$x^3 - N = 0$$

$$\text{let } f(x) = x^3 - N$$

$$f'(x) = 3x^2$$

By Newton Raphson method

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}; i=0,1,2, \dots$$

$$= x_i - \frac{(x_i^3 - N)}{3x_i^2}$$

$$x_{i+1} = \frac{2x_i^3 + N}{3x_i^2}; i=0,1,2, \dots$$

$$x_{i+1} = \frac{1}{N}$$

$$N=18 = \frac{1}{18} = 0.0555$$

$$x_1 = \frac{1}{N} = 0.0555$$

$$x_2 = \frac{1}{N} = 0.0555$$



INTERPOLATION

The word interpolation means the reading in between, about the desired quantity. From known model  $y=f(x)$ , one can tabulate the values of  $y(x)$  for various values of  $x$ .

From a known data  $(x_i, y_i)$ ;  $i=0, 1, 2, 3, \dots$  we have to determine  $y=f(x)$ .

If  $x_i - x_{i-1} = h$ ;  $\forall i=1, 2, 3, \dots$  then the data are called "equally spaced". and if  $x_i - x_{i-1} \neq h$  for atleast one  $i$ , then the data are called "unequally spaced".

Errors in interpolation:-

1) Inherent errors:-

The errors already involved in the statement of the problem are known as inherent errors. These errors arise due to inaccurate data or the limitations of the mathematical tables.

These errors cannot be removed for whatever choice of a numerical method and hence to be minimised.

either by redefining in this statement

2) Rounding errors:-

These errors occur when we round off a number to a certain no. of decimal places. These errors cannot be avoided since we cannot use infinite no. of digits in computational problem.

3) Truncation errors:-

These errors occur if an infinite series truncated to definite no. of steps.

Ex:-  $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$  (say)  $\frac{x^9}{9!} = 1.4 \times 10^{-6}$



$$y_3 = x^3 - \frac{x^3}{3!} + \frac{x^5}{5!}$$

is the truncated series and thus  $y - y_3$  is the truncated error.

4) Absolute relative and percentage errors:-

let  $y_n$  be an approximate value,  $y_e$  be the exact value of  $y$  at  $n$ th state.

(i) absolute error:-

$$\text{the absolute error (A.E)} = |y_n - y_e|$$

(ii) relative error:-

$$\text{the relative error (R.E)} = \frac{|y_n - y_e|}{y_e}$$

(iii) percentage error:-

$$\text{Percentage error (P.E)} = \frac{|y_n - y_e|}{y_e} \times 100$$

example:- The approximate value of  $\log 2$  is computed by numerical methods is  $y_n = 0.702351$ , while the exact value

is  $y_e = 0.693147$ . Find the absolute error, relative error and

percentage error of  $y$ .

Sol:- Given that  $\log 2$ ,  $y_n = 0.702351$ ,  $y_e = 0.693147$

$$(i) \text{ Absolute error (A.E)} = |y_n - y_e|$$

$$= |0.702351 - 0.693147|$$

$$= 0.009204$$

$$(ii) \text{ relative error (R.E)} = \frac{|y_n - y_e|}{y_e}$$

$$= \frac{|0.702351 - 0.693147|}{0.693147}$$

$$= \frac{0.009204}{0.693147}$$

$$(iii) \text{ percentage error (P.E)} = \frac{|y_n - y_e|}{y_e} \times 100$$

$$= 0.013278 \times 100$$

$$= 1.3278$$

Ex-2:- The approximate value of  $\sin(1.5)$  is computed from the first term of the series  $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$  is found to be  $y_3 = 1.00078$ , while the exact value is,  $y_e = 0.997495$ . Find the absolute error, relative error, and percentage error of  $y_3$ .

Sol:- Given that  $y_3 = 1.00078$ ,  $y_e = 0.997495$

$$\text{Absolute error} = |1.00078 - 0.997495|$$

$$= 0.003285$$

$$\text{Relative error} = \frac{|1.00078 - 0.997495|}{0.997495}$$

$$= 0.003293$$

$$\text{Percentage error} = \frac{|1.00078 - 0.997495|}{0.997495} \times 100$$

$$= 0.003293 \times 100 \Rightarrow 0.32932$$

Finite Differences:-

Let  $(x_i, y_i)$   $i = 0, 1, 2, 3, \dots, n$  be the equally spaced data of the unknown function.

Let  $x_0, x_1 = x_0 + h, x_2 = x_0 + 2h, x_3 = x_0 + 3h, \dots, x_n = x_0 + nh$

be an equally spaced points where the function values of  $f(x)$  be  $y_0, y_1, y_2, \dots, y_n$ .

Symbolic operators:-

(i) Forward shift operator ( $E$ ):-

The first order forward shift operator with stepsize  $h$  is, denoted by  $E$  or  $E_h$  and is given by

$$Ef(x) = f(x+h)$$

The second and higher order forward shift operators are given by

$$E^2 f(x) = f(x+2h)$$

$$E^3 f(x) = f(x+3h)$$

$$\dots \dots \dots E^k f(x) = f(x+kh)$$



### Backward shift operator ( $E^{-1}$ )

The first order backward shift operator with step size  $h$  is denoted by  $E^{-1}$  (or)  $E_h^{-1}$ . And is given by

$$E^{-1}f(x) = f(x-h)$$

The second and higher order backward shift operators are given by

$$E^{-2}f(x) = f(x-2h)$$

$$E^{-3}f(x) = f(x-3h)$$

$$E^{-k}f(x) = f(x-kh)$$

### Forward difference operator ( $\Delta$ )

The first order forward difference operator is denoted by  $\Delta$  and is given by

$$\Delta f(x) = f(x+h) - f(x)$$

The second order forward difference operator is given by

$$\Delta^2 f(x) = \Delta(\Delta f(x))$$

$$= \Delta(f(x+h) - f(x))$$

$$= \Delta f(x+h) - \Delta f(x)$$

$$\Delta^2 f(x) = f(x+2h) - f(x+h) - (f(x+h) - f(x))$$

$$\Delta^2 f(x) = f(x+2h) - 2f(x+h) + f(x)$$

$$\Delta^3 f(x) = \Delta(\Delta^2 f(x))$$

$$= \Delta(f(x+2h) - 2f(x+h) + f(x))$$

$$= \Delta f(x+2h) - 2\Delta f(x+h) + \Delta f(x)$$

$$= f(x+3h) - f(x+2h) - 2(f(x+2h) - f(x+h)) + f(x+h) - f(x)$$

$$= f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)$$

### Backward difference operator ( $\nabla$ )

The first order backward difference operator with step size  $h$  is denoted by  $\nabla$  and is given by

$$\nabla f(x) = f(x) - f(x-h)$$

The second order backward difference operator is given by

$$\nabla^2 f(x) = \nabla(\nabla f(x))$$

$$= \nabla(f(x) - f(x-h))$$

$$= \nabla f(x) - \nabla f(x-h)$$

$$= f(x) - f(x-h) - (f(x-h) - f(x-2h))$$

$$\nabla^2 f(x) = f(x) - 2f(x-h) + f(x-2h)$$

$$\nabla^3 f(x) = \nabla(\nabla^2 f(x))$$

$$= \nabla(f(x) - 2f(x-h) + f(x-2h))$$

$$= \nabla f(x) - \nabla 2f(x-h) + \nabla f(x-2h)$$

$$= f(x) - f(x-h) - 2(f(x-h) - f(x-2h)) + f(x-2h) - f(x-3h)$$

$$= f(x) - 3f(x-h) + 3f(x-2h) - f(x-3h)$$

Central difference operator ( $\delta$ ):-

The central difference operator is given by

$$\delta f(x) = f(x+h/2) - f(x-h/2)$$

$$= E^{1/2} f(x) - E^{-1/2} f(x)$$

$$(E^{1/2} - E^{-1/2}) f(x) = (E^{1/2} - E^{-1/2}) f(x)$$

$$\therefore \delta = E^{1/2} - E^{-1/2}$$

Average operator ( $\mu$ ):-

The average operator is given by

$$\mu f(x) = \frac{1}{2} [f(x+h/2) + f(x-h/2)]$$

$$= \frac{1}{2} [E^{1/2} f(x) + E^{-1/2} f(x)]$$

$$\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$$

$$\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$$

$$(E^{1/2} - E^{-1/2}) f(x) = (E^{1/2} - E^{-1/2}) f(x)$$



① show that  $(1+\Delta)(1-\nabla) = -1$  (or)  $1+\Delta = \frac{1}{1-\nabla}$

A L.H.S =  $(1+\Delta)(1-\nabla)$

$$= (1+E-1)(1-(1-E^{-1}))$$

$$= (E)(1-1+E^{-1})$$

$$= EE^{-1}$$

$$= 1 \Rightarrow R.H.S$$

$$\therefore (1+\Delta)(1-\nabla) = -1$$

② show that  $\mu^2 = \frac{\delta^2}{4} + 1$

Sol:- we know  $\mu = \frac{1}{2} [E^{1/2} + E^{-1/2}]$

$$\mu^2 = \frac{1}{4} [E^{1/2} + E^{-1/2}]^2$$

$$= \frac{1}{4} [E + E^{-1} + 2]$$

$$= \frac{1}{4} [(E + E^{-1} - 2) + 4]$$

$$= \frac{1}{4} [\delta^2 + 4] \quad \because \delta = E^{1/2} - E^{-1/2}$$

$$= \frac{\delta^2}{4} + 1$$

$$= R.H.S$$

③ show that  $\Delta \nabla = \delta^2$

Sol:- L.H.S =  $\Delta \nabla$

$$= (E-1)(1-E^{-1})$$

$$= E - 1 - 1 + E^{-1}$$

$$= E + E^{-1} - 2$$

$$R.H.S = \delta^2$$

$$= (E^{1/2} - E^{-1/2})^2$$

$$= E + E^{-1} - 2$$

$$\therefore L.H.S = R.H.S$$

④ show that  $\Delta + \nabla = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$

Sol:-

$$L.H.S = \Delta + \nabla$$

$$= E - 1 + 1 - E^{-1}$$

$$= E - E^{-1}$$

$$\begin{aligned}\frac{\Delta}{\nabla} &= \frac{E-1}{1-E^{-1}} \\ &= \frac{E-1}{1-\frac{1}{E}} \\ &= \frac{E-1}{\frac{E-1}{E}} \\ &= E\end{aligned}$$

$$\frac{\Delta}{\Delta} = \frac{1}{E} = E^{-1}$$

$$R.H.S = E - E^{-1}$$

$$\therefore L.H.S = R.H.S$$

⑤ show that  $\mu\delta = \frac{1}{2}(\Delta + \nabla)$

Sol:- Given  $\mu\delta = \frac{1}{2}(\Delta + \nabla)$

$$L.H.S = \mu\delta$$

$$= \frac{1}{2}[E^{1/2} + E^{-1/2}][E^{1/2} - E^{-1/2}]$$

$$= \frac{1}{2}[E - E^{-1}]$$

$$R.H.S = \frac{1}{2}(\Delta + \nabla)$$

$$= \frac{1}{2}[E - 1 + 1 - E^{-1}]$$

$$= \frac{1}{2}[E - E^{-1}]$$

$$\therefore L.H.S = R.H.S$$

⑥ prove the relation between  $E$  and  $O$ .

Sol:- we know that  $Ef(x) = f(x+h)$

$$= f(x) + \frac{h}{1!}f'(x) + \frac{h^2}{2!}f''(x) + \dots$$

(Taylor's expansion)

$$= f(x) + \frac{h}{1!}Of(x) + \frac{h^2}{2!}O^2f(x) + \dots$$

$$= \left(1 + \frac{hO}{1!} + \frac{(hO)^2}{2!} + \dots\right)f(x)$$

$$Ef(x) = e^{hO}f(x)$$

$$\therefore \boxed{E = e^{hO}}$$

⑦ evaluate  $\Delta \sin x$ .

Sol:- Given that  $\Delta \sin x$



$$\text{let } f(x) = \sin x$$

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta \sin x = \sin(x+h) - \sin x$$

$$= 2 \cos\left(\frac{x+h}{2}\right) \sin\left(\frac{h}{2}\right)$$

$$= 2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)$$

⑧ Evaluate  $\Delta \sin(ax+b)$

Sol: Given that  $\Delta \sin(ax+b)$

$$\text{let } f(x) = \sin(ax+b)$$

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta \sin(ax+b) = \sin(ax+ah+b) - \sin(ax+b)$$

$$\Rightarrow 2 \cos\left(\frac{ax+ax+b+b+ah}{2}\right) \sin\left(\frac{ax+ah+b-ax-b}{2}\right)$$

$$= 2 \cos\left(\frac{2ax+2b+h}{2}\right) \sin\left(\frac{ah}{2}\right)$$

$$= 2 \cos\left((ax+b) + \frac{h}{2}\right) \sin\left(\frac{ah}{2}\right)$$

⑨ Evaluate  $\Delta^2 3e^x$

Sol: Given that  $\Delta^2 3e^x$

$$\text{let } f(x) = 3e^x$$

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta 3e^x = 3e^{x+h} - 3e^x$$

$$\Delta^2 3e^x = \Delta(\Delta 3e^x)$$

$$= \Delta(3e^{x+h} - 3e^x)$$

$$= \Delta 3e^{x+2h} - 3e^{x+h} - (3e^{x+h} - 3e^x)$$

$$= 3e^{x+2h} - 3e^{x+h} + 3e^x$$

$$= 3e^x(e^{2h} - 2e^h + 1)$$

⑩ Evaluate  $\Delta e^{ax} \log bx$

Sol: Given that  $\Delta e^{ax} \log bx$

$$\text{let } f(x) = e^{ax} \log bx$$

$$\Delta f(x) = f(x+h) - f(x)$$

$$\begin{aligned}\Delta e^{ax} \log bx &= e^{a(x+h)} \log b(x+h) - e^{ax} \log bx \\ &= e^{ax+ah} \log b(x+h) - e^{ax} \log bx \\ &= e^{ax} (e^{ah} \log b(x+h) - \log bx)\end{aligned}$$

⑩ Evaluate  $\Delta^2 \cos 2x$

Given that  $\Delta^2 \cos 2x$

$$\text{let } f(x) = \Delta^2 \cos 2x$$

$$\Delta^2 \cos 2x = \cos 2(x+h) - \cos 2x$$

$$\Delta \cos 2x = \cos 2x + 2h - \cos 2x$$

$$\Delta^2 \cos 2x = \Delta(\Delta \cos 2x)$$

$$= \Delta(\cos 2x + 2h - \cos 2x)$$

$$= \cos 2x$$

$$\Delta^2 \cos 2x = \Delta(\Delta \cos 2x)$$

$$= \Delta(\cos 2x + 2h - \cos 2x)$$

$$= \cos 2(x+h) + 2h - (\cos 2x + 2h - \cos 2x)$$

$$= \cos 2x + 2h - \cos 2x - 2h + \cos 2x$$

$$= \cos 2x + 2h - \cos 2x - 2h + \cos 2x$$

$$\therefore \left( 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} \right) = (\cos(2x+2h) + \cos 2x) - 2 \cos(2x+2h)$$

$$= 2 \cos(2x+2h) \cos 2h - 2 \cos(2x+2h)$$

$$= 2 \cos(2x+2h) (\cos 2h - 1)$$

$$= 2 \cos(2x+2h) (\cos 2h - 1)$$



### Forward Difference Table:-

| x     | y     | $\Delta y$   | $\Delta^2 y$   | $\Delta^3 y$   | $\Delta^4 y$   | $\Delta^5 y$   |
|-------|-------|--------------|----------------|----------------|----------------|----------------|
| $x_0$ | $y_0$ |              |                |                |                |                |
| $x_1$ | $y_1$ | $\Delta y_0$ | $\Delta^2 y_0$ |                |                |                |
| $x_2$ | $y_2$ | $\Delta y_1$ | $\Delta^2 y_1$ | $\Delta^3 y_0$ | $\Delta^4 y_0$ |                |
| $x_3$ | $y_3$ | $\Delta y_2$ | $\Delta^2 y_2$ | $\Delta^3 y_1$ | $\Delta^4 y_1$ | $\Delta^5 y_0$ |
| $x_4$ | $y_4$ | $\Delta y_3$ | $\Delta^2 y_3$ | $\Delta^3 y_2$ | $\Delta^4 y_2$ |                |
| $x_5$ | $y_5$ | $\Delta y_4$ | $\Delta^2 y_4$ |                |                |                |

### Backward Difference Table:-

| x     | y     | $\nabla y$   | $\nabla^2 y$   | $\nabla^3 y$   | $\nabla^4 y$   | $\nabla^5 y$   |
|-------|-------|--------------|----------------|----------------|----------------|----------------|
| $x_0$ | $y_0$ |              |                |                |                |                |
| $x_1$ | $y_1$ | $\nabla y_1$ | $\nabla^2 y_2$ |                |                |                |
| $x_2$ | $y_2$ | $\nabla y_2$ | $\nabla^2 y_3$ | $\nabla^3 y_3$ | $\nabla^4 y_4$ |                |
| $x_3$ | $y_3$ | $\nabla y_3$ | $\nabla^2 y_4$ | $\nabla^3 y_4$ | $\nabla^4 y_5$ | $\nabla^5 y_5$ |
| $x_4$ | $y_4$ | $\nabla y_4$ | $\nabla^2 y_5$ | $\nabla^3 y_5$ |                |                |
| $x_5$ | $y_5$ | $\nabla y_5$ |                |                |                |                |

### Central Difference Table:-

| x     | y     | $\delta y$       | $\delta^2 y$   | $\delta^3 y$       | $\delta^4 y$   | $\delta^5 y$       |
|-------|-------|------------------|----------------|--------------------|----------------|--------------------|
| $x_0$ | $y_0$ |                  |                |                    |                |                    |
| $x_1$ | $y_1$ | $\delta y_{1/2}$ | $\delta^2 y_1$ | $\delta^3 y_{3/2}$ | $\delta^4 y_2$ | $\delta^5 y_{5/2}$ |
| $x_2$ | $y_2$ | $\delta y_{3/2}$ | $\delta^2 y_2$ | $\delta^3 y_{5/2}$ | $\delta^4 y_3$ | $\delta^5 y_{7/2}$ |
| $x_3$ | $y_3$ | $\delta y_{5/2}$ | $\delta^2 y_3$ | $\delta^3 y_{7/2}$ | $\delta^4 y_4$ | $\delta^5 y_{9/2}$ |
| $x_4$ | $y_4$ | $\delta y_{7/2}$ | $\delta^2 y_4$ | $\delta^3 y_{9/2}$ |                |                    |
| $x_5$ | $y_5$ | $\delta y_{9/2}$ |                |                    |                |                    |

### Newton forward difference formula:-

In forward difference formula the point  $x$  when  $f(x)$  is estimated is expressed as

$$x = x_0 + p h$$

$$f(x) = f(x_0 + p h)$$

$$= E^p f(x_0)$$

$$= (1 + \Delta)^p f(x_0) \quad (\because \Delta = E - 1 \Rightarrow E = 1 + \Delta)$$

$$= (p_0 + p_1 \Delta + p_2 \Delta^2 + p_3 \Delta^3 + \dots) f(x_0)$$

$$= \left( 1 + p \Delta + \frac{p(p-1)}{2!} \Delta^2 + \frac{p(p-1)(p-2)}{3!} \Delta^3 + \dots \right) f(x_0)$$

$$y(x) = \left( 1 + p \Delta + \frac{p(p-1)}{2!} \Delta^2 + \frac{p(p-1)(p-2)}{3!} \Delta^3 + \dots \right) y_0$$

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

$$\text{where } p = \frac{x - x_0}{h}$$

which is called NFD

### Newton Backward difference formula:-

In backward difference formula the point  $x$  when  $f(x)$  is estimated is expressed as

$$x = x_n + p h$$

$$f(x) = f(x_n + p h)$$

$$= E^p f(x_n)$$

$$= (E^{-1})^p f(x_n)$$

$$= (1 - \nabla)^p f(x_n)$$

$$= \left( 1 + p \nabla + \frac{p(p+1)}{2!} \nabla^2 + \frac{p(p+1)(p+2)}{3!} \nabla^3 + \dots \right) f(x_n)$$

$$y(x) = \left( 1 + p \nabla + \frac{p(p+1)}{2!} \nabla^2 + \frac{p(p+1)(p+2)}{3!} \nabla^3 + \dots \right) y_n$$

$$\therefore y(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$\text{where } p = \frac{x - x_n}{h}$$

which is called NBD



For the following data extrapolate the value of  $y$  at  $1.2$  by using Newton Forward and Backward difference formula.

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| $x$    | 0.2    | 0.4    | 0.6    | 0.8    | 1.0    |
| $y(x)$ | 3.1224 | 3.2499 | 3.3825 | 3.5205 | 3.6642 |

Sol:- difference table:-

| $x$       | $y$          | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|-----------|--------------|------------|--------------|--------------|--------------|
| 0.2 $x_0$ | 3.1224 $y_0$ | 0.1275     |              |              |              |
| 0.4 $x_1$ | 3.2499 $y_1$ | 0.1326     | 0.0051       |              |              |
| 0.6 $x_2$ | 3.3825 $y_2$ | 0.1380     | 0.0054       | 0.0003       |              |
| 0.8 $x_3$ | 3.5205 $y_3$ | 0.1437     | 0.0057       | 0.0003       |              |
| 1.0 $x_4$ | 3.6642 $y_4$ |            |              |              |              |

(i) NEDE:-

$$x = 1.2, x_0 = 0.2, h = 0.2$$

$$p = \frac{x - x_0}{h} = \frac{1.2 - 0.2}{0.2} = 5$$

FOR NEDE

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

$$y(x) = 3.1224 + (5)(0.1275) + \frac{5 \times 4}{2 \times 1} (0.0051) + \frac{5 \times 4 \times 3}{3 \times 2 \times 1} (0.0003) + \dots$$

$$y(x) = 3.8139$$

(ii) NBDE:-

$$x = 1.2, x_4 = 1.0, h = 0.2$$

$$p = \frac{x - x_4}{h} = \frac{1.2 - 1.0}{0.2} = 1$$

FOR NBDE

$$y(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$= 3.6642 + (1 \times 0.1437) + \frac{1 \times 2}{2 \times 1} (0.0057) + \frac{1 \times 2 \times 3}{3 \times 2 \times 1} (0.0003) + \dots$$

$$= 3.8139$$

② using Newton forward difference formula estimate  $y(0.12)$  from the following table

|        |       |       |       |       |       |
|--------|-------|-------|-------|-------|-------|
| $x$    | 0.10  | 0.15  | 0.20  | 0.25  | 0.30  |
| $y(x)$ | 0.656 | 0.522 | 0.410 | 0.316 | 0.240 |

Sol:-

| $x$        | $y$         | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|------------|-------------|------------|--------------|--------------|--------------|
| 0.10 $x_0$ | 0.656 $y_0$ | -0.134     |              |              |              |
| 0.15 $x_1$ | 0.522 $y_1$ | -0.112     | 0.022        |              |              |
| 0.20 $x_2$ | 0.410 $y_2$ | -0.094     | 0.018        | -0.004       |              |
| 0.25 $x_3$ | 0.316 $y_3$ | -0.076     | 0.018        | 0            | 0.004        |
| 0.30 $x_4$ | 0.240 $y_4$ |            |              |              |              |

(i) NEDE:-

$$x = 0.12, x_0 = 0.10, h = 0.05$$

$$p = \frac{x - x_0}{h} = \frac{0.12 - 0.10}{0.05} = \frac{0.02}{0.05} = 0.4$$

For NEDE,

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

$$y(x) = 0.656 + (0.4)(-0.134) + \frac{(0.4)(-0.4)}{2} (0.022) + \frac{(0.4)(-0.4)(-0.8)}{6} (-0.004)$$

$$y(x) = 0.59899$$



Q Fit a polynomial from the given data

x 1 2 3 4

y 2 5 16 41

By using WNFDE UNBOE

Sol:- difference table:-

| x       | y        | $\Delta y$   | $\Delta^2 y$   | $\Delta^3 y$   |
|---------|----------|--------------|----------------|----------------|
| 1 $x_0$ | 2 $y_0$  | $\Delta y_0$ | $\Delta^2 y_0$ | $\Delta^3 y_0$ |
| 2 $x_1$ | 5 $y_1$  | 3            | 8              | 6              |
| 3 $x_2$ | 16 $y_2$ | 11           | 14             | $\Delta^3 y_1$ |
| 4 $x_3$ | 41 $y_3$ | 25           | $\Delta^2 y_1$ |                |

W NFDE:-

$$x = x; x_0 = 1; p = \frac{x - x_0}{h} = \frac{x - 1}{1} = x - 1$$

$$h = 1$$

By NFDE

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

$$y(x) = 2 + 3x - 3 + \frac{(x-1)(x-2)}{2} (8) + \frac{(x-1)(x-2)(x-3)}{6} (6)$$

$$y(x) = 2 + 3x - 3 + 4(x^2 - 3x + 2) + (x^2 - 3x + 2)(x-3)$$

$$y(x) = 3x - 1 + 4x^2 - 12x + 8 + x^3 - 3x^2 + 2x - 3x^2 + 9x - 6$$

$$y(x) = x^3 - 2x^2 + 2x + 1$$

UNBOE:-

$$x = x; x_3 = 4; h = 1; p = \frac{x - x_3}{h} = (x - 4)$$

$$y(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$= 41 + (x-4)(25) + \frac{(x-4)(x-3)}{2} 14 + \frac{(x-4)(x-3)(x-2)}{6} 6 + \dots$$

$$= 41 + (25x - 100) + (x^2 - 7x + 12) 7 + (x^2 - 7x + 12)(x-2) + \dots$$

$$= 41 + 25x - 100 + 7x^2 - 49x + 84 + x^3 - 7x^2 + 12x - 2x^2 + 14x - 24 + \dots$$

$$= x^3 - 2x^2 + 2x + 1$$

④ using Newton forward formula Find the value of  $f(1.6)$  if

|   |      |      |      |     |     |     |
|---|------|------|------|-----|-----|-----|
| x | 1    | 1.4  | 1.8  | 2.2 | 2.6 | 3   |
| y | 3.49 | 4.82 | 5.96 | 6.5 | 7.2 | 8.4 |

Sol:- difference table :-

| x         | y          | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ | $\Delta^5 y$ |
|-----------|------------|------------|--------------|--------------|--------------|--------------|
| 1 $x_0$   | 3.49 $y_0$ | 1.33       |              |              |              |              |
| 1.4 $x_1$ | 4.82 $y_1$ | 1.14       | -0.19        |              |              |              |
| 1.8 $x_2$ | 5.96 $y_2$ | 0.54       | -0.6         | -0.41        |              |              |
| 2.2 $x_3$ | 6.5 $y_3$  | 0.7        | 0.16         | 0.76         | 1.17         |              |
| 2.6 $x_4$ | 7.2 $y_4$  | 1.2        | 0.5          | 0.34         | -0.42        | -1.59        |
| 3 $x_5$   | 8.4 $y_5$  |            |              |              |              |              |

(i) NFOF

$$x=1.6, x_0=1, h=0.4$$

$$p = \frac{x-x_0}{h} = \frac{1.6-1}{0.4} = \frac{0.6}{0.4} = 1.5$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$$y(x) = 3.49 + (1.5)(1.33) + \frac{(1.5)(0.5)}{2 \times 1}(-0.19) + \frac{(1.5)(0.5)(-0.5)}{3 \times 2 \times 1}(-0.41) + \dots$$

$$y(x) = 3.49 + 1.995 - 0.07125 + 0.025625 - 0.001875 + 0.0001875 - \dots$$

$$y(x) = 5.5301$$

$$(p-x) = \frac{0.6-x}{0.4} = 1.5 - x$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$$f(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$



⑤ The population of a town in the decimal senses was given, below estimate the population for the year 1895.

|               |      |      |      |      |      |
|---------------|------|------|------|------|------|
| year(x)       | 1891 | 1901 | 1911 | 1921 | 1931 |
| popularity(y) | 46   | 66   | 81   | 93   | 101  |

sol:- Difference table:-

| year       | population |            |              |              |              |
|------------|------------|------------|--------------|--------------|--------------|
| x          | y          | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
| 1891 $x_0$ | 46 $y_0$   |            |              |              |              |
| 1901 $x_1$ | 66 $y_1$   | 20         | -5           |              |              |
| 1911 $x_2$ | 81 $y_2$   | 15         | -3           | 2            |              |
| 1921 $x_3$ | 93 $y_3$   | 12         | -4           | -1           | -3           |
| 1931 $x_4$ | 101 $y_4$  | 8          |              |              |              |

∴ NBOE

$$x = 1895, x_4 = 1931, h = 10$$

$$p = \frac{x - x_4}{h} = \frac{1895 - 1931}{10} = -3.6$$

FOR NBOE

$$y(x) = y_4 + p \Delta y_4 + \frac{p(p+1)}{2!} \Delta^2 y_4 + \frac{p(p+1)(p+2)}{3!} \Delta^3 y_4 + \dots$$

$$y(x) = 101 + (-3.6)(8) + \frac{(-3.6)(-2.6)}{2 \times 1} (-4) + \frac{(-3.6)(-2.6)(-1.6)}{3 \times 2 \times 1} (-3) + \dots$$

$$y(x) = 101 + (-28.8) + (-18.72) + 2.496$$

$$y(x) = 55.976$$

∴ NFOE

$$x = 1895, x_0 = 1891, h = 10, p = \frac{x - x_0}{h} = \frac{1895 - 1891}{10} = 0.4$$

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

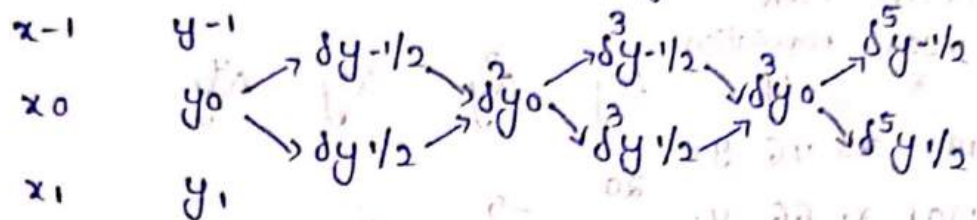
$$y(x) = 46 + (0.4)(20) + \frac{(0.4)(-0.6)}{2 \times 1} (-5) + \frac{(0.4)(-0.6)(-1.6)}{3 \times 2 \times 1} (2) + \dots$$

$$y(x) = 46 + 8 + 0.6 + 0.128$$

$$y(x) = 54.728$$

## Gauss central difference Formula:-

This formula is used to interpolate  $f(x)$  at a point  $x = (x_0 + ph)$  which is in the middle of the table. If  $f(x_0, y_0)$  is the nearest tabulated point to  $x$ , then we central difference formula are better to remember from the following table.



The Gauss difference formula are

G.C.D.F:-

$$y(x) = y_0 + p\delta y_{1/2} + \frac{p(p-1)}{2!}\delta^2 y_0 + \frac{p(p-1)(p+1)}{3!}\delta^3 y_{1/2} + \frac{p(p-1)(p-2)(p+1)}{4!}\delta^4 y_0 + \dots$$

where  $p = \frac{x - x_0}{h}$

G.C.B.D.F:-

$$y(x) = y_0 + p\delta y_{-1/2} + \frac{p(p+1)}{2!}\delta^2 y_0 + \frac{p(p+1)(p-1)}{3!}\delta^3 y_{-1/2} + \frac{p(p+1)(p-2)(p-1)}{4!}\delta^4 y_0 + \dots$$

where  $p = \frac{x - x_0}{h}$

Using gauss backward difference formula find  $y(8)$ . For the following data

| x | 0 | 5  | 10 | 15 | 20 | 25 |
|---|---|----|----|----|----|----|
| y | 7 | 11 | 14 | 18 | 24 | 32 |

Sol:- Difference table:-

| x  | y  | $\delta y$ | $\delta^2 y$ | $\delta^3 y$ | $\delta^4 y$ | $\delta^5 y$ |
|----|----|------------|--------------|--------------|--------------|--------------|
| 0  | 7  |            |              |              |              |              |
| 5  | 11 | 4          |              |              |              |              |
| 10 | 14 | 3          | -1           |              |              |              |
| 15 | 18 | 4          | 1            | -2           |              |              |
| 20 | 24 | 6          | 2            | 0            | 0            |              |
| 25 | 32 | 8          | 2            | 0            | 0            | 0            |



$$x=8; x_0=10; h=5$$

$$p = \frac{x-x_0}{h} = \frac{8-10}{5} = \frac{-2}{5} = -0.4$$

By G.C.B.O.F

$$y(x) = y_0 + p \delta y_{1/2} + \frac{p(p+1)}{2!} \delta^2 y_0 + \frac{p(p+1)(p-1)}{3!} \delta^3 y_{1/2} + \dots$$

$$\therefore y(x) = 14 + (-0.4)(3) + \frac{(-0.4)(0.6)}{2}(11) + \frac{(-0.4)(0.6)(-1.4)}{6}(5) + \frac{(-0.4)(0.6)(-1.4)(-2.4)}{24}(-1)$$

$$= 12.7676$$

② using Gauss forward formula  $y(3.3)$  from the following data

|   |      |      |      |      |      |
|---|------|------|------|------|------|
| x | 1    | 2    | 3    | 4    | 5    |
| y | 15.3 | 15.1 | 15.0 | 14.5 | 14.0 |

| x | y    | $\delta y$ | $\delta^2 y$ | $\delta^3 y$ | $\delta^4 y$ |
|---|------|------------|--------------|--------------|--------------|
| 1 | 15.3 | -0.2       |              |              |              |
| 2 | 15.1 |            | 0.1          |              |              |
| 3 | 15.0 | -0.1       | -0.4         | -0.5         | 0.9          |
| 4 | 14.5 | -0.5       | 0.4          | 0.4          |              |
| 5 | 14.0 | -0.5       |              |              |              |

$$x=3.3; x_0=3; h=1$$

$$p = \frac{x-x_0}{h} = \frac{3.3-3}{1} = 0.3$$

By G.C.E.O.F

$$y(x) = y_0 + p \delta y_{1/2} + \frac{p(p-1)}{2!} \delta^2 y_0 + \frac{p(p-1)(p+1)}{3!} \delta^3 y_{1/2} + \dots$$

$$y(x) = 15 + (0.3)(-0.5) + \frac{(0.3)(0.3-1)}{2!}(-0.4) + \frac{(0.3)(0.3-1)(0.3+1)}{3!}(0.9)$$

$$= 15 + (-0.15) + \frac{(0.3)(-0.7)}{2}(-0.4) + \frac{(0.3)(-0.7)(1.3)}{6}(0.9)$$

$$y(x) = 15 + (-0.15) + (0.042) + (-0.0182) + 0.0174$$

$$y(x) = 14.8912$$

Lagrange's interpolation formula:-

let  $y_0, y_1, y_2, \dots, y_n$  be the values of an unknown function  $y=f(x)$ .

At  $(n+1)$  "unequally spaced points"  $x_0, x_1, x_2, \dots, x_n$ . By using the data  $(x_i, y_i)$  for  $i=0, 1, 2, \dots, n$ .

we approximate the unknown function  $f(x)$  with a polynomial of degree "almost  $n$ " then.

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3) \dots (x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3) \dots (x_1-x_n)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3) \dots (x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3) \dots (x_2-x_n)} y_2 + \dots + y_n$$

this is called lagrange's interpolation formula for unequally spaced point.

① the following table the viscosity of an oil as a function, temperature use lagrange's formula to find the viscosity of oil at temp of  $140^\circ$ .

|           |      |     |     |     |
|-----------|------|-----|-----|-----|
| Temp      | 110  | 130 | 160 | 190 |
| viscosity | 10.8 | 8.1 | 5.5 | 4.8 |

Sol:- Here  $x_0=110$ ;  $x_1=130$ ;  $x_2=160$ ;  $x_3=190$

$y_0=10.8$ ;  $y_1=8.1$ ;  $y_2=5.5$ ;  $y_3=4.8$

$\therefore$  given data is unequally spaced

By lagrang's formula

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$y(x) = \frac{(10)(-20)(-50)}{(-20)(-50)(-80)} (10.8) + \frac{(30)(20)(-50)}{(20)(-30)(-60)} (8.1) + \frac{(30)(10)(-50)}{(50)(30)(-30)} (5.5)$$



$$+ \frac{(30)(10)(-20)}{(80)(60)(30)} (4.8)$$

$$y(x) = -1.35 + 6.75 + 1.83 - 0.2$$

$$y(x) = 7.03 //$$

Q Find the interpolating polynomial  $f(x)$  from the table

$$x \quad 0 \quad 1 \quad 3 \quad 4$$

$$y \quad -12 \quad 0 \quad 6 \quad 12$$

Sol: Here  $x_0 = 0$ ;  $x_1 = 1$ ;  $x_2 = 3$ ;  $x_3 = 4$

$$y_0 = -12; y_1 = 0; y_2 = 6; y_3 = 12$$

$\therefore$  given data is unequally spaced

By Lagrange's formula

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$y(x) = \frac{(x-1)(x-3)(x-4)}{(-1)(-3)(-4)} (-12) + \frac{(x-0)(x-3)(x-4)}{(1)(-2)(-3)} (0) + \frac{(x-0)(x-1)(x-4)}{(3)(2)(-1)} (6) + \frac{(x-0)(x-1)(x-3)}{(4)(3)(1)} (12)$$

$$y(x) = \frac{(x^2-3x-x+3)(x-4)}{(-12)} + \frac{(x^2-3x)(x-4)}{(6)} + \frac{(x^2-x)(x-4)}{(-6)} + \frac{(x^2-x)(x-3)}{(12)}$$

$$y(x) = x^3 - 4x^2 + 3x - 12 + \frac{(x^3 - 3x^2 - 4x + 12)}{6} + \frac{(x^3 - 3x^2 - x + 4)}{(-6)} + \frac{(x^3 - 3x^2 - x + 3)}{12}$$

$$y(x) = x^3 - 4x^2 + 3x - 12 + \frac{x^3 - 3x^2 - 4x + 12}{6} - \frac{x^3 - 3x^2 - x + 4}{6} + \frac{x^3 - 3x^2 - x + 3}{12}$$

$$y(x) = x^3 - 8x^2 + 19x - 12 + x^3 - 5x^2 - 4x + 2 - x^3 + 3x^2 + x - 2 + \frac{x^3 - 3x^2 - x + 3}{12}$$

$$y(x) = x^3 - 7x^2 + 18x - 12$$

③ By using Lagrang's interpolation formula Find  $y(6)$  From the following data

| x | 1  | 2  | 4  | 7   | 8   |
|---|----|----|----|-----|-----|
| y | 22 | 30 | 82 | 106 | 206 |

Sol:- Given  $x_0=1; x_1=2; x_3=4; x_4=7; x_5=8$   
 $y_0=22; y_1=30; y_2=82; y_3=106; y_4=206$   
 $x=6$

Given data is unequally spaced

By Lagrang's Formula

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)(x-x_5)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)(x_0-x_5)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)(x-x_5)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)(x_1-x_5)} y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)(x-x_5)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)(x_2-x_5)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)(x-x_5)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)(x_3-x_5)} y_3 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)(x-x_5)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)(x_4-x_5)} y_4$$

$$y(x) = \frac{(6-2)(6-4)(6-7)(6-8)}{(1-2)(1-4)(1-7)(1-8)} (22) + \frac{(6-1)(6-4)(6-7)(6-8)}{(2-1)(2-4)(2-7)(2-8)} (30) +$$

$$= \frac{(6-1)(6-2)(6-7)(6-8)}{(4-1)(4-2)(4-7)(4-8)} (82) + \frac{(6-1)(6-2)(6-4)(6-8)}{(7-1)(7-2)(7-4)(7-8)} (106) +$$

$$\frac{(6-1)(6-2)(6-4)(6-7)}{(8-1)(8-2)(8-4)(8-7)} (206)$$

$$y(x) = \frac{(4)(2)(-1)(-2)}{(-3)(-9)(-6)(-8)} (22) + \frac{(5)(2)(-1)(-2)}{(1)(-3)(-5)(-6)} (30) + \frac{(5)(4)(-1)(-2)}{(3)(2)(-3)(-4)} (82) +$$

$$= \frac{(5)(4)(2)(-2)}{(6)(5)(3)(-1)} (106) + \frac{(5)(4)(2)(-1)}{(7)(6)(4)(-1)} (206)$$

$$y(x) = 0.788 + (-0.811)x +$$



$$y(x) = 2.79 - 10 + 45.55 + 94.2 - 49.04$$

$$y(x) = 83.5$$

Q) Find the interpolating formula for  $f(x)$

|   |   |   |    |    |
|---|---|---|----|----|
| x | 0 | 1 | 4  | 5  |
| y | 4 | 3 | 24 | 39 |

Sol: Given  $x_0=0$ ;  $x_1=1$ ;  $x_2=4$ ;  $x_3=5$

$$y_0=4$$
;  $y_1=3$ ;  $y_2=24$ ;  $y_3=39$

$$x=x$$

Given data is unequally space d

By Lagrang's Formula

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$y(x) = \frac{(x-1)(x-4)(x-5)}{(0-1)(0-4)(0-5)} (4) + \frac{(x-0)(x-4)(x-5)}{(1-0)(1-4)(1-5)} (3) + \frac{(x-0)(x-1)(x-5)}{(4-0)(4-1)(4-5)} (24) + \frac{(x-0)(x-1)(x-4)}{(5-0)(5-1)(5-4)} (39)$$

$$y(x) = \frac{x^3 - 10x^2 + 29x - 20}{-20} (4) + \frac{x^3 - 9x^2 + 20x}{12} (3) + \frac{x^3 - 6x^2 + 5x}{-12} (24) + \frac{x^3 - 5x^2 + 4x}{20} (39)$$

$$y(x) = \frac{x^3 - 10x^2 + 29x - 20}{-5} + \frac{x^3 - 9x^2 + 20x}{4} + \frac{x^3 - 6x^2 + 5x}{-1} + \frac{x^3 - 5x^2 + 4x}{20} (39)$$

$$y(x) = \frac{x^3 - 10x^2 + 29x - 20}{-5} + \frac{x^3 - 9x^2 + 20x}{4} + (-2x^3 + 12x^2 - 10x) + \frac{x^3 - 5x^2 + 4x}{20} (39)$$

$$y(x) = \frac{-4x^3 + 40x^2 - 116x + 80 + 5x^3 - 45x^2 + 100x - 40x^3 + 240x^2 - 200x + 39x^3 - 195x^2 + 156x}{20}$$

$$y(x) = \frac{40x^2 - 60x + 80}{20}$$

$$y(x) = 2x^2 - 3x + 4$$

$$y(x) = 2x^2 - 3x + 4 //$$

## UNIT-III

### NUMERICAL INTEGRATION AND NUMERICAL SOLUTIONS OF ORDINARY DIFFERENTIAL EQUATIONS

Numerical Integration: Numerical integration is quite useful to evaluate definite integrals especially if the integrand cannot be integrated analytically (or) may be difficult to do.

Let  $(x_i, y_i), i=0, 1, 2, \dots, n$  be  $(n+1)$  tabulated data of the function  $y=f(x)$  in the interval  $a=x_0$  and  $b=x_n$  with stepsize

$$h = \frac{b-a}{n}$$

where  $n$  is the number of sub intervals then

$$\int_a^b f(x) dx \text{ (or) } \int_a^b y dx = \int_{x_0}^{x_n} y dx$$

Evaluated by following rules.

Trapezoidal rule:

$$\int_a^b f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Note: NO restrictions to take number of sub intervals.

Simpson's  $1/3$  rule:

$$\int_a^b f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

In this method we take the number of sub intervals as multiple of 2.

Simpson's  $3/8$  rule:

$$\int_a^b f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

In this method we take the number of sub intervals as multiple of three (3).



Q Evaluate  $\int_0^6 \frac{1}{1+x^2} dx$  by using (i) trapezoidal rule  
(ii) Simpson's  $\frac{1}{3}$ rd rule and  
(iii) Simpson's  $\frac{3}{8}$ th rule

Sol:- Given  $\int_0^6 \frac{1}{1+x^2} dx$

Let  $f(x) = \frac{1}{1+x^2}$

Here  $a=0, b=6, n=6$

$h = \frac{b-a}{n} = \frac{6-0}{6} = 1$

|   |       |       |       |       |       |       |       |
|---|-------|-------|-------|-------|-------|-------|-------|
| x | 0     | 1     | 2     | 3     | 4     | 5     | 6     |
|   | $x_0$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
| y | $y_0$ | 0.5   | 0.2   | 0.1   | 0.056 | 0.038 | 0.027 |
|   |       | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $y_5$ | $y_6$ |

(i) Trapezoidal rule:-

$$\int_0^6 \frac{1}{1+x^2} dx = \frac{h}{2} ((y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5))$$

$$= \frac{1}{2} ((1 + 0.027) + 2(0.5 + 0.2 + 0.1 + 0.056 + 0.038))$$

$$= 1.4095$$

(ii) Simpson's  $\frac{1}{3}$ rd rule:-

$$\int_0^6 \frac{1}{1+x^2} dx = \frac{h}{3} ((y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4))$$

$$= \frac{1}{3} ((1 + 0.027) + 4(0.5 + 0.1 + 0.038) + 2(0.2 + 0.056))$$

$$= \frac{1}{3} \cdot 3.65$$

(iii) Simpson's  $\frac{3}{8}$ th rule:-

$$\int_0^6 \frac{1}{1+x^2} dx = \frac{3h}{8} ((y_0 + y_6) + 3(y_1 + y_2 + y_4 + \dots + y_{n-1}) + 2(y_3 + y_5 + \dots + y_{n-3}))$$

$$= \frac{3h}{8} ((y_0 + y_6) + 3(y_1 + y_2 + y_4) + 2(y_3))$$

$$= \frac{3}{8} ((1 + 0.027) + 3(0.5 + 0.2 + 0.056) + 2(0.1))$$

$$= 1.310$$

- ② Evaluate  $\int_0^{\pi} \frac{\sin x}{x} dx$  by using
- (i) trapezoidal rule
  - (ii) Simpson's  $1/3$  rule
  - (iii) Simpson  $3/8$  rule

Sol:- Given  $f(x) = \int_0^{\pi} \frac{\sin x}{x} dx$

$a=0, b=\pi, n=6$

$h = \frac{b-a}{n} = \frac{\pi}{6}$

$x \quad 0 \quad \pi/6 \quad 2\pi/6 \quad 3\pi/6 \quad 4\pi/6 \quad 5\pi/6 \quad \pi$

$y \quad 1 \quad 0.9549 \quad 0.8269 \quad 0.6369 \quad 0.4134 \quad 0.1909 \quad 0$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

(i) Trapezoidal rule:-

$$\int_0^{\pi} \frac{\sin x}{x} dx = \frac{\pi/6}{2} \left( (y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5) \right)$$

$$= \frac{\pi/6}{2} \left( (1 + 0) + 2(0.9549 + 0.8269 + 0.6369 + 0.4134 + 0.1909) \right)$$

$$= 1.8445$$

(ii) Simpson's  $1/3$  rule:-

$$\int_0^{\pi} \frac{\sin x}{x} dx = \frac{h}{3} \left( (y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) \right)$$

$$= \frac{\pi/6}{3} \left( (1 + 0) + 4(0.9549 + 0.6369 + 0.1909) + 2(0.8269 + 0.4134) \right)$$

$$= 1.8518$$

(iii) Simpson's  $3/8$  rule:-

$$\int_0^{\pi} \frac{\sin x}{x} dx = \frac{3h}{8} \left( (y_0 + y_6) + 3(y_1 + y_2 + y_4) + 2(y_3) \right)$$

$$= \frac{3\pi/6}{8} \left( (1 + 0) + 3(0.9549 + 0.8269 + 0.4134) + 2(0.6369) \right)$$

$$= 1.8518$$



③ Evaluate  $\int_0^1 (1+x^3)^{1/2} dx$  taking  $h=0.1$  using Simpson's  $\frac{1}{3}$  rule.

Given  $\int_0^1 (1+x^3)^{1/2} dx$ ,  $h=0.1$

|   |        |        |        |        |        |        |        |        |        |        |     |
|---|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|
| x | 0      | 0.1    | 0.2    | 0.3    | 0.4    | 0.5    | 0.6    | 0.7    | 0.8    | 0.9    | 1.0 |
| y | 1.0000 | 1.0039 | 1.0134 | 1.0315 | 1.0606 | 1.1021 | 1.1588 | 1.2296 | 1.3149 | 1.4142 |     |

Simpson's  $\frac{1}{3}$  rule

$$\int_0^1 (1+x^3)^{1/2} dx = \frac{h}{3} [(y_0 + y_{10}) + 4(y_1 + y_3 + y_5 + y_7 + y_9) + 2(y_2 + y_4 + y_6 + y_8)]$$

$$= \frac{0.1}{3} [(1 + 1.4142) + 4(1.0000 + 1.0134 + 1.0606 + 1.1588 + 1.3149) + 2(1.0039 + 1.0315 + 1.1021 + 1.2296)]$$

$$= 1.1141$$

Numerical solutions of ordinary differential equations:-  
ordinary differential equations arises when equations involved an independent variable with respect to  $x$ , and dependent variable  $y$  and its derivatives with respect to  $x$ .

mathematical modeling physical laws with respect to that -

- involves rate, velocity, acceleration etc.

This chapter we have to find the numerical solutions of first order differential equations of the form  $\frac{dy}{dx} = f(x, y)$ ;

$$y(x_0) = y_0$$

Taylor's series method:-

To find the numerical solution of the differential equation

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \longrightarrow \textcircled{1}$$

$y(x)$  can be expanded about the point  $x_0$  in a Taylor's series in powers of  $x - x_0$  as

$$y(x) = y(x_0) + \frac{(x-x_0)}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

The value of  $y(x)$  can be obtained if we know the values of its derivative.

Differentiating equation (1) we get  $y', y'', \dots$  and find these values at  $(x_0, y_0)$

$$\text{let } x - x_0 = h$$

$$x = x_0 + h = x_1 \text{ (say)}$$

$$y(x_1) = y(x_0) + \frac{h}{1!} y'(x_0) + \frac{h^2}{2!} y''(x_0) + \frac{h^3}{3!} y'''(x_0) + \dots$$

$$y(x_2) = y(x_1) + \frac{h}{1!} y'(x_1) + \frac{h^2}{2!} y''(x_1) + \frac{h^3}{3!} y'''(x_1) + \dots$$

$$y(x_3) = y(x_2) + \frac{h}{1!} y'(x_2) + \frac{h^2}{2!} y''(x_2) + \frac{h^3}{3!} y'''(x_2) + \dots$$

① Using Taylor's series method compute  $y(0.1), y(0.2)$  of the differential equation  $y' = x - y^2, y(0) = 1$ .

Sol: Given  $y' = x - y^2, y(0) = 1$

$$x_0 = 0, y_0 = 1$$

$$\text{let } x_1 = 0.1, x_2 = 0.2, h = 0.1$$

To find  $y(x_1) = y(0.1)$

$$y'_0 = x_0 - y_0^2$$

$$= 0 - 1 = -1$$

$$y''_0 = 1 - 2y_0 y'_0$$

$$= 1 - 2(1)(-1)$$

$$= 3$$

$$y'''_0 = 0 - 2(y_0 y''_0 + y'_0 y'_0)$$

$$= -2(y_0 y''_0 + (y'_0)^2)$$

$$= -2(1(3) + (-1)^2)$$

$$= -2(3+1)$$

$$= -8$$



By Taylor's method

$$y(x) = y_0 + \frac{h}{1!} y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots$$

$$= 1 + \frac{(0.1)}{1!}(-1) + \frac{(0.1)^2}{2}(3) + \frac{(0.1)^3}{6}(-8) + \dots$$

$$y(0.1) = 0.9136$$

To find  $y(x_2) = y(0.2)$  :-  $x_1 = 0.1, y_1 = 0.9136$

$$y'_1 = x_1 - y_1^2$$

$$= 0.1 - (0.9136)^2$$

$$= -0.7346$$

$$y''_1 = 1 - 2y_1 y'_1$$

$$= 1 - 2(0.9136)(-0.7346)$$

$$= 2.3422$$

$$y'''_1 = 0 - 2(y_1 y''_1 + y'_1 y'_1)$$

$$= 0 - 2((0.9136)(2.3422) + (-0.7346)^2)$$

$$= -5.36091$$

By Taylor's method

$$y(x_2) = y_1 + \frac{h}{1!} y'_1 + \frac{h^2}{2!} y''_1 + \frac{h^3}{3!} y'''_1 + \dots$$

$$y(x_2) = 0.9136 + \frac{0.1}{1!}(-0.7346) + \frac{(0.1)^2}{2!}(2.3422) + \frac{(0.1)^3}{3!}(-5.36091) + \dots$$

$$y(x_2) = 0.8511$$

$$y(0.2) = 0.8511$$

Q. Solve  $y' = x^2 - y, y(0) = 1$  using Taylor's series method and, compute  $y(0.1), y(0.2), y(0.3), y(0.4)$  correct upto 4 decimal places.

Ans: Given that  $y' = x^2 - y, y(0) = 1$

let  $x_0 = 0, y_0 = 1$

$h = 0.1$

To find  $y(x_1)$  i.e.  $y(0.1)$ :-

$$\begin{aligned} y'(x_0) &= x_0^2 - y_0 \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

$$\begin{aligned} y''(x_0) &= 2x_0 - y_0' \\ &= 2(0) - (-1) \\ &= 1 \end{aligned}$$

$$\begin{aligned} y'''(x_0) &= 2 - y_0'' \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \therefore y(x_1) &= y(x_0) + \frac{h}{1!} y'(x_0) + \frac{h^2}{2!} y''(x_0) + \frac{h^3}{3!} y'''(x_0) \\ &= 1 + \frac{0.1}{1!} (-1) + \frac{(0.1)^2}{2!} (1) + \frac{(0.1)^3}{3!} (1) \\ &= 1 - 0.1 + \frac{(0.1)^2}{2} + \frac{(0.1)^3}{6} \end{aligned}$$

$$y(0.1) = 0.9052$$

To find  $y(x_2)$  i.e.  $y(0.2)$ :-

$$\text{let } x_1 = 0.1, y_1 = 0.9052, h = 0.1$$

$$\begin{aligned} y'(x_1) &= (0.1)^2 - 0.9052 \\ &= -0.8952 \end{aligned}$$

$$\begin{aligned} y''(x_1) &= 2(0.1) - (-0.8952) \\ &= 0.2 + 0.8952 \\ &= 1.0952 \end{aligned}$$

$$\begin{aligned} y'''(x_1) &= 2 - 1.0952 \\ &= 0.9048 \end{aligned}$$

$$\begin{aligned} \therefore y(x_2) &= y(x_1) + \frac{h}{1!} y'(x_1) + \frac{h^2}{2!} y''(x_1) + \frac{h^3}{3!} y'''(x_1) \\ &= 0.9052 + \frac{0.1}{1!} (-0.8952) + \frac{(0.1)^2}{2!} (1.0952) + \frac{(0.1)^3}{3!} (0.9048) \end{aligned}$$

$$y(0.2) = 0.8213$$

To find  $y(x_3)$  i.e.  $y(0.3)$ :-

$$\text{let } x_2 = 0.2, y_2 = 0.8213, h = 0.1$$

$$\begin{aligned} y'(x_2) &= (0.2)^2 - 0.8213 \\ &= -0.7813 \end{aligned}$$



$$y''(x_2) = 2(0.1) - (-0.7813)$$

$$= 0.4 + 0.7813$$

$$= 1.1813$$

$$y'''(x_2) = 2 - 1.1813$$

$$= 0.8187$$

$$\therefore y(x_3) = y(x_2) + \frac{h}{1!} y'(x_2) + \frac{h^2}{2!} y''(x_2) + \frac{h^3}{3!} y'''(x_2)$$

$$= 0.8213 + \frac{0.1}{1!} (-0.7813) + \frac{(0.1)^2}{2!} (1.1813) + \frac{(0.1)^3}{3!} (0.8187)$$

$$y(0.3) = 0.7492$$

To find  $y(x_4)$  i.e.  $y(0.4)$ :-

$$\text{let } x_3 = 0.3, y_3 = 0.7492, h = 0.1$$

$$y'(x_3) = (0.3) - (0.7492)$$

$$= -0.6592$$

$$y''(x_3) = 2(0.3) + 0.6592$$

$$= 1.2592$$

$$y'''(x_3) = 2 - 1.2592$$

$$= 0.7408$$

$$\therefore y(x_4) = y(x_3) + \frac{h}{1!} y'(x_3) + \frac{h^2}{2!} y''(x_3) + \frac{h^3}{3!} y'''(x_3)$$

$$= (0.7492) + \frac{(0.1)}{1!} (-0.6592) + \frac{(0.1)^2}{2!} (1.2592) + \frac{(0.1)^3}{3!} (0.7408)$$

$$y(0.4) = 0.6847$$

Exact Solution:-

$$y' = x^2 - y$$

$$y' + y = x^2$$

$$\int dx = x$$

$$y e^x = \int e^x \cdot x^2 dx$$

$$y e^x = e^x (x^2 - 2x + 2) + C$$

$$y = x^2 - 2x + 2 + C e^{-x}$$

$$y(0) = 1$$

$$1 = 0 - 0 + 2 + C$$

$$C = -1$$

$$\therefore y = x^2 - 2x + 2 - e^{-x}$$

③ Find  $y(0.1)$ ,  $y(0.2)$ ,  $z(0.1)$ ,  $z(0.2)$ . Given  $\frac{dy}{dx} = x+z$ ,  $\frac{dz}{dx} = x-y^2$  and  $y(0)=2$ ,  $z(0)=1$  by using Taylor's series method.

Sol:- Given  $\frac{dy}{dx} = x+z$ ;  $\frac{dz}{dx} = x-y^2$

$$y(0)=2 \quad ; \quad z(0)=1$$

$$\text{let } x_1=0.1, x_2=0.2$$

$$\text{let } x_0=0, y_0=2, z_0=1$$

$$y'_0 = x_0 + z_0 = 0 + 1 = 1$$

$$z'_0 = x_0 - y_0^2 = 0 - 2^2 = 0 - 4 = -4$$

$$y''_0 = 1 + z'_0 = 1 - 4 = -3$$

$$z''_0 = 1 - 2y_0 y'_0 = 1 - 2(2)(1) = -3$$

$$y'''_0 = z''_0 = -3$$

$$z'''_0 = -2(y_0 y''_0 + y'^2_0) = -2(2(-3) + 1) = 10$$

To find  $y(x_1) = y(0.1)$  &  $z(x_1) = z(0.1)$ :-

By Taylor's series method,

$$\begin{aligned} y(x_1) &= y(x_0) + \frac{h}{1!} y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots \\ &= 2 + (0.1)(1) + \frac{(0.1)^2}{2}(-3) + \frac{(0.1)^3}{6}(-3) \end{aligned}$$

$$\therefore y(0.1) = 2.0845$$

$$\begin{aligned} z(x_1) &= z(x_0) + \frac{h}{1!} z'_0 + \frac{h^2}{2!} z''_0 + \frac{h^3}{3!} z'''_0 + \dots \\ &= 1 + (0.1)(-4) + \frac{(0.1)^2}{2}(-3) + \frac{(0.1)^3}{6}(10) \end{aligned}$$

$$\therefore z(x_1) = 0.5866$$

$$\therefore z(0.1) = 0.5866$$

To find  $y(x_2) = y(0.2)$  &  $z(x_2) = z(0.2)$ :-

$$y'_1 = x_1 + z_1 = 0.1 + 0.5866 = 0.6866$$

$$z'_1 = x_1 - y_1^2 = 0.1 - 2.0845^2 = -4.2451$$

$$y''_1 = 1 + z'_1 = 1 - 4.2451 = -3.2451$$

$$z''_1 = 1 - 2y_1 y'_1 = 1 - 2(2.0845)(0.6866) = -1.8624$$

$$y'''_1 = z''_1 = -1.8624$$

$$z'''_1 = -2(y_1 y''_1 + y'^2_1) = -2(2.0845(-3.2451) + 0.6866^2) = 12.1557$$

$$(-3.2451) + 0.6866 = 12.1557$$

By Taylor's series method

$$y(x_2) = y(x_1) + \frac{h}{1!} y'_1 + \frac{h^2}{2!} y''_1 + \frac{h^3}{3!} y'''_1 + \dots$$



$$= 2.0845 + (0.1)(0.6866) + \frac{(0.1)^2}{2}(-3.2451) + \frac{(0.1)^3}{6}(-1.8624)$$

$$\therefore y(0.2) = 2.1367$$

$$z(x_2) = z(x_1) + \frac{h}{1!} z'(x_1) + \frac{h^2}{2!} z''(x_1) + \frac{h^3}{3!} z'''(x_1)$$

$$= 0.5866 + (0.1)(-4.2451) + \frac{(0.1)^2}{2}(-1.8624) + \frac{(0.1)^3}{6}(12.1552)$$

$$\therefore z(0.2) = 0.1548$$

Picard's method of successive approximation:-

consider the differential equation  $\frac{dy}{dx} = f(x, y) \rightarrow \textcircled{1}$

and  $y(x_0) = y_0$

$$dy = f(x, y) dx \rightarrow \textcircled{2}$$

integrating equation  $\textcircled{2}$  in the interval  $x_0$  to  $x$ , we get

$$\int_{x_0}^x dy = \int_{x_0}^x f(x, y) dx$$

$$[y]_{x_0}^x = \int_{x_0}^x f(x, y) dx$$

$$y(x) - y(x_0) = \int_{x_0}^x f(x, y) dx$$

$$\therefore y(x) = y_0 + \int_{x_0}^x f(x, y) dx \rightarrow \textcircled{3}$$

Equation  $\textcircled{3}$  is called integral equation. To get the first approximation  $y'(x)$ , put  $y = y_0$  in equation  $\textcircled{3}$

$$y^{(1)}(x) = y_0 + \int_{x_0}^x f(x, y_0) dx \rightarrow \textcircled{4}$$

To get the second approximation  $y^{(2)}(x)$

put  $y = y^{(1)}(x)$  in equation  $\textcircled{3}$

$$y^{(2)}(x) = y_0 + \int_{x_0}^x f(x, y^{(1)}(x)) dx \rightarrow \textcircled{5}$$

similarly

$$y^{(3)}(x) = y_0 + \int_{x_0}^x f(x, y^{(2)}(x)) dx \rightarrow \textcircled{6}$$

$$\dots \dots \dots$$

$$y^{(n)}(x) = y_0 + \int_{x_0}^x f(x, y^{(n-1)}(x)) dx \rightarrow \text{①}$$

At last  $y^{(n)}(x) = y(x)$  is the solution of eqn ①

① Find the value of  $y$  for  $x=0.4$  by picard's method - given that  $\frac{dy}{dx} = x^2 + y^2$ ,  $y(0) = 0$

Sol:- Given differential equation is

$$\frac{dy}{dx} = x^2 + y^2, y(0) = 0$$

$$\text{Let } f(x, y) = x^2 + y^2$$

$$x_0 = 0, y_0 = 0$$

By picard's method, 1<sup>st</sup> successive approximation is

$$y'(x) = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$y(x) = 0 + \int_0^x (x^2 + y_0^2) dx = \int_0^x x^2 dx$$

$$y^{(1)}(x) = \frac{x^3}{3}$$

2<sup>nd</sup> approximation:-

$$y^2(x) = y_0 + \int_{x_0}^x f(x, y^{(1)}(x)) dx$$

$$= 0 + \int_0^x f(x, \frac{x^3}{3}) dx$$

$$= \int_0^x (x^2 + (\frac{x^3}{3})^2) dx$$

$$= \int_0^x (x^2 + \frac{x^6}{9}) dx$$

$$y^{(2)}(x) = \frac{x^3}{3} + \frac{x^7}{63}$$

3<sup>rd</sup> approximation:-

$$y^3(x) = y_0 + \int_{x_0}^x f(x, y^{(2)}(x)) dx$$

$$= 0 + \int_0^x f(x, \frac{x^3}{3} + \frac{x^7}{63}) dx$$

$$= \int_0^x (x^2 + (\frac{x^3}{3} + \frac{x^7}{63})^2) dx$$



$$= \int_0^x (x^2 + \frac{x^6}{9} + \frac{x^{14}}{3969} + \frac{2x^{10}}{189}) dx$$

$$\therefore y^3(x) = \frac{x^3}{3} + \frac{x^7}{63} + \frac{x^{15}}{59535} + \frac{2x^{11}}{2079}$$

$$y(0.4) = 0.0214$$

@ Find the approximate values of  $y$ ,  $x=0.1$ ,  $x=0.2$  if  $\frac{dy}{dx} = x+y$  and  $y=1$  at  $x=0$  by using picard's method.

Sol:- Given differential equation is

$$\frac{dy}{dx} = x+y, y(0) = 1$$

$$\text{let } f(x, y) = x+y$$

$$x_0 = 0, y_0 = 1$$

By picard's method, 1<sup>st</sup> successive approximation is

$$y_1(x) = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$y_1(x) = 1 + \int_0^x (x+y_0) dx$$

$$= 1 + \int_0^x (1+x) dx$$

$$= 1 + x + \frac{x^2}{2}$$

$$y_1(x) = 1 + x + \frac{x^2}{2}$$

2<sup>nd</sup> approximation :-

$$y_2(x) = y_0 + \int_{x_0}^x f(x, y_1(x)) dx$$

$$y_2(x) = 1 + \int_0^x (x + 1 + x + \frac{x^2}{2}) dx$$

$$= 1 + \int_0^x (1 + 2x + \frac{x^2}{2}) dx$$

$$= 1 + x + \frac{2x^2}{2} + \frac{x^3}{6}$$

$$y_2(x) = 1 + x + x^2 + \frac{x^3}{6}$$

3<sup>rd</sup> approximation :-

$$y_3(x) = y_0 + \int_{x_0}^x f(x, y_2(x)) dx$$

$$= 1 + \int_0^x (x + 1 + x + x^2 + \frac{x^3}{6}) dx$$

$$= 1 + \int_0^x (1 + 2x + x^2 + \frac{x^3}{6}) dx$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{24}$$

$$y^3(x) = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}$$

For  $x=0.1$

$$\begin{aligned} y(0.1) &= 1 + 0.1 + (0.1)^2 + \frac{(0.1)^3}{3} + \frac{(0.1)^4}{24} \\ &= 1 + 0.1 + 0.01 + 0.00033 + 0.000004 \\ &= 1.1103 \end{aligned}$$

For  $x=0.2$

$$\begin{aligned} y(0.2) &= 1 + 0.2 + (0.2)^2 + \frac{(0.2)^3}{3} + \frac{(0.2)^4}{24} \\ &= 1 + 0.2 + 0.04 + 0.00266 + 0.000066 \\ &= 1.2427 \end{aligned}$$

③ Find the value of  $y$  for  $x=0.1$  by picard's method given that

$$\frac{dy}{dx} = \frac{y-x}{y+x}, y(0)=1$$

Sol:- Given differential equation is

$$\frac{dy}{dx} = \frac{y-x}{y+x}, y(0)=1$$

$$\text{let } f(x,y) = \frac{y-x}{y+x}$$

$$x_0=0, y_0=1$$

By picard method.

First approximation:-

$$y^{(1)}_x = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$= 1 + \int_0^x \left( \frac{1-x}{1+x} \right) dx$$

$$= 1 + \int_0^x \left( \frac{1-x-1+x}{1+x} \right) dx$$

$$= 1 + \int_0^x \left( \frac{-2x}{1+x} \right) dx$$

$$= 1 + 2(\log(1+x)) - x$$

$$\therefore y(x) = 1 + 2\log(1+x) - x = 0.9827$$



### Euler's method:-

To solve the differential equation  $\frac{dy}{dx} = f(x, y)$ ;  $y(x_0) = y_0$   
by Euler method. The first approximate value

$$y_1 = y(x_1) = y_0 + h f(x_0, y_0)$$

where,  $h$  is step size

The second approximate value

$$y_2 = y(x_2) = y_1 + h f(x_1, y_1)$$

Similarly

$$y_3 = y(x_3) = y_2 + h f(x_2, y_2)$$

$$\vdots$$

$$y_n = y(x_n) = y_{n-1} + h f(x_{n-1}, y_{n-1})$$

① Solve  $y' = x + y$ ,  $y(0) = 1$ , and find  $y(0.3)$  taking step size  $h = 0.1$   
by using Euler's method and compare with exact solution.

sol:- Given  $y' = x + y$ ,  $y(0) = 1$

$$h = 0.1, x_0 = 0, x_1 = 0.1, x_2 = 0.2, x_3 = 0.3$$

$$\text{let } f(x, y) = x + y$$

$$x_0 = 0, y_0 = 1$$

By Euler's single step method

$$y_1 = y(x_1) = y_0 + h f(x_0, y_0)$$

$$= 1 + (0.1)(x_0 + y_0)$$

$$y_1 = y(x_1) = 1 + (0.1)(0 + 1) = 1.1$$

$$y(0.1) = 1.1$$

$$y_2 = y(x_2) = y_1 + h f(x_1, y_1)$$

$$= 1.1 + (0.1)f(x_1, y_1)$$

$$= 1.1 + (0.1)(0.1 + 1.1)$$

$$= 1.1 + (0.1)(1.2)$$

$$y(0.2) = 1.22$$

$$y_3 = y(x_3) = y_2 + h f(x_2, y_2)$$

$$= 1.22 + (0.1)f(x_2, y_2)$$

$$= 1.22 + (0.1)(0.2 + 1.22)$$

$$y(0.3) = 1.362$$

Exact solution:-

$$y' = x + y$$

$$y' - y = x$$

$\frac{dy}{dx} - y = x$  is linear differential equation

$$P = -1, Q(x) = x$$

$$I.F = e^{\int P dx} = e^{-x}$$

$$y e^{-x} = \int x e^{-x} dx + C$$

$$y e^{-x} = -e^{-x}(1+x) + C$$

$$y(0) = 1$$

$$y = -(1+x) + C e^x$$

$$1 = -(1+0) + C$$

$$C = 1 + 1$$

$$C = 2$$

$$E.y = e^x - x + 2e^x$$

$$y(0.3) = 1.39997$$

Q using Euler's method solve for 'y' at  $x=2$  from  $\frac{dy}{dx} = 3x^2 + 1$

$y(1) = 2$ , taking stepsize (i)  $h = 0.5$  (ii)  $h = 0.25$

Sol:- Given  $\frac{dy}{dx} = 3x^2 + 1$ ,  $y(1) = 2$

$$\text{let } f(x, y) = 3x^2 + 1$$

$$x_0 = 1, y_0 = 2 \quad (\text{stepsize } h = 0.5) \quad x_1 = x_0 + h = 1 + 0.5 = 1.5$$

(i)  $h = 0.5$  by Euler's single step method

$$y_1 = y(x_1) = y_0 + h f(x_0, y_0)$$

$$y_1 = y(1.5) = 2 + 0.5 f(1, 2)$$

$$y_1 = 2 + 0.5(4) = 4$$

$$y_2 = y(x_2) = y_1 + h f(x_1, y_1)$$

$$y_2 = 4 + 0.5 f(1.5, 4) = 6.1$$

$$y_2 = 7.875$$

$$(0.5 \times 3x^2 + 1) + 4 = 6.1$$



$$(ii) h=0.25$$

$$y_1 = y(1.25) = 2 + 0.25 f(1, 2) = 2 + 0.25 [4] = 3$$

$$y_2 = y_1 + h f(x_1, y_1)$$

$$y_2 = y(1.5) = 3 + 0.25 [3(1.25)^2 + 1] = 4.42188$$

$$y_3 = y(1.75) = y_2 + h f(x_2, y_2)$$

$$y_3 = y(1.75) = 4.42188 + 0.25 [3(1.5)^2 + 1] = 6.35938$$

$$y_4 = y(2) = y_3 + h f(x_3, y_3)$$

$$y_4 = y(2) = 6.35938 + 0.25 [3(1.75)^2 + 1]$$

$$= 8.90626$$

Euler's modified method:-

to solve  $\frac{dy}{dx} = f(x, y)$ ,  $y(x_0) = y_0$

to find  $y(x_1) = y_1$ :-

$$y_1(0) = y_0 + h f(x_0, y_0)$$

$$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

$$y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$y_1^{(k)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(k-1)})] \quad k=1, 2, \dots, k$$

If two successive value of  $y_1^{(k)}$ ,  $y_1^{(k-1)}$  are sufficiently close to one another.

to find  $y(x_2) = y_2$ :-

$$y_2^{(0)} = y_1 + h f(x_1, y_1)$$

$$y_2^{(1)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(0)})]$$

$$y_2^{(k)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(k-1)})]$$

If two successive values of  $y_2^{(k)}$  and  $y_2^{(k-1)}$  are closer to each other, then taking  $y_2$  as  $y_2^{(k)}$

To find  $y(x_3) = y_3$ :-

$$y_3^{(0)} = y_2 + h f(x_2, y_2)$$

$$y_3^{(k)} = y_2 + \frac{h}{2} [f(x_2, y_2) + f(x_2, y_3^{(k-1)})] \quad k=1, 2, \dots, n$$

To find  $y_n = y(x_n)$ :-

$$y_n^{(0)} = y_{n-1} + h f(x_{n-1}, y_{n-1})$$

$$y_n^{(k)} = y_{n-1} + \frac{h}{2} [f(x_{n-1}, y_{n-1}) + f(x_n, y_n^{(k-1)})] \quad k=1, 2, \dots, n$$

① solve numerically  $y' = y + e^x$ ,  $y(0) = 0$  from  $x = 0.2, 0.4$  by Euler's modified method.

Sol: Given differential equation is

$$y' = y + e^x, \quad y(0) = 0$$

$$\text{here } f(x, y) = y + e^x, \quad x_0 = 0, y_0 = 0$$

$$h = 0.2, \quad x_1 = 0.2, \quad x_2 = 0.4$$

To find  $y(x_1) = y_1$ :-

by Euler's modified method

$$y_1^{(0)} = y_0 + h f(x_0, y_0)$$

$$= 0 + 0.2 f(0, 0)$$

$$= 0 + 0.2 (0 + e^0)$$

$$= 0.2(1)$$

$$y_1^{(0)} = 0.2$$

$$y_1^{(1)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

$$= 0 + \frac{0.2}{2} [f(0, 0) + f(0.2, 0.2)]$$

$$= 0.1 [0 + 1.4214]$$

$$= 0.2421$$

$$y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 0 + \frac{0.2}{2} [0 + 1 + 0.2421 + e^{0.2}]$$

$$= 0.2464$$



$$y_1^{(3)} = (0.1)[1 + 0.2464 + e^{0.2}]$$

$$= 0.2468$$

$$y_1^{(4)} = (0.1)[1 + 0.2468 + e^{0.2}]$$

$$= 0.2468$$

$$\text{Hence } y_1^{(3)} = y_1^{(4)} = y_1$$

$$\therefore y_1 = 0.2468$$

To Find  $y_2(0.4) = y_2$  :-

$$y_2^{(0)} = y_1 + h f(x_1, y_1)$$

$$= 0.2468 + (0.2) f(0.2, 0.2468)$$

$$= 0.2468 + (0.2) (0.2468 + e^{0.2})$$

$$= 0.5404$$

$$y_2^{(1)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(0)})]$$

$$= 0.2468 + 0.1 [0.2468 + e^{0.2} + 0.5404 + e^{0.4}]$$

$$= 0.5968$$

$$y_2^{(2)} = 0.2468 + 0.1 [0.2468 + e^{0.2} + 0.5968 + e^{0.4}]$$

$$= 0.6025$$

$$y_2^{(3)} = 0.2468 + 0.1 [0.2468 + e^{0.2} + 0.6025 + e^{0.4}]$$

$$= 0.6031$$

$$y_2^{(4)} = 0.6031$$

$$\therefore y = 0.6031$$

Runge - kutta method :-

1. Runge - kutta first order method :-

To solve  $y' = f(x, y)$ ,  $y(x_0) = y_0$

To find  $y_1$  :-

$$y_1 = y_0 + hf(x_0, y_0)$$

$$y_2 = y_1 + hf(x_1, y_1)$$

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$$

2. Runge - kutta second order method :-

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h, y_0 + k_1)$$

$$y_2 = y_1 + \frac{1}{2}(k_1 + k_2)$$

where

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf(x_1 + h, y_1 + k_1)$$

$$y_n = y_{n-1} + \frac{1}{2}(k_1 + k_2)$$

where

$$k_1 = hf(x_{n-1}, y_{n-1})$$

$$k_2 = hf(x_{n-1} + h, y_{n-1} + k_1)$$

3. Runge - kutta third order method :-

$$y_1 = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$k_3 = hf(x_0 + h, y_0 + 2k_2 - k_1)$$

$$y_2 = y_1 + \frac{1}{6}(k_1 + 4k_2 + k_3)$$



where

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf(x_1 + h/2, y_1 + k_1/2)$$

$$k_3 = hf(x_1 + h, y_1 + 2k_2 - k_1)$$

$$k_4 = hf(x_1 + 3h/2, y_1 + 3k_3 - 3k_2 + k_1)$$

similarly

find  $y_3, y_4, \dots, y_n$  by above procedure

4. Runge-kutta fourth order method:-

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf(x_1 + h/2, y_1 + k_1/2)$$

$$k_3 = hf(x_1 + h/2, y_1 + k_2/2)$$

$$k_4 = hf(x_1 + h, y_1 + k_3)$$

similarly

find  $y_3, y_4, \dots$  by above procedure.

Q Find  $y(0.1), y(0.2)$  using Runge-kutta 4th order method given that  $y' = x^2 - y, y(0) = 1$ .

Sol:- Given differential equation is  $y' = x^2 - y, y(0) = 1$

here

$$x_0 = 0, y_0 = 1$$

$$f(x, y) = x^2 - y$$

$$\text{let } h = 0.1, x_1 = 0.1, x_2 = 0.2$$

To find  $y_1, y(0.1)$ :-

By Runge-kutta fourth order method

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_1 = (0.1)f(0, 1)$$

$$= (0.1)(0^2 - 1)$$

$$k_1 = -0.1$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$= (0.1)f(0.05, 0.95)$$

$$= 0.1(-0.9475)$$

$$k_2 = -0.0948$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$$

$$= (0.1)f(0.05, 0.9526)$$

$$k_3 = -0.095$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$= 0.1f(0.1, 0.905)$$

$$= 0.1(-0.8950)$$

$$= -0.0895$$

$$\therefore y_1 = 1 + \frac{1}{6} (-0.1 + 2(-0.0948) + 2(-0.095) - 0.0895)$$

$$y_1 = 0.9052$$

to find  $y_2$  i.e.  $y(0.2)$ :-

By Runge-Kutta fourth order method

$$y_2 = y_1 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\text{here } k_1 = hf(x_1, y_1)$$

$$= hf(0.1, 0.9052)$$

$$= (0.1)(-0.8952)$$

$$= -0.0895$$

$$k_2 = hf(x_1 + h/2, y_1 + k_1/2)$$

$$= (0.1)f(0.15, 0.8605)$$

$$= (0.1)(-0.8388)$$

$$= -0.0838$$



$$k_3 = hf(x_1 + h/2, y_1 + k_2/2)$$

$$= (0.1)f(0.15, 0.8633)$$

$$= (0.1)((0.15)^2 - 0.8633)$$

$$= -0.0841$$

$$k_4 = hf(x_1 + h, y_1 + k_3)$$

$$= (0.1)f(0.2, 0.8211)$$

$$= -0.0781$$

$$\therefore y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= 0.9052 + \frac{1}{6}[-0.0893 + 2(-0.0833) + 2(-0.0841) - 0.0781]$$

$$y_2 = 0.8213$$

Q solve  $y' = xy$  using R.K method for  $x=0.2$  given that  $h=0.2$   
 $y(0)=1$ .

Sol:- Given differential equation  $y' = xy, y(0)=1$

here  $x_0=0, y_0=1$  and  $h=0.2$

$$f(x, y) = xy$$

To find  $y, y(0.2)$ :-

By Runge kutta method

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(x_0, y_0)$$

$$k_1 = 0.2f(0, 1)$$

$$k_1 = 0.2(0)$$

$$k_1 = 0$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$k_2 = (0.2)(0 + 0.2/2, 1 + 0/2)$$

$$k_2 = (0.2)(0.1, 1) \Rightarrow (0.2)(0.1)$$

$$k_2 = 0.02$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$$

$$k_3 = (0.2) \left( 0 + \frac{0.2}{2}, 1 + \frac{0.02}{2} \right)$$

$$k_3 = (0.2) (0.1, 1.01)$$

$$k_3 = (0.2) (0.101) \Rightarrow 0.0202$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$k_4 = (0.2) f(0 + 0.2, 1 + 0.0202)$$

$$k_4 = (0.2) (0.2, 1.0202)$$

$$k_4 = (0.2) (0.20404) \Rightarrow 0.040808$$

By R-K method

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$= 1 + \frac{1}{6} (0 + 2(0.02) + 2(0.0202) + 0.040808)$$

$$= 1.0202$$



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UNIT - IVSPECIAL FUNCTIONSBeta and Gamma Functions :-\* Beta Function :-

The definite integral  $\int_0^1 x^{m-1} (1-x)^{n-1} dx$  is called the beta function and is denoted by  $\beta(m, n)$ . and read as "beta m, n." The above converges for  $m > 0$ ,  $n > 0$ . Therefore,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

where,  $m > 0$  and  $n > 0$ .

$\beta$  function is also called eulerian integral of the first kind.

Properties of  $\beta$  function :-

1. Symmetry of beta ( $\beta$ ) function

1. Prove that  $\beta(m, n) = \beta(n, m)$

Proof :-

We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{put } 1-x = y \Rightarrow -dx = dy$$

$$1-y = x \Rightarrow dx = -dy$$

If  $x=0 \Rightarrow y=1$

$x=1 \Rightarrow y=0$

$$\beta(m, n) = \int_1^0 (1-y)^{m-1} (y)^{n-1} (-dy)$$

$$= - \int_0^1 (1-y)^{m-1} (y)^{n-1} dy$$

$$= \int_0^1 y^{n-1} (1-y)^{m-1} dy$$

$$= \beta(n, m)$$

$$\therefore \beta(m, n) = \beta(n, m)$$

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v.v.v.  
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2. Prove that  $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

Proof :-

We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{put } x = \sin^2 \theta$$

$$dx = 2 \sin \theta \cos \theta d\theta$$

$$\text{If } x=0 \Rightarrow \theta=0$$

$$x=1 \Rightarrow \theta = \pi/2$$

$$\beta(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} \sin^{2m-2} \theta (\cos^2 \theta)^{n-1} \sin \theta \cos \theta d\theta$$



$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

$$\therefore \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

3. Prove that  $\beta(m, n) = \beta(m+1, n) + \beta(m, n+1)$

Proof :-

By Beta definition

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{--- (1)}$$

Consider

$$\beta(m+1, n) + \beta(m, n+1) = \int_0^1 x^{(m+1)-1} (1-x)^{n-1} dx + \int_0^1 x^{m-1} (1-x)^{(n+1)-1} dx$$

$$= \int_0^1 x^m (1-x)^{n-1} dx + \int_0^1 x^{m-1} (1-x)^n dx$$

$$= \int_0^1 x^{m-1} (1-x)^{n-1} [x + (1-x)] dx$$

$$= \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \beta(m, n)$$

$$\therefore \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

4. Prove that  $\beta(m, 1) = 1/m$ .

Proof :-

By the beta definition

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Consider

$$\beta(m, 1) = \int_0^1 x^{m-1} (1-x)^{1-1} dx$$

$$= \int_0^1 x^{m-1} dx$$

$$= \left[ \frac{x^{m-1+1}}{m-1+1} \right]_0^1$$

$$= \left[ \frac{1^m}{m} - 0 \right] = 1/m$$

$$\therefore \boxed{\beta(m, 1) = 1/m}$$

5. Prove that  $\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2 \int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta$

Proof :-

By using second property,

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

$$\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2 \int_0^{\pi/2} \sin^{2\left(\frac{m+1}{2}\right)-1} \theta \cos^{2\left(\frac{n+1}{2}\right)-1} \theta d\theta$$



5. Prove that  $\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2 \int_0^{\pi/2} \sin^m \theta \cdot \cos^n \theta d\theta$

$$\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2 \int_0^{\pi/2} \sin^{m+1-1} \theta \cos^{n+1-1} \theta d\theta$$

$$\therefore \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2 \int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta$$

6. Prove that  $\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$

Proof :-

By using second property

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = 2 \int_0^{\pi/2} \sin^{2\left(\frac{1}{2}\right)-1} \theta \cdot \cos^{2\left(\frac{1}{2}\right)-1} \theta d\theta$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = 2 \int_0^{\pi/2} \sin \theta \cos \theta d\theta$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = 2 [0]_0^{\pi/2}$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = 2 \left[ \frac{\pi}{2} - 0 \right]$$

$$\boxed{\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \pi}$$

4. If  $m$  and  $n$  are positive integers then

$$\beta(m, n)$$

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Other forms of Beta Function :-Form - I :-

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\beta(m, n) = \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy$$

(or)

$$\beta(p, q) = \int_0^{\infty} \frac{y^{p-1}}{(1+y)^{p+q}} dy$$

Proof :-

By beta definition

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{put } x = \frac{1}{1+y} \Rightarrow dx = \left( \frac{-1}{(1+y)^2} \right) dy$$

$$\text{If } x=0 \Rightarrow 0 = \frac{1}{1+y} \Rightarrow y \rightarrow \infty$$

$$x=1 \Rightarrow 1 = \frac{1}{1+y} \Rightarrow y=0$$

$$\beta(m, n) = \int_{\infty}^0 \left( \frac{1}{1+y} \right)^{m-1} \left( 1 - \frac{1}{1+y} \right)^{n-1} \left( \frac{-1}{(1+y)^2} \right) dy$$

$$= - \int_{\infty}^0 \left( \frac{1}{1+y} \right)^{m-1} \left( \frac{1+y-1}{1+y} \right)^{n-1} \left( \frac{1}{(1+y)^2} \right) dy$$



$$\beta(m, n) = \int_0^{\infty} \left( \frac{1}{(1+y)} \right)^{m-1} \frac{y^{n-1}}{(1+y)^{n-1}} \frac{1}{(1+y)^2} dy$$

$$\beta(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m-1+n+2}} dy$$

$$\beta(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy$$

Put  $y = x$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\beta(n, m) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx \quad \left[ \because \text{by symmetry property} \right. \\ \left. \beta(m, n) = \beta(n, m) \right]$$

Form - II :-

To show that  $\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$

$$\beta(p, q) = \int_0^1 \frac{x^{p-1} + x^{q-1}}{(1+x)^{p+q}} dx$$

Proof :-

We know that

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$\beta(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_1^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx \quad \text{--- (1)}$$

$$\text{put } x = \frac{1}{y} \Rightarrow dx = \left( \frac{-1}{y^2} \right) dy$$

$$\text{If } x = 1 \Rightarrow y = 1$$

$$x \rightarrow \infty \Rightarrow y = 0$$

$$\int_1^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_1^0 \frac{\left(\frac{1}{y}\right)^{m-1}}{\left(1+\frac{1}{y}\right)^{m+n}} \left(\frac{-1}{y^2}\right) dy$$

$$= - \int_1^0 \frac{\left(\frac{1}{y}\right)^{m-1}}{\frac{(y+1)^{m+n}}{y^{m+n}}} \left(\frac{1}{y^2}\right) dy$$

$$= - \int_1^0 \frac{y^{m+n}}{(y+1)^{m+n}} y^{m-1} \left(\frac{1}{y^2}\right) dy$$

$$= \int_0^1 \frac{y^{m+n-m+1-2}}{1+y^{m+n}} dy$$

$$= \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}} dy$$



$$\beta(m, n) = \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

From equation ①

$$\beta(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx.$$

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Form - III :-

To show that  $\beta(m, n) = a^m b^n \int_0^{\infty} \frac{x^{m-1}}{(ax+b)^{m+n}} dx$

Proof :-

$$\begin{aligned} \text{R.H.S} &= a^m b^n \int_0^{\infty} \frac{x^{m-1}}{(ax+b)^{m+n}} dx \\ &= a^m b^n \int_0^{\infty} \frac{x^{m-1}}{b^{m+n} \left(\frac{a}{b}x + 1\right)^{m+n}} dx \end{aligned}$$

Put  $\frac{a}{b}x = y$

$$\frac{a}{b} dx = dy \Rightarrow dx = \frac{b}{a} dy$$

$$x = \frac{b}{a} y$$

$$x=0 \Rightarrow y=0$$

$$x \rightarrow \infty \Rightarrow y \rightarrow \infty$$

$$\text{R.H.S} = a^m b^n \int_0^{\infty} \frac{\left(\frac{b}{a}y\right)^{m-1}}{b^{m+n}(y+1)^{m+n}} dy$$

$$= a^m b^n \int_0^\infty \frac{b^{m-1} y^{m-1} \left(\frac{b}{a}\right) dy}{b^{m+n} (y+1)^{m+n} \left(\frac{b}{a}\right) dy}$$

$$\Rightarrow a^m b^n \int_0^\infty \frac{y^{m-1}}{a^m \cdot a b^{m-1} b \cdot b^{m+n-1} (y+1)^{m+n}} dy$$

$$\Rightarrow a^m b^n \int_0^\infty \frac{y^{m-1}}{a^m b^n (y+1)^{m+n}} dy$$

$$\Rightarrow \int_0^\infty \frac{y^{m-1}}{(y+1)^{m+n}} dy \quad (\because \text{form - I})$$

$$= B(m, n).$$

Form - IV

$$\text{To show that } \int_0^1 \frac{x^{m-1} (1-x)^{n-1}}{(x+a)^{m+n}} = \frac{B(m, n)}{a^n (1+a)^m}$$

Proof:-

$$\text{w.k.T } B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \rightarrow \textcircled{1}$$

$$\text{put } x = \frac{(1+a)t}{(t+a)} \Rightarrow dx = (1+a) \left( \frac{(t+a)(1) - t(1+a)}{(t+a)^2} \right) dt$$

$$\text{If } x=0 \Rightarrow t=0$$

$$x=1 \Rightarrow t=1$$

$$= (1+a) \left[ \frac{t+a-t}{(t+a)^2} \right] dt$$

$$= dx = (1+a) \left( \frac{a}{(t+a)^2} \right) dt$$

from eqn (1)

$$B(m, n) = \int_0^1 \left( \frac{(1+a)t}{(t+a)} \right)^{m-1} \left( 1 - \frac{(1+a)t}{(t+a)} \right)^{n-1} (1+a) \left( \frac{a}{(t+a)^2} \right) dt$$

$$= \int_0^1 \left( \frac{(1+a)t}{t+a} \right)^{m-1} \left( \frac{t+a-t-at}{t+a} \right)^{n-1} (1+a) \left( \frac{a}{(t+a)^2} \right) dt$$



$$= \int_0^1 \frac{((1+a)t)^{m-1}}{(t+a)^{m-1}} \frac{a^{n-1}(1-t)^{n-1}}{(t+a)^{n-1}} \left( (1+a) \left( \frac{a}{(t+a)^2} \right) \right) dt$$

$$= (1+a)^{m-1} a^{n-1} (1+a) a \int_0^1 \frac{t^{m-1} (1-t)^{n-1}}{(t+a)^{n-1} (t+a)^2 (t+a)^{m-1}} dt$$

$$= (1+a)^m \cdot a^n \cdot \int_0^1 \frac{t^{m-1} (1-t)^{n-1}}{(t+a)^{m+n}} dt$$

Put  $t = x$

$$\therefore \beta(m, n) = (1+a)^m a^n \int_0^1 \frac{x^{m-1} (1-x)^{n-1}}{(x+a)^{m+n}} dx$$

$$\frac{\beta(m, n)}{(1+a)^m a^n} = \int_0^1 \frac{x^{m-1} (1-x)^{n-1}}{(x+a)^{m+n}} dx$$

19/11/23 Form - V:-

To show that  $\int_a^b (x-b)^{m-1} (a-x)^{n-1} dx = (a-b)^{m+n} \beta(m, n)$ ,  $m > 0$ ,  $n > 0$ .

Proof:-

We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{Put } x = \frac{t-b}{a-b} \Rightarrow dx = \frac{dt}{a-b}$$

$$\text{If } x=0 \Rightarrow t=b$$

$$\text{If } x=1 \Rightarrow t=a$$

$$\beta(m, n) = \int_b^a \left( \frac{t-b}{a-b} \right)^{m-1} \left( 1 - \frac{t-b}{a-b} \right)^{n-1} \left( \frac{dt}{a-b} \right)$$

$$= \int_b^a \frac{(t-b)^{m-1}}{(a-b)^{m-1}} \left( \frac{a-b-t+b}{a-b} \right)^{n-1} \left( \frac{dt}{a-b} \right)$$

$$= \int_b^a \frac{(t-b)^{m-1}}{(a-b)^{m-1}} \frac{(a-t)^{n-1}}{(a-b)^{n-1}} \frac{dt}{(a-b)}$$

$$= \int_b^a \frac{(t-b)^{m-1} (a-t)^{n-1}}{(a-b)^{m-1+n-1}} dt$$

$$\beta(m, n) = \int_b^a \frac{(t-b)^{m-1} (a-t)^{n-1}}{(a-b)^{m+n-1}} dt$$

Put  $t = x$

$$\beta(m, n) = \int_b^a \frac{(x-b)^{m-1} (a-x)^{n-1}}{(a-b)^{m+n-1}} dx$$

$$\beta(m, n) (a-b)^{m+n-1} = \int_b^a (x-b)^{m-1} (a-x)^{n-1} dx$$

Problems :-

1. Prove that  $\int_0^1 \frac{x dx}{\sqrt{1-x^5}} = \frac{1}{5} \beta\left(\frac{2}{5}, \frac{1}{2}\right)$

2. Put  $x^5 = y$

$x = y^{1/5}$



$$dx = \frac{1}{5} y^{\frac{1}{5}-1} dy$$

$$dx = \frac{1}{5} y^{-4/5} dy$$

consider

$$\int_0^1 \frac{x dx}{\sqrt{1-x^5}} = \int_0^1 \frac{y^{1/5}}{\sqrt{1-y}}$$

$$= \int_0^1 \frac{y^{1/5}}{\sqrt{1-y}} \left( \frac{1}{5} y^{-4/5} \right) dy$$

$$= \frac{1}{5} \int_0^1 y^{-3/5} (1-y)^{-1/2} dy$$

$$= \frac{1}{5} \int_0^1 y^{2/5-1} (1-y)^{\frac{1}{2}-1} dy$$

$$\int_0^1 \frac{x dx}{\sqrt{1-x^5}} = \frac{1}{5} \beta \left( \frac{2}{5}, \frac{1}{2} \right)$$

2. Prove that  $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4} \beta \left( \frac{1}{4}, \frac{1}{2} \right)$

So

put  $x^4 = y$

$$x = y^{1/4}$$

$$dx = \frac{1}{4} y^{1/4-1} dy$$

$$dx = \frac{1}{4} y^{-3/4} dy$$

consider  $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \int_0^1 \frac{y^{1/4} \cdot \frac{1}{4} y^{-3/4}}{\sqrt{1-y}} dy$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4} \int_0^1 \frac{1}{\sqrt{1-y}} \cdot \frac{1}{4} y^{-3/4} dy$$

$$= \frac{1}{4} \int_0^1 y^{-3/4} (1-y)^{-1/2} dy$$

$$= \frac{1}{4} \int_0^1 y^{\frac{1}{4}-1} (1-y)^{\frac{1}{2}-1} dy$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4} \beta\left(\frac{1}{4}, \frac{1}{2}\right)$$

3 Prove that  $\int_0^1 x^3 (1-\sqrt{x})^5 dx = 2\beta(8,6)$

sol put  $\sqrt{x} = y$

$$x = y^2$$

$$dx = 2y^{2-1} dy$$

$$dx = 2y^1 dy$$

Consider  $\int_0^1 x^3 (1-\sqrt{x})^5 dx = \int_0^1 (y^2)^3 (1-y)^5 2y dy$

$$= 2 \int_0^1 y^6 (1-y)^5 y dy$$

$$= 2 \int_0^1 y^7 (1-y)^5 dy$$

$$= 2 \int_0^1 y^{8-1} (1-y)^{6-1} dy$$

$$= 2\beta(8,6)$$



4. Prove that  $\int_0^{\infty} \frac{x^{10} - x^{18}}{(1+x)^{30}} dx = 0$

Sol Given that  $\int_0^{\infty} \frac{x^{10} - x^{18}}{(1+x)^{30}} dx = 0$

$$\Rightarrow \int_0^{\infty} \frac{x^{10}}{(1+x)^{30}} dx - \int_0^{\infty} \frac{x^{18}}{(1+x)^{30}} dx$$

$$\Rightarrow \int_0^{\infty} \frac{x^{11-1}}{(1+x)^{11+19}} dx - \int_0^{\infty} \frac{x^{19-1}}{(1+x)^{19+11}} dx$$

$$\Rightarrow \beta(11, 19) - \beta(19, 11) \quad (\because \text{form I})$$

$$\Rightarrow 0$$

$\therefore$  Hence Proved.

5. Show that  $\int_0^1 \frac{(1-x^4)^{3/4}}{(1+x^4)^2} dx = \frac{1}{2^{9/2}} \beta\left(\frac{7}{4}, \frac{1}{4}\right)$

6. Express the following integrals in terms of  $\beta$  function.

(i)  $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx$

(ii)  $\int_0^{3/4} \frac{dx}{\sqrt{9-x^2}}$

(iii)  $\int_5^7 (x-5)^6 (7-x)^3 dx = 2^{10} \beta(7, 4)$

(iv)  $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^5}}$

(v)  $\int_0^1 \frac{x^3}{\sqrt{1-x^5}} dx$

20/4/23

## Gamma Function

The definite integral  $\int_0^{\infty} e^{-x} x^{n-1} dx$ ,  $n > 0$  is called the gamma function and is denoted by  $\Gamma(n)$  and it read as "Gamma of  $n$ ".

The integral converges only for  $n > 0$ .

$$\therefore \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, n > 0$$

Gamma function is called eulerian integral of the second kind.

The integral  $\int_0^{\infty} e^{-x} x^{n-1} dx$  does not converges if  $n \leq 0$ .

### Properties of Gamma Function :-

1. To show that  $\Gamma(1) = 1$ .

sol We know that

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, n > 0$$

$$\Gamma(1) = \int_0^{\infty} e^{-x} x^{1-1} dx$$

$$= \int_0^{\infty} e^{-x} dx$$

$$\int e^{-x} = -e^{-x}$$

$$= [-e^{-x}]_0^{\infty}$$

$$= -[e^{-\infty} - e^{-0}] = -[0 - 1] = 1 \Rightarrow \Gamma(1) = 1$$

notion



2. To show that  $\Gamma(n) = (n-1)\Gamma(n-1)$ , where  $n > 1$ .

Sol. We know that

$$\begin{aligned}\Gamma n &= \int_0^{\infty} e^{-x} x^{n-1} dx \\&= \int_0^{\infty} x^{n-1} d(-e^{-x}) \\&= \left[ x^{n-1} (-e^{-x}) \right]_0^{\infty} - \int_0^{\infty} (-e^{-x}) (n-1) x^{n-2} dx \\&= [0-0] + (n-1) \int_0^{\infty} e^{-x} x^{(n-1)-1} dx \\&= (n-1) \int_0^{\infty} e^{-x} x^{(n-1)-1} dx\end{aligned}$$

$$\boxed{\Gamma n = (n-1) \Gamma(n-1)}$$

Note :-

1.  $\Gamma n+1 = n \Gamma n$

2. If  $n$  is +ve fraction

$$\Gamma n = (n-1)(n-2)(n-3) \dots (n-r) \Gamma(n-r), \quad (n-r) > 0$$

3. If  $n$  is non-negative integer then  $\Gamma(n+1) = n!$

Sol. We know that  $\Gamma n = (n-1) \Gamma(n-1)$

$$\Gamma n+1 = n \Gamma n$$

$$\Gamma n+1 = n(n-1) \Gamma(n-1)$$

$$\begin{aligned}
 \Gamma_{n+1} &= n(n-1)(n-2) \Gamma_{(n-2)} \\
 &= n(n-1)(n-2)(n-3) \Gamma_{(n-3)} \\
 &= n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1 \\
 &= n!
 \end{aligned}$$

$$\boxed{\Gamma_{n+1} = n!}$$

\* Relation between Beta and Gamma Functions :-

$$\beta(m, n) = \frac{\Gamma_m \Gamma_n}{\Gamma_{m+n}}, \text{ where } m > 0, n > 0$$

Proof :-

$$\text{We know that } \Gamma_m = \int_0^{\infty} e^{-x} x^{m-1} dx \quad \text{--- (1)}$$

$$\text{put } x = yt \Rightarrow dx = y dt$$

$$\text{If } x = 0 \Rightarrow y = 0$$

$$x \rightarrow \infty \Rightarrow y \rightarrow \infty$$

$$\Gamma_m = \int_0^{\infty} e^{-yt} (yt)^{m-1} y dt$$

$$= \int_0^{\infty} e^{-yt} y^{m-1} t^{m-1} y dt$$

$$= \int_0^{\infty} e^{-yt} y^m t^{m-1} dt$$



put  $t=x$  in above equation we get

$$\Gamma_m = \int_0^{\infty} e^{-yx} y^m x^{m-1} dx \quad \text{--- (2)}$$

$$\frac{\Gamma_m}{y^m} = \int_0^{\infty} e^{-yx} x^{m-1} dx \quad \text{--- (3)}$$

Multiplying equation (3) on both sides  $e^{-y}$

$e^{-y} \cdot y^{m+n-1}$  we get

$$\frac{\Gamma_m}{y^m} e^{-y} y^{m+n-1} = e^{-y} y^{m+n-1} \int_0^{\infty} e^{-yx} x^{m-1} dx$$

$$\Gamma_m e^{-y} y^{n-1} = \int_0^{\infty} y \cdot e^{-y(1+x)} y^{m+n-1} x^{m-1} dx$$

Integrating on both sides w.r. to  $y$  to 0 to  $\infty$ .  
We get

$$\int_0^{\infty} \Gamma_m e^{-y} y^{n-1} dy = \int_0^{\infty} \left[ \int_0^{\infty} e^{-y(1+x)} y^{m+n-1} x^{m-1} dx \right] dy$$

$$= \int_0^{\infty} \left[ \int_0^{\infty} e^{-y(1+x)} y^{m+n-1} dy \right] x^{m-1} dx$$

$$\Gamma_m \Gamma_n = \int_0^{\infty} \frac{\Gamma_{m+n}}{(1+x)^{m+n}} x^{m-1} dx$$

$$\Gamma_m \Gamma_n = \Gamma_{m+n} \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \Gamma_{m+n} \beta(m, n)$$

$$\beta(m, n) = \frac{\Gamma_m \cdot \Gamma_n}{\Gamma_{m+n}}$$

Prove that  $\Gamma_{\frac{1}{2}} = \sqrt{\pi}$ .

We know that

$$\beta(m, n) = \frac{\Gamma_m \Gamma_n}{\Gamma_{m+n}}$$

Taking  $m = n = \frac{1}{2}$  in above equation, we get

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\Gamma_{\frac{1}{2}} \Gamma_{\frac{1}{2}}}{\Gamma_{\frac{1}{2} + \frac{1}{2}}}$$

$$= \frac{(\Gamma_{\frac{1}{2}})^2}{\Gamma_1}$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = (\Gamma_{\frac{1}{2}})^2 \quad \text{--- (1)} \quad (\because \Gamma_1 = 1)$$

We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^1 x^{\frac{1}{2}-1} (1-x)^{\frac{1}{2}-1} dx$$

$$= \int_0^1 x^{-\frac{1}{2}} (1-x)^{-\frac{1}{2}} dx$$

$$\text{put } x = \sin^2 \theta$$

$$dx = 2 \sin \theta \cos \theta d\theta$$

$$\text{If } x = 0 \Rightarrow \theta = 0$$

$$x = 1 \Rightarrow \theta = \pi/2$$



$$\begin{aligned}
 B\left(\frac{1}{2}, \frac{1}{2}\right) &= \int_0^{\pi/2} (\sin^2 \theta)^{-1/2} (1 - \sin^2 \theta)^{-1/2} 2 \sin \theta \cos \theta d\theta \\
 &= 2 \int_0^{\pi/2} \cancel{\sin \theta} (\cos^2 \theta)^{-1/2} \cancel{\cos \theta} d\theta \\
 &= 2 \int_0^{\pi/2} d\theta \\
 &= 2 [\theta]_0^{\pi/2} \\
 &= 2 \left[ \frac{\pi}{2} - 0 \right]
 \end{aligned}$$

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$$

From eq ①  $B\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\Gamma\left(\frac{1}{2}\right)\right)^2 = \sqrt{\pi} \Rightarrow \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$

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2. To show that  $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

Sol We know that

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx$$

put  $n = \frac{1}{2}$

$$\Gamma \frac{1}{2} = \int_0^{\infty} e^{-x} x^{\frac{1}{2}-1} dx$$

$$\Gamma \frac{1}{2} = \int_0^{\infty} e^{-x} x^{-\frac{1}{2}} dx$$

put  $x = t^2$

$$dx = 2t dt$$

If  $x=0 \Rightarrow t=0$

$x \rightarrow \infty \Rightarrow t \rightarrow \infty$

$$\begin{aligned}\Gamma_{\frac{1}{2}} &= \int_0^{\infty} e^{-t^2} (t^2)^{-\frac{1}{2}} 2t \, dt \\ &= 2 \int_0^{\infty} e^{-t^2} \cancel{t^{-1}} t \, dt\end{aligned}$$

$$\Gamma_{\frac{1}{2}} = \sqrt{\pi} = 2 \int_0^{\infty} e^{-t^2} dt$$

put  $t = x$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$\frac{\Gamma_{\frac{1}{2}}}{2} = \sqrt{\pi}$

3. To show that  $\Gamma n = \int_0^1 (\log \frac{1}{y})^{n-1} dy \quad (n > 0)$

so We know that

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx \quad \text{--- (1)}$$

put  $e^{-x} = y$

$$-e^{-x} dx = dy$$

$$-y dx = dy$$

$$dx = -\frac{1}{y} dy$$

If  $x=0 \Rightarrow y=1$

$x \rightarrow \infty \Rightarrow y=0$



From eq ①

$$\Gamma n = \int_1^0 \left( \log\left(\frac{1}{y}\right) \right)^{n-1} y \left(-\frac{1}{y}\right) dy$$

Answer

$$\Gamma n = - \int_1^0 \log\left(\frac{1}{y}\right)^{n-1} dy$$

$$\Gamma n = \int_0^1 \log\left(\frac{1}{y}\right)^{n-1} dy$$

4. Express the following integrals in terms of Gamma function.

a)  $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$

So, put  $x^2 = \sin \theta \Rightarrow x = \sqrt{\sin \theta} = \sin^{1/2} \theta$

$$2x dx = \cos \theta d\theta$$

$$dx = \frac{\cos \theta}{2x} d\theta$$

$$dx = \frac{\cos \theta}{2 \sin^{1/2} \theta} d\theta$$

If  $x=0 \Rightarrow \theta=0$

$x=1 \Rightarrow \theta=\pi/2$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \int_0^{\pi/2} \frac{1}{\sqrt{1-\sin^2 \theta}} \cdot \frac{\cos \theta}{2 \sin^{1/2} \theta} d\theta$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{2} \int_0^{\pi/2} \sin^{-1/2} \theta \, d\theta$$

$$\text{From } \Gamma(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta$$

$$= \frac{1}{4} 2 \int_0^{\pi/2} \sin^{2(\frac{1}{4})-1} \theta \cos^{2(\frac{1}{2})-1} \theta \, d\theta$$

$$= \frac{1}{4} \beta\left(\frac{1}{4}, \frac{1}{2}\right) \quad (\because 3^{rd} \text{ property})$$

$$= \frac{1}{4} \frac{\Gamma\left(\frac{1}{4}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4} + \frac{1}{2}\right)} \quad (\because \text{Relation b/w } \beta \text{ \& } \Gamma)$$

$$= \frac{1}{4} \frac{\Gamma\left(\frac{1}{4}\right) \cdot \sqrt{\pi}}{\Gamma\left(\frac{3}{4}\right)}$$

$$2m-1 = -\frac{1}{2}$$

$$2m = \frac{1}{2}$$

$$\Rightarrow m = \frac{1}{4}$$

$$2n-1 = 0$$

$$2n = 1$$

$$\Rightarrow n = \frac{1}{2}$$

$$\boxed{\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}}{4} \frac{\left(\Gamma\left(\frac{1}{4}\right)\right)}{\left(\Gamma\left(\frac{3}{4}\right)\right)}}$$

$$b) \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta$$

$$\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta = \int_0^{\pi/2} \sqrt{\frac{\sin \theta}{\cos \theta}} \, d\theta \quad \left(\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta\right)$$

$$= \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta \, d\theta$$

$$= \int_0^{\pi/2} \sin^{2(\frac{3}{4})-1} \theta \cos^{2(\frac{1}{4})-1} \theta \, d\theta$$



$$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \frac{1}{2} \beta\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$= \frac{1}{2} \cdot \frac{\Gamma \frac{3}{4} \cdot \Gamma \frac{1}{4}}{\Gamma \frac{3}{4} + \frac{1}{4}} \quad \left( \because \text{Relation b/w } \beta \text{ and } \Gamma \right)$$

$$= \frac{1}{2} \frac{\Gamma \frac{3}{4} \Gamma \frac{1}{4}}{\Gamma 1}$$

$$2m-1 = \frac{1}{2}$$

$$2m = \frac{3}{2}$$

$$m = \frac{3}{4}$$

$$2n-1 = -\frac{1}{2}$$

$$2n = \frac{1}{2}$$

$$n = \frac{1}{4}$$

$$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \frac{1}{2} \Gamma \frac{3}{4} \Gamma \frac{1}{4}$$

$$c) \int_0^1 x^4 \left(\log \frac{1}{x}\right)^3 dx$$

$$\text{Given } \int_0^1 x^4 \left(\log \frac{1}{x}\right)^3 dx \rightarrow \textcircled{1}$$

$$\text{put } \log \left(\frac{1}{x}\right) = \frac{t}{5}$$

$$\frac{1}{x} = e^{t/5}$$

$$x = e^{-t/5}$$

Differentiating on both sides

$$dx = -e^{-t/5} \left(\frac{1}{5}\right) dt$$

$$dx = -\frac{1}{5} e^{-t/5} dt$$

$$\text{If } x=0 \Rightarrow t=\infty$$

$$\text{If } x=1 \Rightarrow t=0$$

$\therefore$  Equating ① becomes,

$$\int_0^1 (e^{-t/5})^4 \left(\frac{t}{5}\right)^3 \cdot \frac{1}{5} (e^{-t/5}) dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \int_0^1 (e^{-t/5})^{4+1} \left(\frac{t}{5}\right)^3 \frac{1}{5} dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{5} \int_0^1 (e^{-t/5})^5 \left(\frac{t}{5}\right)^3 dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{5} \int_0^1 e^{-t} \left(\frac{t}{5}\right)^3 dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{5} \int_0^1 e^{-t} \frac{t^3}{5^3} dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{5} \cdot \frac{1}{125} \int_0^1 e^{-t} t^3 dt$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} \int_0^1 e^{-t} t^4 dt$$

Put  $t=x$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} \int_0^1 e^{-x} x^4 dt$$

$$= \frac{1}{625} \Gamma_4$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} \int_0^1 e^{-x} x^4 dt$$

$$= \frac{1}{625} \Gamma_4$$



$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} \Gamma(3+1)$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} (3!)$$

$$\int_0^1 x^4 (\log \frac{1}{x})^3 dx = \frac{1}{625} (6)$$

$$\therefore \int_0^1 x^4 (\log \frac{1}{x}) dx = \frac{6}{625}$$

d)  $\int_0^{\infty} x^{p-1} e^{-kx} dx \quad (k > 0)$

Sol Put  $Kx = y$

$$x = \frac{y}{K}$$

$$dx = \frac{1}{K} dy$$

If  $x = 0 \Rightarrow y = 0$

$x \rightarrow \infty \Rightarrow y \rightarrow \infty$

$$\int_0^{\infty} x^{p-1} e^{-kx} dx = \int_0^{\infty} \left(\frac{y}{K}\right)^{p-1} e^{-y} \frac{1}{K} dy$$

$$= \int_0^{\infty} \frac{y^{p-1}}{K^{p-1}} e^{-y} \frac{1}{K} dy$$

$$= \int_0^{\infty} y^{p-1} K^{-(p-1)} K^{-1} e^{-y} dy$$

$$= \int_0^{\infty} e^{-y} y^{p-1} K^{-p} dy$$

$$\int_0^{\infty} x^{p-1} e^{-ax} dx = \frac{1}{a^p} \int_0^{\infty} e^{-y} y^{p-1} dy$$

$$= \frac{1}{a^p} \Gamma(p) \quad \left( \because \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \right)$$

$$i) \int_0^{\infty} \sqrt{x} e^{-x^3} dx$$

$$\text{put } x^3 = y$$

$$x = y^{1/3}$$

$$dx = \frac{1}{3} y^{-2/3} dy$$

$$dx = \frac{1}{3} y^{-2/3} dy$$

$$\text{If } x=0 \Rightarrow y=0$$

$$x \rightarrow \infty \Rightarrow y \rightarrow \infty$$

$$\int_0^{\infty} \sqrt{x} e^{-x^3} dx = \int_0^{\infty} x^{1/2} e^{-y} \frac{1}{3} y^{-2/3} dy$$

$$= \frac{1}{3} \int_0^{\infty} (y^{1/3})^{1/2} e^{-y} y^{-2/3} dy$$

$$= \frac{1}{3} \int_0^{\infty} (y^{1/6}) e^{-y} y^{-2/3} dy$$

$$= \frac{1}{3} \int_0^{\infty} e^{-y} y^{-1/2} dy$$

$$= \frac{1}{3} \int_0^{\infty} e^{-y} y^{1/2-1} dy = \frac{1}{3} \Gamma \frac{1}{2}$$



$$= \frac{1}{3} \sqrt{\pi}$$

$$= \sqrt{\pi}/3$$

$$(\because \Gamma_n = \int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma_{\frac{1}{2}} = \sqrt{\pi})$$

Ex: compute (i)  $\Gamma_{3.5}$  (ii)  $\Gamma_{4.5}$  (iii)  $\Gamma_{\frac{1}{4}}$

(iv)  $\beta(0.5, 1.5)$  (v)  $\beta(\frac{9}{2}, \frac{7}{2})$

i)  $\Gamma_{3.5}$

$$\Gamma_{3.5} = \Gamma_{\frac{7}{2}}$$

$$= \left(\frac{7}{2} - 1\right) \Gamma_{\left(\frac{7}{2} - 1\right)}$$

$$= \frac{5}{2} \Gamma_{\frac{5}{2}}$$

$$= \frac{5}{2} \left(\frac{5}{2} - 1\right) \Gamma_{\left(\frac{5}{2} - 1\right)}$$

$$= \frac{5}{2} \left(\frac{3}{2}\right) \Gamma_{\left(\frac{3}{2}\right)}$$

$$= \frac{5}{2} \left(\frac{3}{2}\right) \left(\frac{3}{2} - 1\right) \left(\Gamma_{\left(\frac{3}{2} - 1\right)}\right)$$

$$= \frac{5}{2} \left(\frac{3}{2}\right) \left(\frac{1}{2}\right) \Gamma_{\frac{1}{2}}$$

$$= \frac{5}{2} \left(\frac{3}{2}\right) \left(\frac{1}{2}\right) (\sqrt{\pi})$$

$$\Gamma_{3.5} = \frac{15 \sqrt{\pi}}{8}$$

$$\Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\Gamma 4.5$$

$$\Gamma 4.5 = \Gamma \frac{9}{2}$$

$$= \left(\frac{9}{2} - 1\right) \Gamma\left(\frac{9}{2} - 1\right)$$

$$= \frac{7}{2} \Gamma \frac{7}{2}$$

$$= \frac{7}{2} \left(\frac{7}{2} - 1\right) \Gamma\left(\frac{7}{2} - 1\right)$$

$$= \frac{7}{2} \left(\frac{5}{2}\right) \Gamma\left(\frac{5}{2}\right)$$

$$= \frac{7}{2} \left(\frac{5}{2}\right) \left(\frac{5}{2} - 1\right) \Gamma\left(\frac{5}{2} - 1\right)$$

$$= \left(\frac{7}{2}\right) \left(\frac{5}{2}\right) \left(\frac{3}{2}\right) \Gamma \frac{3}{2}$$

$$= \left(\frac{7}{2}\right) \left(\frac{5}{2}\right) \left(\frac{3}{2} - 1\right) \Gamma\left(\frac{3}{2} - 1\right)$$

$$= \left(\frac{7}{2}\right) \left(\frac{5}{2}\right) \left(\frac{3}{2}\right) \Gamma\left(\frac{1}{2}\right)$$

$$= \left(\frac{7}{2}\right) \left(\frac{5}{2}\right) \left(\frac{3}{2}\right) \left(\frac{1}{2}\right) (\sqrt{\pi})$$

$$\Gamma 4.5 = \frac{105}{16} \sqrt{\pi}$$

$$(iv) \beta(2.5, 1.5)$$

We know that

$$\beta(m, n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$$



$$\beta(2.5, 1.5) = \frac{\Gamma 2.5 \Gamma 1.5}{\Gamma (2.5 + 1.5)}$$

$$= \frac{\Gamma \frac{5}{2} \Gamma \frac{3}{2}}{\Gamma \left( \frac{5}{2} + \frac{3}{2} \right)}$$

$$= \frac{\Gamma \frac{3}{2} + 1 \Gamma \frac{1}{2} + 1}{\Gamma 4}$$

$$= \frac{\frac{3}{2} \Gamma \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}}{\Gamma 4}$$

$$= \frac{\frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2} + 1 \sqrt{\pi}}{\Gamma 4}$$

$$= \frac{\frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\pi}}{\Gamma 4}$$

$$= \frac{\frac{3}{8} \pi}{\Gamma 4} = \frac{3\pi}{8 \times 6}$$

$$\beta(2.5, 1.5) = \frac{\pi}{16}$$

$$(v) \beta \left( \frac{9}{2}, \frac{7}{2} \right)$$

So We know that

$$5) \text{ Ans } \frac{5\pi}{2048}$$

$$\beta(m, n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$$

$$\beta(4.5, 3.5) = \frac{\Gamma 4.5 + \Gamma 3.5}{\Gamma(4.5 + 3.5)}$$

$$= \frac{\Gamma \frac{9}{2} \Gamma \frac{7}{2}}{\Gamma(\frac{9}{2} + \frac{7}{2})}$$

$$= \Gamma(\frac{7}{2} + 1)$$

$$= \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}}{\Gamma 8}$$

$$= \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \sqrt{\pi}}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$$= \frac{5 \pi}{15 \cdot 24}$$

$$\beta(\frac{9}{2}, \frac{7}{2}) = \frac{5 \pi}{2048}$$



2. Show that  $\int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$ .

$$\begin{aligned} \text{Sol} \quad \int_0^{\pi/2} \sqrt{\cot \theta} d\theta &= \int_0^{\pi/2} \sqrt{\frac{\cos \theta}{\sin \theta}} d\theta \\ &= \int_0^{\pi/2} \cos^{1/2} \theta \cdot \sin^{-1/2} \theta d\theta \\ &= 2 \frac{1}{2} \int_0^{\pi/2} \sin^{2(\frac{1}{4})-1} \theta \cos^{2(\frac{3}{4})-1} \theta d\theta \\ &= \frac{1}{2} B\left(\frac{1}{4}, \frac{3}{4}\right) \end{aligned}$$

$$2m-1 = -\frac{1}{2}$$

$$2m = -\frac{1}{2} + 1$$

$$2m = \frac{1}{2}$$

$$m = \frac{1}{4}$$

$$2n-1 = \frac{1}{2}$$

$$n = \frac{3}{4}$$

$$= \frac{\frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right) + \Gamma\left(\frac{3}{4}\right)} \quad (\because \text{Relation b/w } B \text{ \& } \Gamma)$$

$$= \frac{\frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)}{1}$$

$$\int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)$$

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3. Prove that  $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$

$$\text{Sol} \quad \int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$$

$$\int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \int_0^{\pi/2} \sin^{1/2} \theta d\theta$$

$$2m-1 = \frac{1}{2}$$

$$2m = \frac{3}{2}$$

$$m = \frac{3}{4}$$

$$\int_0^{\pi/2} \sqrt{\sin \theta} \, d\theta = \int_0^{\pi/2} \sin^{2(\frac{3}{4})-1} \theta \cos^{2(\frac{1}{2})-1} \theta \, d\theta \quad \left| \begin{array}{l} 2n-1=0 \\ 2n=1 \\ n=\frac{1}{2} \end{array} \right.$$

$$= \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{2}\right)$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4} + \frac{1}{2}\right)} \quad (\because \text{Relation b/w } B \& \Gamma)$$

$$\int_0^{\pi/2} \sqrt{\sin \theta} \, d\theta = \frac{1}{2} \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{5}{4}\right)} \quad \text{--- (1)}$$

$$\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \int_0^{\pi/2} \sin^{-1/2} \theta \, d\theta$$

$$= \int_0^{\pi/2} \sin^{2(\frac{1}{4})-1} \theta \cos^{2(\frac{1}{2})-1} \theta \, d\theta \quad \left| \begin{array}{l} 2m-1 = \frac{1}{2} \\ 2m = \frac{1}{2} \\ m = \frac{1}{4} \\ 2n-1=0 \\ n = \frac{1}{2} \end{array} \right.$$

$$= \frac{1}{2} B\left(\frac{1}{4}, \frac{1}{2}\right)$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4} + \frac{1}{2}\right)} \quad (\because \text{Relation b/w } B \& \Gamma)$$

$$\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \frac{1}{2} \frac{\Gamma\left(\frac{1}{4}\right) \sqrt{\pi}}{\Gamma\left(\frac{3}{4}\right)} \quad \text{--- (2)}$$

$$\int_0^{\pi/2} \sqrt{\sin \theta} \, d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \frac{1}{2} \frac{\Gamma\left(\frac{3}{4}\right) \sqrt{\pi}}{\Gamma\left(\frac{5}{4}\right)} \times \frac{1}{2} \frac{\Gamma\left(\frac{1}{4}\right) \sqrt{\pi}}{\Gamma\left(\frac{3}{4}\right)}$$



$$\begin{aligned}
 \int_0^{\pi/2} \sqrt{\sin \theta} \, d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} &= \frac{1}{2} \frac{\sqrt{\pi}}{\left(\frac{5}{4}-1\right)\Gamma\left(\frac{5}{4}-1\right)} \times \frac{1}{2} \Gamma\frac{1}{4} \cdot \sqrt{\pi} \\
 &= \frac{1}{2} \frac{\sqrt{\pi}}{\frac{1}{4}\Gamma\frac{1}{4}} \times \frac{1}{2} \cdot \Gamma\frac{1}{4} \cdot \sqrt{\pi} \\
 &= (\sqrt{\pi})^2
 \end{aligned}$$

$$\therefore \int_0^{\pi/2} \sqrt{\sin \theta} \, d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$$

4. Evaluate (i)  $\int_0^1 x^5 (1-x)^3 \, dx$

(ii)  $\int_0^1 x^4 (1-x)^2 \, dx$

(iii)  $\int_0^2 x (8-x^3)^{1/3} \, dx$

(iv)  $\int_0^1 x^{5/2} (1+x^2)^{3/2} \, dx \Rightarrow x^2 = y$

so (i)  $\int_0^1 x^5 (1-x^3) \, dx = \int_0^1 x^{6-1} (1-x)^{4-1} \, dx$

$= \beta(6, 4)$  ( $\because$  by  $\beta$ -function)

$= \frac{\Gamma 6 \Gamma 4}{\Gamma 6+4}$  ( $\because$  the relation b/w  $\beta$  &  $\Gamma$ )

$= \frac{5! 3!}{9!}$  [ $\Gamma n = (n-1)!$ ]

$$\Gamma \frac{1}{4} \cdot \sqrt{\pi}$$

$$\frac{1}{4} \cdot \sqrt{\pi}$$

$$\int_0^1 x^5 (1-x^3) dx = \frac{5! \times 3 \times 2 \times 1!}{9 \times 8 \times 7 \times 6 \times 5!}$$

$$= \frac{1}{9 \times 8 \times 7}$$

$$\int_0^1 x^5 (1-x^3) dx = \frac{1}{504}$$

$$(ii) \int_0^1 x^4 (1-x^2) dx$$

$$\int_0^1 x^4 (1-x^2) dx = \int_0^1 x^{5-1} (1-x)^{3-1} dx$$

$$= \beta(5, 3) \quad (\because \text{by } \beta\text{-function})$$

$$= \frac{\Gamma 5 \Gamma 3}{\Gamma 5+3}$$

$$= \frac{4! 2!}{8! 7!}$$

$$= \frac{5! 3!}{8 \times 7 \times 6 \times 5 \times 4 \times 3!}$$

$$= \frac{1}{8 \times 7} \frac{5! 3!}{7 \times 6 \times 5!}$$

$$= \frac{4! 2!}{7!} = \frac{4! 2!}{7 \times 6 \times 5 \times 4!}$$

$$\int_0^1 x^4 (1-x^2) dx = \frac{1}{105}$$



(iii)  $\int_0^2 x(8-x^3)^{1/3} dx$

Sol: put  $x^3 = 8y$

$$x = (8y)^{1/3}$$

$$= (8)^{1/3} y^{1/3}$$

$$= (2^3)^{1/3} y^{1/3}$$

$$\boxed{x = 2 \cdot y^{1/3}}$$

$$dx = 2 \cdot \frac{1}{3} y^{-2/3} dy$$

$$dx = \frac{2}{3} y^{-2/3} dy$$

If  $x=0 \Rightarrow y=0$

$x=2 \Rightarrow y=1$

$$\int_0^2 x(8-x^3)^{1/3} dx = \int_0^1 (2y)^{1/3} (8-8y)^{1/3} \frac{2}{3} y^{-2/3} dy$$

$$= \int_0^1 2^{1/3} \cdot y^{1/3} \cdot y^{-2/3} \cdot 8^{1/3} (1-y)^{1/3} \left(\frac{2}{3}\right) dy$$

$$= \frac{2}{3} \int_0^1 16^{1/3} y^{-1/3} (1-y)^{1/3} dy$$

$$= \frac{2}{3} \cdot 16^{1/3} \int_0^1 y^{-1/3} (1-y)^{1/3} dy$$

$$\int_0^2 x(8-x^3)^{1/3} dx = \frac{2}{3} \cdot 16^{1/3} \int_0^1 y^{\frac{2}{3}-1} (1-y)^{1/3} dy$$

$$\int_0^2 x(8-x)^{1/3} dx = \frac{2}{3} \cdot 16^{1/3} \beta\left(\frac{2}{3}, \frac{4}{3}\right)$$

$$= \frac{2}{3} \cdot 16^{1/3} \frac{\Gamma\frac{2}{3} \Gamma\frac{4}{3}}{\Gamma\frac{2}{3} + \frac{4}{3}}$$

$$= \frac{2}{3} \cdot 16^{1/3} \frac{\Gamma\frac{2}{3} \left(\frac{4}{3} - 1\right) \Gamma\frac{4}{3} - 1!}{\Gamma 2} \quad \Gamma n + 1 = \Gamma n \cdot n!$$

$$\Gamma 1 = 1$$

$$= \frac{2}{3} \cdot 16^{1/3} \frac{\Gamma\frac{1}{3} \cdot \frac{1}{3} \Gamma\frac{2}{3}}{\Gamma 1}$$

$$= \frac{2}{3} \cdot 16^{1/3} \Gamma\frac{2}{3} \cdot \frac{1}{3} \Gamma 1 - \frac{1}{3} \quad \left(\because n \Gamma 1 - n = \frac{\pi}{\sin n}\right)$$

$$= \frac{2}{3} \cdot 16^{1/3} \Gamma\frac{2}{3} \cdot \frac{\pi}{\sin \frac{\pi}{3}}$$

$$= \frac{2}{3} \cdot 16^{1/3} \Gamma\frac{2}{3} \cdot \frac{\pi}{\frac{\sqrt{3}}{2}}$$

$$\boxed{\int_0^2 x(8-x)^{1/3} dx = \frac{2}{3} \cdot 16^{1/3} \Gamma\frac{2}{3} \cdot \frac{2\pi}{\sqrt{3}}}$$

$$(i) \int_0^1 x^{5/2} (1-x^2)^{3/2} dx$$

$$\text{Put } x^2 = y$$



5. Evaluate (i)  $\int_0^{\infty} x^6 e^{-2x} dx$

(ii)  $\int_0^{\infty} e^{-4x} x^{3/2} dx$

Sol (i)  $\int_0^{\infty} x^6 e^{-2x} dx \Rightarrow$

put  $2x = y$

$x = \frac{y}{2}$

$dx = \frac{1}{2} dy$

If  $x = 0 \Rightarrow y = 0$

$x \rightarrow \infty \Rightarrow y \rightarrow \infty$

$\int_0^{\infty} x^6 e^{-2x} dx = \int_0^{\infty} \left(\frac{y}{2}\right)^6 e^{-y} \left(\frac{1}{2}\right) dy$

$= \int_0^{\infty} \frac{y^6}{2^7} e^{-y} \left(\frac{1}{2}\right) dy$

$= \frac{1}{2^7} \int_0^{\infty} y^6 e^{-y} dy$

$= \frac{1}{2^7} \int_0^{\infty} y^{7-1} e^{-y} dy$

$= \frac{1}{2^7} \Gamma 7$

$= \frac{1}{2^7} 6!$

$= \frac{1}{128} 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$\int_0^{\infty} x^6 e^{-2x} dx = \frac{45}{8}$

## Functions of Complex Variables

Any number of the form  $x+iy$  where  $x$  and  $y$  are real numbers and  $i = \sqrt{-1}$  is called a complex number. Here  $x$  is called real part of  $x+iy$  and is written as  $\text{Re}(x+iy)$ ,  $y$  is called imaginary part of  $x+iy$  and is written as  $\text{Im}(x+iy)$ .

- \* If  $z = x+iy$  then  $\text{Re}(z) = x$  and  $\text{Im}(z) = y$
- \* If  $z = x+iy$  then the number  $x-iy$  is called conjugate of  $x+iy$  and is denoted by  $\bar{z}$ .
- \* If  $z = x+iy$  then the number  $r = \sqrt{x^2+y^2}$  is called modulus value of  $x+iy$  and is denoted by  $|z|$  or  $\text{mod}(x+iy)$  or  $|x+iy|$
- \* The angle  $\theta = \tan^{-1} \frac{y}{x}$  is called amplitude or argument of  $x+iy$  and is denoted by  $\text{amp}(x+iy)$  or  $\text{arg}(x+iy)$ .
- \* Let  $z = x+iy$  put  $x = r \cos \theta$  and  $y = r \sin \theta$ . then we get  $z = r \cos \theta + i r \sin \theta = r (\cos \theta + i \sin \theta)$ .  
ie.,  $z = r e^{i\theta}$  is called polar form of the complex number  $x+iy$ .

## Functions of Complex Variables

If for each value of the complex variable  $z (= x+iy)$  in given region  $R$ , we assigns one or more values of  $w (= u+iv)$ , then  $w$  is said to be a complex function of  $z$  and we write  $w = f(z) = u(x,y) + i v(x,y)$  where  $u(x,y)$  and  $v(x,y)$  are real functions of  $x$  &  $y$ .



examp:- let  $w = f(z) = z^2 + z$  is complex function

put  $z = x + iy$  then we get

$$w = (x + iy)^2 + (x + iy)$$

$$= x^2 - y^2 + 2ixy + x + iy$$

$$w = x^2 - y^2 + 2ixy + x + iy$$

$$w = (x^2 - y^2 + x) + i(2xy + y)$$

$$= u(x, y) + i v(x, y)$$

Here  $u(x, y) = x^2 - y^2 + x$  - real part of  $w$

$v(x, y) = 2xy + y$  - imaginary part of  $w$

$$\text{If } z = 1 + i = x + iy$$

$$x = 1, y = 1$$

$$\text{then } w = 1 + 3i$$

$$\text{If } z = 2 - 3i = x + iy$$

$$x = 2, y = -3$$

$$x = 1, y = 1$$

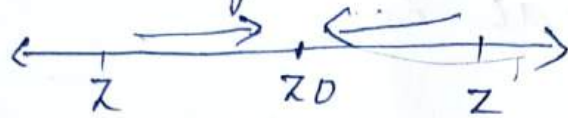
$$w = 1 + 3i$$

Neighbourhood of point:-

Neighbourhood of a point  $z_0$  is the set of all points  $z$  for which  $|z - z_0| < \epsilon$  where  $\epsilon$  is positive constant.

Limit of a function:- A function  $f(z)$  is said to tend to limit 'l' as  $z$  approaches a point  $z_0$  if for every real  $\epsilon$ , we can find a positive  $\delta$  such that  $|f(z) - l| < \epsilon$  for  $0 < |z - z_0| < \delta$ .

Note:- In case of real system  $z$  can approach  $z_0$  only along the real line either from its left or from right its right.



In complex plane, complex variables  $z$  may approach  $z_0$  from any direction



Problem:- Find  $\lim_{z \rightarrow 1} \frac{z^2 - 1}{z - 1}$

Sol:-  $\lim_{z \rightarrow 1} \frac{z^2 - 1}{z - 1} = \lim_{z \rightarrow 1} \frac{(z-1)(z+1)}{z-1}$   
 $= \lim_{z \rightarrow 1} z+1 = 1+1 = 2$

Continuity:- A function  $f(z)$  is said to be continuous at  $z = z_0$  if  $f(z_0)$  is defined and  $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

Note:-  $f(z)$  is said to be continuous in a domain if it is continuous at every point of that domain.

Properties:- (i) If  $f(z)$  and  $g(z)$  are continuous at  $z_0$ , then

(i)  $f(z) + g(z)$  (ii)  $f(z) - g(z)$  (iii)  $f(z) \cdot g(z)$  (iv)  $\frac{f(z)}{g(z)}$ ,  $g(z) \neq 0$   
 are also continuous at  $z_0$ .

(2) Any polynomial function is continuous for all  $z$

(3) Any irrational function is continuous whenever its denominator is non-zero.



- (4) All trigonometric and exponential functions are continuous.
- (5) If  $f(z)$  is continuous at  $z_0$  then  $|f(z)|$  is also continuous at  $z_0$ .

$$\lim_{\substack{z \rightarrow z_0 \\ y \rightarrow y_0}} f(z), \quad \lim_{\substack{y \rightarrow y_0 \\ z \rightarrow z_0}} f(z), \quad \lim_{\substack{y \rightarrow my \\ x \rightarrow x_0}} f(z), \quad \lim_{\substack{x \rightarrow mx \\ y \rightarrow y_0}} f(z), \quad \dots$$

It is denoted by  $\lim_{z \rightarrow z_0} f(z) = l$  otherwise the limit does not exist.

Ex:- Is the function  $f(z) = \begin{cases} \frac{z^2 + 3iz - 2}{(z+i)} & \text{for } z \neq -i \\ b & z = -i \end{cases}$

continuous.

Sol:- Here  $z_0 = -i = 0 + i(-1) = x_0 + iy_0$

$$x_0 = 0, y_0 = -1$$

$$\lim_{\substack{z \rightarrow z_0 \\ x \rightarrow x_0 \\ y \rightarrow y_0}} f(z) = \lim_{\substack{z \rightarrow z_0 \\ x \rightarrow x_0 \\ y \rightarrow y_0}} f(z) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow -1}} \frac{-(x+iy)^2 + 3i(x+iy) - 2}{x+iy+i}$$

$$= \lim_{y \rightarrow -1} \frac{(iy)^2 + 3i(iy) - 2}{iy+1}$$

$$= \frac{-[y^2 + 3y - 2]}{iy+1}$$

## Differentiability:

A function  $f(z)$  is said to be differentiable at a point  $z_0$  if the limit  $\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$

exists. It is denoted by  $f'(z_0)$  taking  $z - z_0 = \Delta z$

$$\text{Then } f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

## Analytic Function

A function  $f(z)$  is said to be Analytic at a point  $z = z_0$  if  $f(z)$  is derivable at every point in an  $\varepsilon$ -neighbourhood of  $z_0$ . ( $|z - z_0| < \varepsilon$ )

Let  $D$  be the domain of complex number, if  $f(z)$  is analytic at every point  $z$  in  $D$ , then  $f(z)$  is said to be analytic in the domain  $D$ .

If  $f(z)$  is analytic at every point  $z$  on the complex plane,  $f(z)$  is said to be entire function or integral function.

An analytic function is also known as regular or holomorphic function.

## Singular Point:-

A point  $z$  at which a complex function  $w = f(z)$  fails to be analytic is called a singular point of  $f(z)$ .



## Cauchy - Riemann (C-R) Equation

Cauchy - Riemann Equations are used to determine whether a complex function is analytic or not.

Cartesian form:- The necessary and Sufficient Condition for the derivative of a function  $f(z)$

$f(z) = w = u(x, y) + i v(x, y)$  to exist for all values of  $z$  in domain  $R$  are

(i)  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  are continuous functions of  $x$  and  $y$  in  $R$ .

(ii)  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  and  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ .

The Equations (ii) are known as Cauchy-Riemann Equations.

Note:- C-R Equations are necessary but not sufficient. i.e., If the function  $f(z) = w = u(x, y) + i v(x, y)$  is differentiable then it must satisfy C-R Equations.

But the converse is not true.

i.e., If a function satisfies C-R Equations at a point but it is not differentiable at that point.

Polar form:- Let  $x = r \cos \theta$  and  $y = r \sin \theta$ .

$$z = x + iy = r \cos \theta + i r \sin \theta$$

$$= r(\cos \theta + i \sin \theta)$$

$= r e^{i\theta}$ , is the polar form of a complex number

C-R Equations are  $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$  and  $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$

## Properties of Analytic Function

(14)

~~Prove~~

(1) If  $f(z)$  and  $g(z)$  are analytic functions, then

(a)  $f(z) \pm g(z)$  is an analytic function

(b)  $f(z) \cdot g(z)$  is an analytic function.

(c)  $\frac{f(z)}{g(z)}$  is analytic provided  $g(z) \neq 0$  in  $D$ .

(2) Analytic function of an analytic function is analytic

(3) An entire function of an entire function is entire

C-R Equation at a point  $z = z_0$  :-

The Cauchy-Riemann Equations at point  $z = z_0 = x_0 + iy_0$  are given by

$$\left(\frac{\partial u}{\partial x}\right)_{(x_0, y_0)} = \lim_{x \rightarrow x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0}$$

$$\left(\frac{\partial u}{\partial y}\right)_{(x_0, y_0)} = \lim_{y \rightarrow y_0} \frac{u(x_0, y) - u(x_0, y_0)}{y - y_0}$$

$$\left(\frac{\partial v}{\partial x}\right)_{(x_0, y_0)} = \lim_{x \rightarrow x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0}$$

$$\left(\frac{\partial v}{\partial y}\right)_{(x_0, y_0)} = \lim_{y \rightarrow y_0} \frac{v(x_0, y) - v(x_0, y_0)}{y - y_0}$$



Problems:- check whether the polynomial function  $f(z) = z^2 + z$  is analytic or not for all  $z$ ?

Sol:- Given  $f(z) = z^2 + z$   
$$= (x+iy)^2 + (x+iy)$$
$$= x^2 - y^2 + 2ixy + x + iy$$
$$= x^2 - y^2 + x + i(2xy + y)$$
$$= u(x, y) + iv(x, y) \text{ say}$$

where  $u(x, y) = x^2 - y^2 + x$ ,  $v(x, y) = 2xy + y$

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [x^2 - y^2 + x] = 2x + 1$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [x^2 - y^2 + x] = -2y$$

$$v_x = \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} [2xy + y] = 2y$$

$$v_y = \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} [2xy + y] = 2x + 1$$

From above  $u_x = v_y$  and  $u_y = -v_x$

C-R Equations are satisfied and all partial derivatives are continuous.

$\therefore$  Given function is analytic function

Problem:- Check whether the polynomial function.

$f(z) = (2x^2 + y) + i(y^2 - x)$  is analytic or not.  $\forall z$

Sol:- Given  $f(z) = (2x^2 + y) + i(y^2 - x)$

which is of the form  $f(z) = u(x, y) + iv(x, y)$

where  $u(x, y) = 2x^2 + y$

$v(x, y) = y^2 - x$

$$\left. \begin{aligned} u_x &= \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(2x^2 + y) = 4x \\ u_y &= \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(2x^2 + y) = 1 \\ v_x &= \frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(y^2 - x) = -1 \\ v_y &= \frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(y^2 - x) = 2y \end{aligned} \right\}$$

$$u_x \neq v_y, \quad u_y \neq -v_x$$

C-R Equations are not Satisfied

$\therefore$  Given function is not an analytic function

Problem: Find all values of  $k$  such that  $f(z)$ ,

$f(z) = e^z [\cos ky + i \sin ky]$  is analytic.

Sol:- Given  $f(z) = e^z [\cos ky + i \sin ky]$

$$= e^x \cos ky + i e^x \sin ky$$

which is of the form  $f(z) = u(x, y) + i v(x, y)$

$$u(x, y) = e^x \cos ky, \quad v(x, y) = e^x \sin ky$$

$$\frac{\partial u}{\partial x} = u_x = e^x \cos ky, \quad \frac{\partial v}{\partial x} = v_x = e^x \sin ky$$

$$\frac{\partial u}{\partial y} = u_y = -e^x \sin ky \cdot k, \quad \frac{\partial v}{\partial y} = v_y = \cos ky \cdot k e^x$$

$$= -k e^x \sin ky, \quad \frac{\partial v}{\partial y} = k e^x \cos ky$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow e^x \cos ky = k e^x \cos ky$$

$$\Rightarrow k = 1$$

$$+ k e^x \sin ky = -e^x \sin ky$$

$$\Rightarrow k = -1$$

$$\therefore k = \pm 1$$



Problem:- prove that the function  $f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is continuous and the Cauchy-Riemann Equations are satisfied at the origin but  $f'(0)$  does not exist.

Sol:- Here  $z_0 = 0 \Rightarrow x_0 + iy_0 = 0 + i0$

$$x_0 = 0$$

$$y_0 = 0$$

Given  $f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

Continuity at origin

path 1:-  $\lim_{z \rightarrow z_0} f(z) = \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(z) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}$

$$= \lim_{y \rightarrow 0} \frac{-y^3(1-i)}{y^2} = 0$$

path 2:-  $\lim_{z \rightarrow z_0} f(z) = \lim_{\substack{y \rightarrow y_0 \\ x \rightarrow x_0}} f(z) = \lim_{\substack{y \rightarrow 0 \\ x \rightarrow 0}} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} = 0$

path 3:-  $\lim_{z \rightarrow z_0} f(z) = \lim_{\substack{y \rightarrow my \\ x \rightarrow x_0}} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} = \lim_{x \rightarrow x_0} \frac{x^3(1+i) - m^3x^3(1-i)}{x^2 + m^2x^2}$

path 4:-  $\lim_{z \rightarrow z_0} f(z) = \lim_{\substack{x \rightarrow mx \\ y \rightarrow y_0}} \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} = \lim_{y \rightarrow y_0} \frac{m^3y^3(1+i) - y^3(1-i)}{m^2y^2 + y^2} = 0$

Therefore,  $\lim_{z \rightarrow 0} f(z)$  exists and is equal to  $f(0)$

hence  $f(z)$  is continuous at origin.

Cauchy-Riemann Equations at origin (6)

$$\text{Given } f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2} = \frac{(x^3+y^3) + i(x^3-y^3)}{x^2+y^2}$$

$$= \frac{x^3+y^3}{x^2+y^2} + i \frac{x^3-y^3}{x^2+y^2}$$

$$\therefore f(z) = u(x,y) + i v(x,y) \text{ form}$$

$$u(x,y) = \frac{x^3+y^3}{x^2+y^2}, \quad v(x,y) = \frac{x^3-y^3}{x^2+y^2}$$

$$\text{Now } u_x = \frac{\partial u}{\partial x} = \lim_{x \rightarrow 0} \frac{u(x,0) - u(0,0)}{x-0} = \lim_{x \rightarrow 0} \frac{x-0}{x} = 1$$

$$u_y = \frac{\partial u}{\partial y} = \lim_{y \rightarrow 0} \frac{u(0,y) - u(0,0)}{y-0} = \lim_{y \rightarrow 0} \frac{-y-0}{y} = -1$$

$$v_x = \frac{\partial v}{\partial x} = \lim_{x \rightarrow 0} \frac{v(x,0) - v(0,0)}{x-0} = \lim_{x \rightarrow 0} \frac{x-0}{x} = 1$$

$$v_y = \frac{\partial v}{\partial y} = \lim_{y \rightarrow 0} \frac{v(0,y) - v(0,0)}{y-0} = \lim_{y \rightarrow 0} \frac{y-0}{y} = 1$$

Hence  $u_x = v_y$  and  $u_y = -v_x$ .

Therefore the given function satisfies C-R Equations at origin.

Differentiability at origin

$$\text{Now } f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z-0} = \lim_{z \rightarrow 0} \frac{(x^3+y^3) + i(x^3-y^3)}{(x^2+y^2)(x+iy)}$$

Let  $z \rightarrow 0$  along the path  $y = mx$ .

$$\begin{aligned} \therefore f'(0) &= \lim_{\substack{y \rightarrow mx \\ x \rightarrow 0}} \frac{(x^3+y^3) + i(x^3-y^3)}{(x^2+y^2)(x+iy)} = \lim_{x \rightarrow 0} \frac{(x^3-m^3x^3) + i(x^3-x^3m^3)}{(x^2+m^2x^2)(x+imx)} \\ &= \frac{(1-m^3) + i(1+m^3)}{(1+m^2)(1+im)} \end{aligned}$$



which depends on the value of  $m$ , hence it is not unique. Thus  $f'(z)$  does not exist at  $(0,0)$ .  
Hence  $f'(z)$  is not analytic at  $z=0$ .

Problem:- Show that the function  $f(z) = z\bar{z}$  is differentiable but not analytic at  $z=0$ .

Sol:- Given that  $f(z) = z\bar{z}$

where  $z = x+iy$   $\bar{z} = x-iy$

$$f(z) = z\bar{z} = (x+iy)(x-iy) = x^2 + y^2$$

$$\text{ie, } f(z) = (x^2 + y^2) + i(0) = u(x,y) + i v(x,y) \text{ Say}$$

$$\text{Here } u(x,y) = x^2 + y^2, \quad v(x,y) = 0$$

$$u_x = \frac{\partial u}{\partial x} = 2x, \quad v_x = \frac{\partial v}{\partial x} = 0$$

$$u_y = \frac{\partial u}{\partial y} = 2y, \quad v_y = \frac{\partial v}{\partial y} = 0$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} x+y^2 = 0$$

$$\therefore u_x = v_y \text{ only when } x=0$$

$$u_y = -v_x \text{ only when } y=0$$

Hence the C-R Equations are satisfied only at origin.

Hence  $f(z)$  has a derivative at  $z=0$ .

But the C-R Equations are not satisfied for  $z \neq 0$ .

Thus  $f(z)$  is not analytic at  $z=0$ .

$\therefore f(z)$  is differentiable but not analytic at  $z=0$ .

## Laplace Equation

A function is said to satisfy Laplace's Equation if  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$  i.e.,  $\nabla^2 u = 0$

Problem: Show that both the real and imaginary parts of an analytic function satisfy Laplace's Equation.

Solu - Let  $f(z) = u(x, y) + iv(x, y)$  is analytic function.

Since  $f(z)$  is analytic, it must satisfy C-R Equations

$$\text{i.e., } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad , \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{--- (2)}$$

Now differentiate (1) w.r.t  $x$  partially,

$$\text{we get } \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{--- (3)}$$

Differentiate (2) w.r.t  $y$  partially

$$\frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x} \quad \text{--- (4)}$$

(3) + (4)

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x} = 0$$

$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ . Hence  $u$  satisfies Laplace

Equation

Now diff (1) w.r.t  $y$  partially, we get

$$\frac{\partial u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2} \quad \text{--- (5)}$$

Diff (2) w.r.t  $x$  partially we get

$$\frac{\partial u}{\partial x \partial y} = -\frac{\partial^2 v}{\partial x^2} \Rightarrow \frac{\partial^2 v}{\partial x^2} = -\frac{\partial u}{\partial x \partial y} \quad \text{--- (6)}$$



$$(5) + (6) \Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y \partial x} = 0$$

$$\therefore \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Here  $v$  satisfies Laplace's Equation

Here both the real and imaginary parts of an analytic function satisfy Laplace's Equation.

## Harmonic Functions

A function which satisfies Laplace's Equation is called a Harmonic function.

i.e. if  $\nabla^2 u = 0$  then  $u$  is called a Harmonic function

Note:- The real and imaginary parts of an analytic function are Harmonic.

i.e., if  $f(z) = u + iv$  then  $u$  and  $v$  satisfy the Laplace's Equation. Here  $u$  is called conjugate harmonic of  $v$ . Similarly  $v$  is called conjugate harmonic function of  $u$ .

Example:- Show that  $\left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] \left[ \text{Real } f(z) \right]^2 = 2 |f'(z)|^2$  where  $f(z)$  is analytic.

Sol:- Let  $f(z) = u(x, y) + iv(x, y)$

$$\text{Now } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

Since  $f(z)$  is analytic

So it satisfies C-R Equations  $\left[ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \right]$  (6)

$$\text{Now } |f'(z)| = \sqrt{u_x^2 + u_y^2}$$

$$\text{Since } |z| = |x + iy| = \sqrt{x^2 + y^2}$$

$$\Rightarrow |f'(z)|^2 = u_x^2 + u_y^2$$

$$\text{Now } \frac{\partial}{\partial x} u^2 = 2u \frac{\partial u}{\partial x}$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} [u^2] = \frac{\partial}{\partial x} \left[ \frac{\partial}{\partial x} u^2 \right] = \frac{\partial}{\partial x} \left[ 2u \frac{\partial u}{\partial x} \right]$$

$$= 2 \left[ u \frac{\partial^2 u}{\partial x^2} + \left( \frac{\partial u}{\partial x} \right)^2 \right] \quad \text{--- (1)}$$

$$\text{Similarly } \frac{\partial^2}{\partial y^2} [u^2] = 2 \left[ u \frac{\partial^2 u}{\partial y^2} + \left( \frac{\partial u}{\partial y} \right)^2 \right] \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow \frac{\partial^2 u^2}{\partial x^2} + \frac{\partial^2 u^2}{\partial y^2} = 2 \left[ u \frac{\partial^2 u}{\partial x^2} + \left( \frac{\partial u}{\partial x} \right)^2 + u \frac{\partial^2 u}{\partial y^2} + \left( \frac{\partial u}{\partial y} \right)^2 \right]$$

$$\Rightarrow \left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] u^2 = 2 \left[ \left( \frac{\partial u}{\partial x} \right)^2 + \left( \frac{\partial u}{\partial y} \right)^2 + u \left[ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right] \right]$$

Since  $f(z)$  is analytic function hence real part  $u$  satisfies the Laplace's Equation

$$\text{i.e., } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\Rightarrow \left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] [\text{real } f(z)]^2 = 2 \left[ \left( \frac{\partial u}{\partial x} \right)^2 + \left( \frac{\partial u}{\partial y} \right)^2 \right]$$

$$\Rightarrow \left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] [\text{real } f(z)]^2 = 2 [u_x^2 + u_y^2]$$
$$= 2 |f'(z)|^2$$

$$\therefore \left[ \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] (\text{real } f(z))^2 = 2 |f'(z)|^2$$



Problem:- Show that  $xy^2$  cannot be a real part of an analytic function.

Sol:- Let  $u(x,y) = xy^2$

If  $u(x,y)$  is a real part of an analytic function  $f(z)$  then  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

$$\text{Now } \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(xy^2) = y^2$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(xy^2) = 2xy$$

$$\text{and } \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial x}\right] = \frac{\partial}{\partial x}(y^2) = 0$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial y}\right] = \frac{\partial}{\partial y}(2xy) = 2x$$

$$\text{Clearly } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 + 2x = 2x \neq 0$$

Hence  $u(x,y) = xy^2$  cannot be a real part of an analytic function.

problem:-

Find the most general analytic function whose real part is  $x^2 - y^2 - x$ .

Sol:- Let  $u(x,y) = x^2 - y^2 - x$

$$\text{Now } \frac{\partial u}{\partial x} = 2x - 1$$

$$\frac{\partial u}{\partial y} = -2y$$

Let  $v(x,y)$  be the conjugate harmonic function of  $u(x,y)$

Then by C-R Equations

2

(9)

We have  $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -[-2y] = 2y$

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 2x - 1$$

Since  $v(x, y)$  is a function of  $x, y$  so we write

$$\begin{aligned} dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \\ dv &= 2y dx + (2x - 1) dy \\ &= M dx + N dy \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} M &= 2y, \quad N = 2x - 1 \\ \frac{\partial M}{\partial y} &= 2, \quad \frac{\partial N}{\partial x} = 2 \end{aligned}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

The general solution of Eq (1) Exact D.E is

$$v = \int M dx + \int (\text{terms of } N \text{ not containing } x) dy = c$$

$$v = \int 2y dx + \int (-1) dy = c$$

$$v = 2yx - y$$

$$v(x, y) = 2xy - y$$

Hence  $f(z) = u(x, y) + i v(x, y)$

$$= \{x^2 - y^2 - x\} + i \{2xy - y\} \text{ is the required}$$

analytic function.



## Milne Thomson's Method

problem:- If  $f(z) = u + iv$  is an analytic function of  $z$  and if  $u - v = e^x [\cos y - \sin y]$ , find  $f(z)$  in terms of  $z$ .

Sol:- Given that  $u - v = e^x [\cos y - \sin y]$  — (1)

diff (1) partially w.r.t  $x$  and  $y$   
we get

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} = e^x [\cos y - \sin y] \text{ — (2)}$$

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} = e^x [-\sin y - \cos y] \text{ — (3)}$$

$$\text{But } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Since  $f(z)$  is analytic

so eq (3) becomes

$$-\frac{\partial v}{\partial x} - \frac{\partial v}{\partial y} = -e^x [\sin y + \cos y]$$

$$\Rightarrow \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} = e^x [\sin y + \cos y]$$

$$\Rightarrow \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x} = e^x [\sin y + \cos y] \text{ — (4)}$$

$$(2) + (4) \Rightarrow 2 \frac{\partial u}{\partial x} = 2 e^x [\cos y]$$

$$\Rightarrow \frac{\partial u}{\partial x} = e^x \cos y$$

$$(4) - (2) \Rightarrow 2 \frac{\partial v}{\partial x} = 2 e^x \sin y$$

$$\Rightarrow \frac{\partial v}{\partial x} = e^x \sin y$$

$$\text{Now } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x [\cos y + i \sin y]$$

$$\cos y - \sin y + i$$

By Milne-Thomson Method,

Express  $f'(z)$  in terms of  $z$  by replacing  $x$  by  $z$  and  $y$  by 0

$$f'(z) = e^z$$

Integrating on b.s, we get

$$f(z) = e^z + c$$

This is the required analytic function.

→ Find the regular function whose imaginary part is  $e^{-x}[x \cos y + y \sin y]$ .

sol:- Let  $v(x, y) = e^{-x}[x \cos y + y \sin y]$

$$\text{Now } \frac{\partial v}{\partial x} = \cos y [-x e^{-x} + e^{-x}] + y \sin y (-e^{-x})$$

$$= -x e^{-x} \cos y + e^{-x} \cos y - e^{-x} y \sin y$$

$$= e^{-x} [-x \cos y + \cos y - y \sin y]$$

$$\frac{\partial v}{\partial y} = -x e^{-x} \sin y + e^{-x} [y \cos y + \sin y]$$

$$= -x e^{-x} \sin y + e^{-x} y \cos y + e^{-x} \sin y$$

$$= e^{-x} [-x \sin y + y \cos y + \sin y]$$

$$\text{Now } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial y}$$

$$= \frac{\partial v}{\partial y} - i \frac{\partial v}{\partial x}$$

$$f(z) = u + iv$$
$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$
$$= \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$



→

Prove that  $z^n$  is analytic. and hence find its derivative  
sol: let  $z = re^{i\theta}$

$$f(z) = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$$

$$= u + iv$$

$$\text{where } u = r^n \cos n\theta \quad v = r^n \sin n\theta$$

$$\frac{\partial u}{\partial r} = nr^{n-1} \cos n\theta$$

$$\frac{\partial v}{\partial r} = nr^{n-1} \sin n\theta$$

$$\frac{\partial u}{\partial \theta} = -nr^n \sin n\theta$$

$$\frac{\partial v}{\partial \theta} = nr^n \cos n\theta$$

$$\text{Now } \frac{1}{r} \frac{\partial v}{\partial \theta} = \frac{1}{r} \cdot nr^n \cos n\theta = nr^{n-1} \cos n\theta = \frac{\partial u}{\partial r}$$

$$\frac{1}{r} \frac{\partial u}{\partial \theta} = \frac{1}{r} (-nr^n \sin n\theta) = -nr^{n-1} \sin n\theta = -\frac{\partial v}{\partial r}$$

Since C-R are satisfied, the partial derivatives are continuous,  $\therefore f(z) = z^n$  is analytic.

Differentiating  $f(z) = u + iv$  w.r.t  $r$  we get

$$\frac{\partial f}{\partial r} = f'(z) \frac{\partial z}{\partial r} = f'(z) e^{i\theta}$$

$$f'(z) = \frac{1}{e^{i\theta}} \frac{\partial f}{\partial r} = \frac{1}{e^{i\theta}} \left[ \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right]$$

$$= e^{-i\theta} [nr^{n-1} \cos n\theta + i nr^{n-1} \sin n\theta]$$

$$= e^{-i\theta} nr^{n-1} [\cos n\theta + i \sin n\theta]$$

$$= nr^{n-1} e^{-i\theta} e^{in\theta} = n (re^{i\theta})^{n-1} = n z^{n-1}$$

Problem- If  $f(z) = u + iv = z^3$  prove that  $u(x, y) = C_1$  and  $v(x, y) = C_2$  where  $C_1$  and  $C_2$  are constants cut each other orthogonally. (11) ①

9, 26, 36, 61, 62

Sol:- Given, that  $f(z) = u + iv = z^3$

$$= (x + iy)^3$$

$$= x^3 - iy^3 + 3x^2iy - 3xy^2$$

ie,  $f(z) = (x^3 - 3xy^2) + i(3x^2y - y^3)$

Here  $u = x^3 - 3xy^2$  &  $v = 3x^2y - y^3$

we have two family of curves

$x^3 - 3xy^2 = C_1$  — (1) and  $3x^2y - y^3 = C_2$  — (2)

Let these two curves intersects at the point  $(x_0, y_0)$

Now differentiate (1) w.r.t  $x$ , we get

$$3x^2 - 3y^2 \left[ x \cdot 2y \frac{dy}{dx} + y^2 \cdot 2 \right] = 0$$

$$\Rightarrow 3x^2 - 6xy \frac{dy}{dx} - 3y^2 = 0$$

$$\Rightarrow 3x^2 - 3y^2 = 6xy \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3(x^2 - y^2)}{6xy} = \frac{x^2 - y^2}{2xy}$$

Slope of the tangent at  $(x_0, y_0)$  for the first curve is  $m_1 = \left[ \frac{dy}{dx} \right]_{(x_0, y_0)} = \frac{x_0^2 - y_0^2}{2x_0y_0}$

Now diff (2) w.r.t  $x$ , we get

$$3 \left[ x^2 \frac{dy}{dx} + y^2 \cdot 2 \right] - 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (3x^2 - 3y^2) \frac{dy}{dx} = -6xy \Rightarrow \frac{dy}{dx} = \frac{-6xy}{3(x^2 - y^2)} = \frac{-2xy}{x^2 - y^2}$$



(2)

Slope of the tangent at  $(x_0, y_0)$  for the second

Curve  $m_2 = \left[ \frac{dy}{dx} \right]_{(x_0, y_0)} = \frac{-2x_0y_0}{x_0^2 - y_0^2}$

Now  $m_1 \cdot m_2 = \left[ \frac{x_0^2 - y_0^2}{2x_0y_0} \right] \cdot \left[ \frac{-2x_0y_0}{x_0^2 - y_0^2} \right]$   
 $= -1$

Hence the two families of the curves intersect orthogonally.

## Complex Integration

Line Integral:-

Integration along any line or curve is called Line Integral and this is denoted by  $\int_C f(z) dz$

Where  $C$  is any curve or line. If  $C$  is closed curve then the line integral is denoted by  $\oint_C f(z) dz$ .

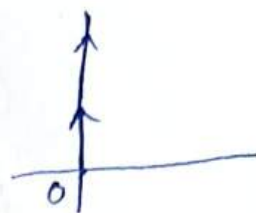
Remember:-

- (1)  $\int dx = x$
- (2)  $\int x^n dx = \frac{x^{n+1}}{n+1}$
- (3)  $\int e^{ax} dx = \frac{e^{ax}}{a}$
- (4)  $\int \sin ax dx = -\frac{\cos ax}{a}$
- (5)  $\int \cos ax dx = \frac{\sin ax}{a}$
- (6)  $\int \frac{1}{x} dx = \log x$
- (7)  $\int u v dx = u \int v dx - \int [u' \int v dx] dx$
- (8)  $\int u v dx = u v_1 - u' v_2 + u'' v_3 - u''' v_4 + \dots$

where  $v_1 = \int v dx$   
 $v_2 = \int v_1 dx$   
 $v_3 = \int v_2 dx$   
 (9)  $\frac{d}{dx} x^n = n x^{n-1}$

Problem:- Evaluate  $\int_C (x+y)dx + xy^2dy$  along  $y=3x$  between  $(0,0)$  and  $(3,9)$ . (12) (3)

Sol:- Given curve  $C$  is  $y=3x$   
 $\Rightarrow dy=3dx$



Here  $x$  varies from 0 to 3

$$\begin{aligned} \text{Now } \int_C (x+y)dx + xy^2dy &= \int_0^3 (x+3x)dx + x(9x^2)(3dx) \\ &= \int_0^3 4x dx + \int_0^3 27x^3 dx \\ &= 4 \left[ \frac{x^2}{2} \right]_0^3 + 27 \left[ \frac{x^4}{4} \right]_0^3 \\ &= 4 \left[ \frac{9}{2} \right] + 27 \left[ \frac{81}{4} \right] \\ &= 18 + \frac{729}{4} = \frac{801}{4} \end{aligned}$$

$$\begin{array}{r} 18 \\ 72 \\ \hline 27 \\ 81 \\ \hline 216 \\ 2187 \\ \hline 2403 \end{array}$$

problem:- Evaluate  $\int_C z^2 dz$  where  $C$  is the straight line segment from  $O(z=0)$  and  $A(z=2+i)$ .  $z=0=0+i0$   
 $z=2+i$

Sol:- Given that  $C$  is the straight line joining the points  $(0,0)$  to  $(2,1)$

The Equation of the st line  $OA$  is  $\frac{x-0}{2-0} = \frac{y-0}{1-0}$

$$\left[ \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} \right]$$

$$\text{i.e. } \frac{x}{2} = y \Rightarrow x=2y$$

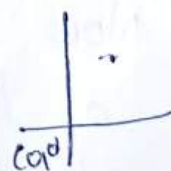
$$dx = 2dy$$

Here  $y$  varies from 0 to 1

$$\text{Let } f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + 2ixy, \quad z = x+iy$$

$$dz = dx + i dy$$

$$\begin{aligned} \text{Now } f(z)dz &= z^2 dz = [x^2 - y^2 + i2xy][dx + i dy] \\ &= (x^2 - y^2)dx + i(x^2 - y^2)dy + 2xydy + i2xydx \end{aligned}$$





$$= [(2y)^2 - y^2] [2dy] + i [(2y)^2 - y^2] dy + i 2 \cdot 2y \cdot y 2dy - 2 \cdot 2y \cdot y \cdot dy \quad (4)$$

$$= 6y^2 dy + i 3y^2 dy + i 8y^2 dy - 4y^2 dy$$

$$= 2y^2 dy + i 11y^2 dy$$

$$\text{Now } \int_C f(z) dz = \int_0^1 2y^2 dy + \int_0^1 i 11y^2 dy$$

$$= 2 \left[ \frac{y^3}{3} \right]_0^1 + i 11 \left[ \frac{y^3}{3} \right]_0^1$$

$$= 2 \cdot \frac{1}{3} + i 11 \cdot \frac{1}{3}$$

$$= \frac{2}{3} + i \frac{11}{3}$$

$$= \frac{1}{3} [2 + i 11]$$

Problem: - Evaluate  $\int (y^2 + 2xy) dx + (x^2 - 2xy) dy$  where  $C$  is boundary of the region by  $y = x^2$ ,  $x = y^2$

Sol: - Given  $C$  is the boundary of the region by the curves  $y = x^2$  — (1)

$$x = y^2 \text{ — (2)}$$

Now we have to find the intersecting points of these curves

For this sub (2) in (1)

$$y = x^2 = (y^2)^2 = y^4$$

$$\Rightarrow y^4 - y = 0$$

$$\Rightarrow y(y^3 - 1) = 0$$

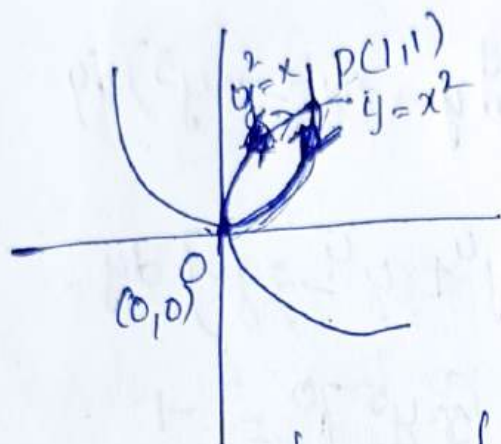
$$\therefore y = 0 \text{ or } y = 1$$

(13)  
(5)

When  $y=0$  from (2) we have  $x=0$

$y=1$  from (2) we have  $x=1$

Hence these two curves intersect at  $(0,0)$  and  $(1,1)$



From figure, we have  $\int_{OP} f(z) dz = \int_{OP} f(z) dz + \int_{PO} f(z) dz$

Along OP, we have  $y = x^2$   
 $dy = 2x dx$

$x$  varies from 0 to 1

$$\int_{OP} f(z) dz = \int_{OP} (y^2 + 2xy) dx + (x^2 - 2xy) dy$$

$$= \int_{x=0}^1 ((x^2)^2 + 2x \cdot x^2) dx + (x^2 - 2x \cdot x^2)(2x dx)$$

$$= \int_0^1 (x^4 + 2x^3) dx + (2x^3 - 4x^4) dx$$

$$= \int_0^1 (3x^4 + 6x^3) dx$$

$$= \left[ -3 \cdot \frac{x^5}{5} + 6 \cdot \frac{x^4}{4} \right]_0^1 = -3 \left[ \frac{1}{5} - 0 \right] + 6 \left[ \frac{1}{4} - 0 \right]$$

$$= -\frac{3}{5} + \frac{6}{4} = \frac{-12 + 30}{20} = \frac{18}{20} = \frac{9}{10} = -\frac{3}{5} + 1 = \frac{-3+5}{2} = \frac{2}{5}$$



Along PO we have  $x = y^2$   
 $\Rightarrow dx = 2y dy$

and  $y$  varies from 1 to 0

$$\begin{aligned} \int_{PO} f(z) dz &= \int_{y=1}^0 (y^2 + 2xy) dx + (x^2 - 2xy) dy \\ &= \int_{y=1}^0 (y^2 + 2y^3) dy + (y^4 - 2y^3) dy \\ &= \int_{y=1}^0 [2y^3 + 4y^4 + y^4 - 2y^3] dy \\ &= \int_{y=1}^0 5y^4 dy = \left[ \frac{5y^5}{5} \right]_1^0 = -1 \end{aligned}$$

$$\text{Hence } \int (y^2 + 2xy) dx + (x^2 - 2xy) dy = \frac{2}{5} - 1 = -\frac{3}{5}$$

Problem:- Evaluate  $\int_{0 \rightarrow 2+i} z^2 dz$ .

(1) along the real axis to 2 and then vertically to  $2+i$

(2) along the imaginary axis to  $i$  then horizontally to  $2+i$

Sol:-

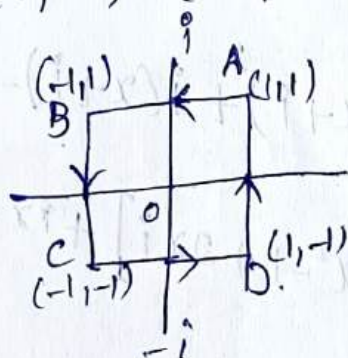
## Cauchy's Integral Theorem

(14)  
7

Let  $f(z) = u(x, y) + i v(x, y)$  be analytic on within a closed contour  $C$  and let  $f'(z)$  be continuous there. Then  $\int_C f(z) dz = 0$ .

Problem:- Verify Cauchy's theorem for the function  $f(z) = 3z^2 + iz - 4$  if  $C$  is square with vertices at  $1 \pm i$  and  $-1 \pm i$

Sol:- Given that  $C$  is the boundary of the square with vertices at  $1 \pm i$  and  $-1 \pm i$   
i.e.,  $C$  is the square with vertices at  $(1, 1)$ ,  $(1, -1)$ ,  $(-1, 1)$  and  $(-1, -1)$



$1+i$   
 $1-i$   
 $-1+i$   
 $-1-i$

Given  $f(z) = 3z^2 + iz - 4$

Clearly  $f(z)$  is a polynomial function and it is analytic every where  
Hence  $f(z)$  is analytic on and within the given region.

Hence Cauchy's Integral theorem, we have

$$\int_C f(z) dz = 0$$

Now we have to verify



Given that  $f(z) = 3z^2 + iz - 4$

$$f(z)dz = (3z^2 + iz - 4)dz$$

$$= [3(x+iy)^2 + i(x+iy) - 4] [dx + i dy]$$

From figure, we have.

$$\int_C f(z)dz = \int_{AB} f(z)dz + \int_{BC} f(z)dz + \int_{CD} f(z)dz + \int_{DA} f(z)dz$$

$$\begin{aligned} H &= \tan^2 \phi = \sec^2 \phi \\ t^2 &= \sec^2 \phi \\ \sec \phi &= \frac{1}{\cos \phi} \\ \frac{1}{\cos \phi} &= \frac{\sqrt{\sin^2 \phi + \cos^2 \phi}}{\cos \phi} \\ &= \frac{\sqrt{\sin^2 \phi}}{\cos \phi} \\ &= \frac{\sin \phi}{\cos \phi} \\ &= \tan \phi \end{aligned}$$

(i) along AB, we have  $y=1$   
 $dy=0$

and  $x$  varies from 1 to -1

$$\int_{AB} f(z)dz = \int_{AB} [3(x+iy)^2 + i(x+iy) - 4] [dx + i dy]$$

$$= \int_{x=1}^{-1} 3[(x+i)^2 + i(x+i) - 4] [dx]$$

$$= \int_{x=1}^{-1} [3(x^2 - 1 + 2xi) + ix - 1 - 4] dx.$$

$$= \int_{x=1}^{-1} (3x^2 - 8 + 7i x) dx$$

$$= \left[ 3 \frac{x^3}{3} - 8x + 7i \frac{x^2}{2} \right]_{-1}^1$$

$$= \left[ -1 + 8 + \frac{7i}{2} \right] - \left[ 1 - 8 + i \frac{7}{2} \right]$$

$$= 14.$$

(ii) Along BC we have  $x=-1$  and  $y$  varies from 0 to -1  
 $dx=0$

$$\int_{BC} f(z)dz = \int_{BC} [3(x+iy)^2 + i(x+iy) - 4] [dx + i dy]$$

$$= \int_{y=-1}^{-1} [3(-1+iy)^2 + i(-1+iy) - 4] i dy$$

$$= i \int_{-1}^{-1} [3(1-y^2-2iy) - 1-y-4] dy$$

$$= i \int_{-1}^{-1} [-3y^2 - 6iy - i - y - 1] dy$$

$$= i \left[ -3 \frac{y^3}{3} - 6i \frac{y^2}{2} - i y - \frac{y^2}{2} - y \right]_{-1}^{-1}$$

$$= -i \left[ y^3 + 3iy^2 + iy + \frac{y^2}{2} + y \right]_{-1}^{-1}$$

$$= -i \left[ \left\{ -1 + 3i - i + \frac{1}{2} - 1 \right\} - \left\{ 1 + 3i + 1 + \frac{1}{2} + 1 \right\} \right]$$

$$= -i \left[ -\frac{3}{2} + 2i - \frac{5}{2} - 4i \right] = -i [-4 - 2i] = -2 + 4i$$

Along CD we have  $y = -1 \Rightarrow dy = 0$   $x$  varies from  $-1$  to  $1$

$$\int_{CD} f(z) dz = \int_{CD} [3(x+iy)^2 + i(x+iy) - 4] [dx + i dy]$$

$$= \int_{-1}^1 [3(x-i)^2 + i(x-i) - 4] [dx]$$

$$= \int_{-1}^1 [3(x^2 - 1 - 2xi) + ix - 1 - 4] dx$$

$$= \int_{-1}^1 [3x^2 - 3 - 6xi + ix - 1 - 4] dx$$

$$= \int_{-1}^1 (3x^2 - 5xi - 6) dx = \left[ x \frac{3x^2}{3} - 5i \frac{x^2}{2} - 6x \right]_{-1}^1$$

$$= \left[ 1 - \frac{5i}{2} - 6 \right] - \left[ -1 - \frac{5i}{2} + 6 \right]$$

$$= 1 - \frac{5i}{2} - 6 + 1 + \frac{5i}{2} - 6 = 2 - 12 = -10$$



Along DA we have  $x=1 \Rightarrow dx=0$  and  $y$  varies from  $-1$  to  $1$  10

$$\int_{DA} f(z) dz = \int_{DA} [3(x+iy)^2 + i(x+iy) - 4] [dx + i dy]$$

$$= \int_{-1}^1 [3[1+iy]^2 + i[1+iy] - 4] [i dy]$$

$$= i \int_{-1}^1 [3[1+2iy-y^2] + i - y - 4] dy$$

$$= i \int_{-1}^1 [3 + 6iy - 3y^2 + i - y - 4] dy$$

$$= i \int_{-1}^1 [-3y^2 - y + 6iy + i - 1] dy$$

$$= i \left[ -\frac{3y^3}{3} - \frac{y^2}{2} + \frac{6iy^2}{2} + iy - y \right]_{-1}^1$$

$$= i \left[ \left( -1 - \frac{1}{2} + 3i + i - 1 \right) - \left( +1 - \frac{1}{2} + \frac{3i}{2} - i + 1 \right) \right]$$

$$= i \left[ \left( -2 - \frac{1}{2} + \frac{5i}{2} \right) - \left( 2 - \frac{1}{2} + \frac{i}{2} \right) \right]$$

$$= i \left[ -2 - \frac{1}{2} + \frac{5i}{2} - 2 + \frac{1}{2} - \frac{i}{2} \right] = i \left[ -4 + \frac{4i}{2} \right] = -4i - 2$$

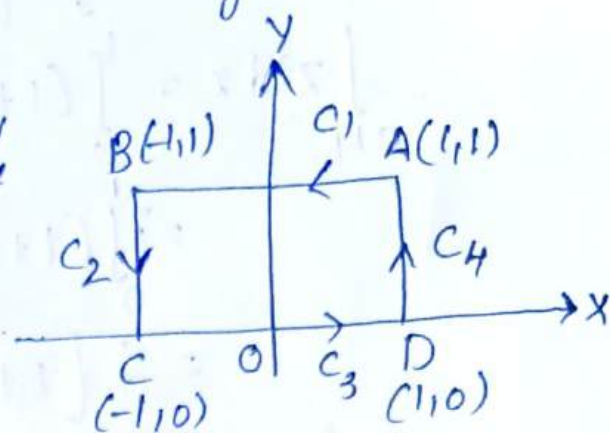
$$\therefore \int_C f(z) dz = 14 + 4i - 2 - 10 - 4i - 2 = 14 - 14 = 0.$$

$$\therefore \int_C f(z) dz = 0.$$

Hence Cauchy's Integral Theorem Verified.

Problem:- Verify Cauchy's theorem for the integral of  $z^3$  taken over the boundary of the rectangle with vertices  $-1, 1, 1+i, -1+i$  (11)

Solution:- The boundary of rectangle  $C$  consists of curves  $C_1, C_2, C_3$ , and  $C_4$ . So



$$\int_C f(z) dz = \int_C z^3 dz$$

$$= \int_{C_1} z^3 dz + \int_{C_2} z^3 dz + \int_{C_3} z^3 dz + \int_{C_4} z^3 dz$$

Along  $C_1$ :-  $y=1, -1 \leq x \leq 1$   $dz = dx$  [ $\because dy=0$ ]

$$\begin{aligned} \int_{C_1} z^3 dz &= \int_{-1}^1 (x+iy)^3 (dx+idy) \quad [\because dz = dx+idy \\ &= dx+i(0) = dx] \\ &= \int_{-1}^1 (x+i)^3 (dx) = \int_{-1}^1 (x^3 - i + 3ix^2 - 3x) dx \\ &= \left[ \frac{x^4}{4} - ix + 3i \frac{x^3}{3} - \frac{3x^2}{2} \right]_{-1}^1 = 0 \end{aligned}$$

Along  $C_2$ :-  $x=-1, 0 \leq y \leq 1$

$$dz = i dy \quad [\because dx=0]$$

$$\begin{aligned} \int_{C_2} z^3 dz &= \int_0^1 (-1+iy)^3 i dy = \int_0^1 (-1-iy^3+3iy+3y^2) dy \\ &= 5/4 \end{aligned}$$

Along  $C_3$ :-  $y=0, -1 \leq x \leq 1$

$$dz = dx \quad [\because dy=0]$$

$$\int_{C_3} z^3 dz = \int_{-1}^1 x^3 dx = 0 \quad [\because x^3 \text{ is odd function}]$$



Along  $C_4$ :  $x=1, 0 \leq y \leq 1$

$$dz = i dy \quad \therefore dx = 0$$

$$\int_{C_4} z^3 dz = \int_0^1 (1+iy)^3 (i dy)$$

$$= i \int_0^1 [1 + 3iy - 3y^2 - iy^3] dy$$

$$= i \left[ 1 + \frac{3i}{2} - 1 - \frac{i}{4} \right] = -\frac{5}{4}$$

$$\text{So } \int_C f(z) dz = \int_C z^3 dz = 0 + \frac{5}{4} + 0 - \frac{5}{4} = 0$$

Hence theorem is verified.

### Cauchy's Integral Formula

Let  $f(z)$  be an analytic function everywhere on and within a closed contour  $C$ . If  $z=a$  is any point within  $C$ , then.

$$f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z-a} dz$$

Where the integral is taken in the positive sense around  $C$ .

### Generalization of Cauchy's Integral Formula.

If  $f(z)$  is analytic on and within a simple closed curve  $C$  and if  $a$  is any point within  $C$  then

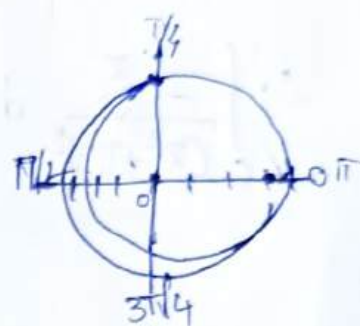
$$f^n(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz$$

Singular point:- A point at which a function  $f(z)$  is not analytic. It is called a singular point or singularity of  $f(z)$ .

Problem: Use Cauchy's integral formula to Evaluate  $\int_C \frac{e^z}{(z^2 + \pi^2)^2} dz$  where  $C$  is the circle  $|z| = 4$  (2) (13)

Sol: Given  $f(z) = \frac{e^z}{(z^2 + \pi^2)^2} = \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2}$

$f(z) = e^z$  is analytic within the circle  $|z| = 4$  and the two singular points  $z = \pm \pi i$  lies inside  $C$ .



Let  $\frac{1}{(z^2 + \pi^2)^2} = \frac{1}{(z + \pi i)^2 (z - \pi i)^2}$

$\Rightarrow \frac{1}{(z^2 + \pi^2)^2} = \frac{A}{z + \pi i} + \frac{B}{(z + \pi i)^2} + \frac{C}{z - \pi i} + \frac{D}{(z - \pi i)^2}$

$\Rightarrow 1 = A(z - \pi i)^2 (z + \pi i) + B(z - \pi i)^2 + C(z - \pi i)(z + \pi i)^2 + D(z + \pi i)^2$  — (1)

Put  $z = \pi i$  | Put  $z = -\pi i$

$1 = D(2\pi i)^2$

$\Rightarrow 1 = D(4\pi^2(-1))$

$\Rightarrow D = -\frac{1}{4\pi^2}$

$1 = B(-2\pi i)^2$

$\Rightarrow B = -\frac{1}{4\pi^2}$

Eq (1) can be written as

$1 = A(z^2 + \pi^2 - 2z\pi i)(z + \pi i) + B(z^2 - \pi^2 - 2z\pi i) + C(z - \pi i)(z^2 - \pi^2 + 2z\pi i) + D(z^2 + \pi^2 + 2z\pi i)$

Comparing the coefficients  $z^3, z^2$

$0 = A + C, 0 = -2\pi i A + B + 2C\pi i + D$

$\Rightarrow 0 = -2\pi i A - \frac{1}{4\pi^2} + 2C\pi i - \frac{1}{4\pi^2}$

$\Rightarrow +2\pi i A - 2C\pi i = -\frac{2}{4\pi^2}$

$\Rightarrow \pi i(A - C) = -\frac{1}{4\pi^2}$



$$A + C = \frac{-1}{4\pi^2} \times \frac{1}{\pi i} = \frac{-1}{4\pi^3 i}$$

$$A - C = -\frac{1}{4\pi^3 i}$$

$$A + C = 0$$

(14)

and  $A = \frac{7}{2\pi^3 i}$ ,  $C = \frac{-7}{2\pi^3 i}$ ,  $B = \frac{1}{4\pi^2}$ ,  $D = \frac{-1}{4\pi^2}$

$$\therefore \int_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{7}{2\pi^3 i} \int_C \frac{e^z}{(z + \pi i)} dz - \frac{1}{4\pi^2} \int_C \frac{e^z}{(z + \pi i)^2} dz$$

$$- \frac{7}{2\pi^3 i} \int_C \frac{e^z}{z - \pi i} dz - \frac{1}{4\pi^2} \int_C \frac{e^z}{(z - \pi i)^2} dz$$

By Cauchy's Integral formula

$$\int_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{7}{2\pi^3 i} 2\pi i f(-\pi i) - \frac{1}{4\pi^2} 2\pi i f'(-\pi i)$$

$$= -\frac{7}{2\pi^3 i} 2\pi i f(\pi i) - \frac{1}{4\pi^2} 2\pi i f'(\pi i)$$

$$= \frac{7}{\pi^2} e^{-i\pi} - \frac{i}{2\pi} e^{-i\pi} - \frac{7}{\pi^2} e^{i\pi} - \frac{i}{2\pi} e^{i\pi} = \frac{i}{\pi}$$

→ Using Cauchy's integral formula, Evaluate

$$\int_C \frac{z^4}{(z+1)(z-i)^2} dz \text{ where } C \text{ is the ellipse } 9x^2 + 4y^2 = 36$$

sol:- Given ellipse is  $9x^2 + 4y^2 = 36$

$$\text{i.e., } \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$$

Its Center is (0,0)

$f(z) = z^4$  is analytic within the ellipse  $C$   
 $9x^2 + 4y^2 = 36$  and the two singularities points  
 $z = -1$ ,  $z = i$  lie inside  $C$ .

Consider  $\frac{1}{(z+1)(z-i)^2}$

Let  $\frac{1}{(z+1)(z-i)^2} = \frac{A}{z+1} + \frac{B}{z-i} + \frac{C}{(z-i)^2}$  — (1)

Then  $1 = A(z-i)^2 + B(z+1)(z-i) + C(z+1)$  — (2)  
 $= (A+B)z^2 + (-2iA - iB + B + C)z + (-A - iB + C)$

put  $z=i$  in (2)

$\therefore 1 = C(i+1)$   
 $\Rightarrow C = \frac{1}{1+i} = \frac{1-i}{2}$  [Rationalising]

put  $z=-i$  in (2)

$1 = A(-1-i)^2$   
 $\Rightarrow A = \frac{1}{(1+i)^2} = \frac{1}{2i} = -\frac{i}{2}$  [Rationalising]

Comparing Constant term.

$-A - iB + C = 1 \Rightarrow iB = -A + C - 1$

$= \frac{i}{2} + \frac{1-i}{2} - 1$

$= \frac{1}{2}(i+1-i-2) = -\frac{1}{2}$

$\therefore B = -\frac{1}{2i} = \frac{i}{2} = \frac{1}{2}$

Substituting the values of A, B and C in (1),

$\frac{1}{(z+1)(z-i)^2} = \frac{-i}{2(z+1)} + \frac{i}{2(z-i)} + \frac{1-i}{2(z-i)^2}$

Hence  $\int_C \frac{z^4}{(z+1)(z-i)^2} dz = \frac{i}{2} \int_C \frac{z^4}{z+1} dz + \frac{1}{2} \int_C \frac{z^4}{z-i} dz + \frac{1-i}{2} \int_C \frac{z^4}{(z-i)^2} dz$



$$= \frac{-i}{2} 2\pi i \cdot f(-1) + \frac{i}{2} 2\pi i \cdot f(1) + \frac{1-i}{2} 2\pi i f'(1)$$

$$= \pi f(-1) - \pi f(1) + \pi(i+1)f'(1)$$

$$= \pi [f(-1) - f(1) + (i+1)f'(1)]$$

$$= \pi [-4i(i+1)] = 4\pi [-i^2 - i]$$

$$= 4\pi(1-i)$$

$$1 + (1+i)4i^2 \\ (1+i)4i^3 \\ -(1+i)4i$$

## Laurent Series Expansion

Let  $C_1$  and  $C_2$  be two circles given by  $|z'-a|=r_1$  and  $|z'-a|=r_2$  respectively where  $r_2 < r_1$  and  $z'$  is any point on  $C_1$  or  $C_2$

Let  $f(z)$  be analytic on  $C_1$  and  $C_2$  and throughout the region between the two circles. Let  $z$  be any point in the ring shaped region between the two circles  $C_1$  and  $C_2$ . Then

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n} \quad \text{--- (1)}$$

$$\text{where } a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z')}{(z'-a)^{n+1}} dz' \quad \text{--- (2)}$$

$$b_n = \frac{1}{2\pi i} \oint_{C_2} \frac{f(z')}{(z'-a)^{n+1}} dz' \quad \text{--- (3)}$$

where the integrals are taken around  $C_1$  and  $C_2$  in the anticlockwise sense. Then series (1) is called Laurent's Series Expansion of  $f(z)$  around  $z=z_0$ .

Problem:- Expand  $f(z) = \frac{1}{z^2 - 3z + 2}$  in the region

(i)  $0 < |z-1| < 1$  (ii)  $1 < |z| < 2$

Sol:- we have

$$f(z) = \frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} = \frac{(z-1) - (z-2)}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$$

$\therefore z=1, z=2$  are the singular points of  $f(z)$

① The function  $f(z)$  is analytic in the ring shaped region.  $0 < r_2 < |z-1| < r_1 < 1$ .  $0 < |z-1| < 1$

Put  $z-1 = w$  ✓

$$\Rightarrow \boxed{z = 1+w}$$

$$\Rightarrow z-2 = 1+w-2 = w-1$$

$$\therefore f(z) = \frac{1}{w-1} - \frac{1}{w} = -\frac{1}{w-1} + \frac{1}{w}$$

$$= -\frac{1}{w} + \frac{1}{1-w} = -\frac{1}{w} + [1 + w + w^2 + \dots]$$

$$= -\frac{1}{z-1} + \sum_{n=0}^{\infty} (z-1)^n \text{ if } 0 < |z-1| < 1$$

$$= -\sum_{n=-1}^{\infty} (z-1)^n \text{ if } 0 < |z-1| < 1$$



② Given  $1 < |z| < 2$  i.e.  $1 < |z|$  and  $|z| < 2$

(or)  $|\frac{1}{z}| < 1$  and  $|\frac{z}{2}| < 1$

$$\therefore f(z) = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{-2[1-\frac{z}{2}]} - \frac{1}{z[1-\frac{1}{z}]}$$

$$= \frac{1}{2} \left[ 1 - \frac{z}{2} \right]^{-1} - \frac{1}{z} \left[ 1 - \frac{1}{z} \right]^{-1}$$

$(z-1)^{-1}$   
 $1 < |z|$   
 $\frac{1}{z} > \frac{1}{|z|}$   
 $\frac{1}{|z|} < 1$



$$= -\frac{1}{2} \left[ 1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right] - \frac{1}{2} \left[ 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right] \quad (18)$$

$$= -\frac{1}{2} \left[ 1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right] - \left[ \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right]$$

→ Expand  $f(z) = \frac{e^{2z}}{(z-1)^3}$  about  $z=1$  as a Laurent's Series. Also find the region of convergence.

sol:-  $f(z) = \frac{e^{2z}}{(z-1)^3}$

We want the Laurent's Series Expansion of  $f(z)$  around  $z=1$

put  $z-1=w$

$\Rightarrow z=1+w$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\therefore f(z) = \frac{e^{2(1+w)}}{(w)^3} = e^2 \cdot \frac{e^{2w}}{w^3}$$

$$= e^2 \cdot \frac{1}{w^3} \left[ 1 + (2w) + \frac{(2w)^2}{2!} + \frac{(2w)^3}{3!} + \dots \right]$$

$$= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} [w^{n-3}] \text{ , if } w \neq 0$$

$$= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} (z-1)^{n-3} \text{ , if } z-1 \neq 0$$

$$= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} (z-1)^{n-3} \text{ , if } |z-1| > 0.$$

## Taylor Series:-

$$f(z) = f(a) + \sum_{n=1}^{\infty} \frac{(z-a)^n}{n!} f^n(a)$$

$$= \sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (z-a)^n$$

$$\begin{aligned} z &= a \\ (z-a) &= 0 \end{aligned}$$

This is Taylor's Series.

If we take  $a=0$  in Taylor's Series Expansion we get

$$\begin{aligned} f(z) &= f(0) + f'(0)z + \frac{f''(0)}{2!} z^2 + \frac{f'''(0)}{3!} z^3 + \dots \\ &= f(0) + \sum_{n=1}^{\infty} \frac{f^n(0)}{n!} z^n \end{aligned}$$

This is Maclaurin's Series expansion of  $f(z)$

## Some important results

$$(a) e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots + \frac{z^n}{n!} + \dots = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad \forall z \in \mathbb{C}$$

$$(b) \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad \forall z \in \mathbb{C}$$

$$(c) \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad \text{for } |z| < \infty$$

$$(d) \sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \quad \text{for } |z| < \infty$$

$$(e) \cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} \quad \text{for } |z| < \infty$$

$$(f) \frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n, \quad \text{for } |z| < 1$$

$$(g) \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad \text{for } |z| < 1$$

$$(h) \frac{1}{(1-z)^m} = \sum_{n=0}^{\infty} \frac{(m+n-1)!}{(m-1)!(n)!} z^n, \quad \text{for } |z| < 1$$



$$(i) \frac{1}{(1+z)^m} = \sum_{n=0}^{\infty} \binom{-m}{n} z^n = 1 - mz + \frac{m(m+1)}{2!} z^2 - \frac{m(m+1)(m+2)}{3!} z^3 + \dots \quad (29)$$

Since  $e^z$ ,  $\sin z$ ,  $\cos z$ ,  $\sinh z$ ,  $\cosh z$  are all entire functions [i.e., analytic anywhere], their Maclaurin's Series Expansions are valid for all  $z$ .

### Important suggestion for working problems in Taylor's Series

To obtain the Taylor's series Expansion of  $f(z)$  around/about  $z=a$ , put  $z-a=w$ . Then

$$f(z) = f(w+a) = \phi(w) \text{ say.}$$

Write Maclaurin's Series of  $\phi(w)$  either by using standard expansions or directly.

Then substitute  $w=z-a$ . We get the required Taylor's Series Expansion.

Problem: Obtain the Taylor's Series to represent the function  $\frac{z^2-1}{(z+2)(z+3)}$  in the region  $|z| < 2$ . (2)

Sol: Let  $f(z) = \frac{z^2-1}{(z+2)(z+3)}$

$$\frac{f(z)}{z} = \frac{z^2-1}{z(z+2)(z+3)}$$

Resolve into partial fractions  
Here numerator and denominator have same degree i.e., 2  
so it can be written as

$$f(z) = 1 - \frac{5z+7}{z^2+5z+6}$$

Now  $\frac{5z+7}{z^2+5z+6}$  into partial fraction.

$$\frac{5z+7}{z^2+5z+6} = \frac{A}{z+2} + \frac{B}{z+3}$$

$$\text{put } 5z+7 = A(z+3) + B(z+2)$$

$$\text{put } z=-3 \text{ and } z=-2$$

$$\Rightarrow B=8 \text{ and } A=-3$$

$$\therefore \frac{5z+7}{z^2+5z+6} = \frac{-3}{z+2} + \frac{8}{z+3}$$

$$\therefore \frac{z^2-1}{(z+2)(z+3)} = 1 - \left[ \frac{-3}{z+2} + \frac{8}{z+3} \right]$$

$$= 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

$$= 1 + \frac{3}{2\left[1+\frac{z}{2}\right]} - \frac{8}{3\left[1+\frac{z}{3}\right]}$$



$$f(z) = 1 + \frac{3}{2} \left[1 + \frac{z}{2}\right]^{-1} - \frac{8}{3} \left[1 + \frac{z}{3}\right]^{-1}$$

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Expanding by binomial series,

$$f(z) = 1 + \frac{3}{2} \left[1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots\right] - \frac{8}{3} \left[1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots\right]$$

$$= 1 + \frac{3}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} \cdot z^n - \frac{8}{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n} z^n$$

$$\therefore f(z) = 1 + \sum_{n=0}^{\infty} (-1)^n \left[ \frac{3}{2^{n+1}} - \frac{8}{3^{n+1}} \right] z^n$$

→ Expand  $f(z) = \frac{1}{(z+1)^2}$  in Taylor's Series about  $z = -i$

Sol:- The Taylor's Series Expansion of  $f(z)$  in powers of  $z+i$

$$\text{put } z+i = w$$

$$\Rightarrow z = w - i$$

$$f(z) = \frac{1}{(z+1)^2} = \frac{1}{(w-i+1)^2}$$

$$= \frac{1}{((1-i)+w)^2} = \frac{1}{(1-i)^2 \left[1 + \frac{w}{(1-i)}\right]^2}$$

Expanding by binomial expansion

$$f(z) = \frac{1}{(1-i)^2} \left[1 + \frac{w}{1-i}\right]^{-2}$$

$$= \frac{1}{2i} \left[1 - \frac{2w}{1-i} + \frac{3w^2}{(1-i)^2} - \frac{4w^3}{(1-i)^3} + \dots\right]$$

$$= \frac{i}{2} \left[1 - \frac{2}{1-i}(z+i) + \frac{3}{(1-i)^2}(z+i)^2 - \frac{4}{(1-i)^3}(z+i)^3 + \dots\right]$$

This is the required expansion

# Singularities and Residue Theorem

③  
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## Definitions

### (1) Zeros of an analytic function

A zero of an analytic function  $f(z)$  is a value of  $z$  such that  $f(z)=0$ . particularly, a point ' $a$ ' is called a zero of an analytic function  $f(z)$  if  $f(a)=0$ .

Zero of  $m^{\text{th}}$  order: If an analytic function  $f(z)$  can be expressed in the form  $f(z) = (z-a)^m \phi(z)$

where  $\phi(z)$  is analytic and  $\phi(a) \neq 0$ , then  $z=a$  is called zero of  $m^{\text{th}}$  order of the function  $f(z)$ .

Ex: If  $f(z) = (z-1)^3$ , then  $z=1$  is a zero of order 3 of  $f$ .

②  $f(z) = \frac{1}{1-z}$ , then  $z=\infty$  is a simple zero of  $f(z)$ .

### (2) Singular Points:

A singular point of a function  $f(z)$  is the point at which the function  $f(z)$  ceases to be analytic.

### Different types of singularities

#### Isolated Singularity:

A point  $z=a$  is called an Isolated singularity of an analytic function  $f(z)$ , if

(a)  $f(z)$  is not analytic at the point  $z=a$ .

(b)  $f(z)$  is analytic in the deleted neighbourhood of  $z=a$ .



ie., there exists a neighbourhood of the point  $z=a$  which contains no other singularity (24)

Ex:- If  $f(z) = \frac{e^z}{z^2+1}$ , then  $z = \pm i$  are two isolated singular points of  $f(z)$ .

Essential singularity:

If there exists no finite value of  $n$  such that  $\lim_{z \rightarrow a} (z-a)^n f(z) = \text{finite non-zero constant}$ , then  $z=a$  is called essential singularity.

Removable Singularity:

If a single-valued function  $f(z)$  is not defined at  $z=a$  but  $\lim_{z \rightarrow a} f(z)$  exists, then the singularity  $z=a$  is known as a removable singularity.

Singularities at infinity:

Taking  $z = \frac{1}{t}$  in  $f(z)$  we obtain  $f\left(\frac{1}{t}\right) = F(t)$ .

Then the nature of the singularity at  $z=\infty$  is defined to be the same as that of  $F(t)$  at  $t=0$ .

\* Poles of an Analytic Function

If  $z=a$  is an isolated singular point of an analytic function  $f(z)$ , then  $f(z)$  can be expanded in Laurent's series about  $z=a$ ,

$$\text{ie., } f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n} \quad \text{--- (1)}$$

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The series of negative integral powers of  $(z-a)$  namely  $\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}$  is known as the principal part of the Laurent's series of  $f(z)$ .

If the principal part contains a finite number of terms, say  $m$ , (ie.,  $b_n = 0 \forall n$  such that  $n > m$ ), then the singular point  $z=a$  is called a pole of order  $m$  of  $f(z)$ .

ie., if all the negative powers of  $(z-a)$  in (1) after the  $n^{\text{th}}$  term are missing, then the singularity at  $z=a$  is called a pole of order  $n$ .

Simple pole  $\therefore$  Simple pole is a pole of order one.

Ex:- If  $f(z) = \frac{z^2}{(z-1)(z+2)^2}$  then  $z=1$  is a

Simple pole and  $z=-2$  is a pole of order 2.

If there exists a positive integer  $n$  such that  $\lim_{z \rightarrow a} (z-a)^n f(z) = A \neq 0$  then  $z=a$  is called a pole of order  $n$ .

Residues :-

The coefficient of  $\frac{1}{z-a}$  in the Laurent series expansion of  $f(z)$  about the isolated singularity  $z=a$  is called the residue of  $f(z)$  at that point. Thus the residue of  $f(z)$  at  $z=a$  is  $b_1$ .



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From Laurent series, we know that the coefficient  $b_1$  is given by

$$b_1 = \frac{1}{2\pi i} \int_C f(z) dz$$

$$\therefore \int_C f(z) dz = 2\pi i b_1 = 2\pi i \times [\text{Residue of } f(z) \text{ at } z=a]$$

$$= 2\pi i [\text{Res } f(z)]_{z=a}$$

where  $C$  is a closed curve containing the point  $z=a$ .

### Residue at a Pole

The residue of  $f$  at  $z=z_0$  is defined as the coefficient of  $(z-z_0)^{-1}$  which is  $a_{-1}$  in the Laurent's series expansion of  $f(z)$  and is denoted by  $\text{Res}(f; z_0)$  or  $[\text{Res } f(z)]_{z=z_0}$ .

Thus, if  $f$  has a pole at  $z_0$  then

$$\text{Res}(f; z_0) = a_{-1} = \frac{1}{2\pi i} \int_C f(z) dz.$$

A pole of order one is called a Simple pole.

### Cauchy's Residue Theorem

If  $f(z)$  is analytic within and on a closed curve  $C$ , except at a finite number of poles  $z_1, z_2, z_3, \dots, z_n$  within  $C$  and  $R_1, R_2, \dots, R_n$  be the residues of  $f(z)$  at these poles then

$$\int_C f(z) dz = 2\pi i [R_1 + R_2 + \dots + R_n]$$

$$(or) \int_C f(z) dz = 2\pi i \times (\text{sum of the residues at the poles within } C).$$

Problem:- Find the poles and residues of  $\frac{1}{z^2-1}$

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Sol:- Let  $f(z) = \frac{1}{z^2-1} = \frac{1}{(z-1)(z+1)}$

$$= \frac{1}{2} \left[ \frac{1}{z-1} - \frac{1}{z+1} \right] \quad \left\{ \because \text{using partial fractions} \right\}$$

poles of  $f(z)$  are given by.

$$(z-1)(z+1) = 0$$

$$\text{ie., } z = \pm 1$$

These are simple poles.

To find the residue at  $z=1$  and  $z=-1$

At  $z=1$

We expand the function in a Laurent series in powers of  $z-1$ .

To Expand in powers of  $z-1$ ,

we write

$$f(z) = \frac{1}{2} \left[ \frac{1}{z-1} - \frac{1}{z+1} \right]$$

$$= \frac{1}{2} \left[ \frac{1}{z-1} - \frac{1}{z-1+2} \right]$$

$$= \frac{1}{2} \left[ \frac{1}{u} - \frac{1}{u+2} \right] \quad \left\{ \because \text{put } 1-z=u \right\}$$

$$= \frac{1}{2u} - \frac{1}{2} \left[ \frac{1}{u+2} \right] = \frac{1}{2u} - \frac{1}{4} \left[ \frac{1}{1+\frac{u}{2}} \right]$$

$$= \frac{1}{2u} - \frac{1}{4} \left[ 1 + \frac{u}{2} \right]^{-1}$$

$$= \frac{1}{2u} - \frac{1}{4} \left[ 1 - \frac{u}{2} + \frac{u^2}{4} - \frac{u^3}{8} + \dots \right]$$

$$= \frac{1}{2u} - \frac{1}{4} + \frac{u}{8} - \frac{u^2}{16} + \frac{u^3}{32} - \dots$$

$$= -\frac{1}{4} + \frac{1}{2(z-1)} + \frac{z-1}{8} - \frac{(z-1)^2}{16} + \frac{(z-1)^3}{32} + \dots$$

Coefficient of  $\frac{1}{z-1} = \frac{1}{2}$  and hence

Residue at  $z=1$  is  $\frac{1}{2}$ .



To find residue at  $z = -1$

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We Expand  $f(z)$  in a Laurent Series in powers of  $(z+1)$ .

$$\begin{aligned}\text{We have } f(z) &= \frac{1}{2} \left[ \frac{1}{z-1} - \frac{1}{z+1} \right] \\ &= \frac{1}{2} \left[ \frac{1}{z+1-2} - \frac{1}{z+1} \right] \\ &= \frac{1}{2} \left[ \frac{1}{u-2} - \frac{1}{u} \right] \quad \left[ \because \text{put } z+1=u \right] \\ &= -\frac{1}{2} \left[ \frac{1}{u-2} \right] - \frac{1}{2u} \\ &= -\frac{1}{4} \left[ 1 - \frac{u}{2} \right]^{-1} - \frac{1}{2u} \\ &= -\frac{1}{4} \left[ 1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right] - \frac{1}{2u} \\ &= -\frac{1}{4} \left[ 1 + \frac{z+1}{2} + \frac{(z+1)^2}{4} + \frac{(z+1)^3}{8} + \dots \right] - \frac{1}{2(z+1)}\end{aligned}$$

Thus the coefficient of  $\frac{1}{z+1} = -\frac{1}{2}$ .

$\therefore$  Residue at  $z = -1$  is  $-\frac{1}{2}$

→ Find the poles of the function

$$\textcircled{1} f(z) = \frac{z^2}{(z-1)(z-2)^2} \quad \textcircled{2} f(z) = \frac{z^2}{(z-1)(z+2)^2}$$

Sol:- The given function has a simple pole at  $z=1$  and another pole of order 2 at  $z=2$ .

$\therefore$  Residue of  $f(z)$  at  $z=1$  is

$$\lim_{z \rightarrow 1} \{ (z-1) f(z) \} = \lim_{z \rightarrow 1} \frac{z^2}{(z-2)^2} = \lim_{z \rightarrow 1} \frac{z^2}{(z-2)^2} = \frac{1}{1} = 1$$

Also Residue of  $f(z)$  at  $z=2$  is

$$\begin{aligned}\lim_{z \rightarrow 2} \frac{d}{dz} \{ (z-2)^2 f(z) \} &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[ (z-2)^2 \cdot \frac{z^2}{(z-1)(z-2)^2} \right] = \lim_{z \rightarrow 2} \frac{d}{dz} \frac{z^2}{z-1} \\ &= \lim_{z \rightarrow 2} \frac{(z-1)2z - z^2 \cdot 1}{(z-1)^2} = \lim_{z \rightarrow 2} \frac{z^2 - 2z}{(z-1)^2} = 0\end{aligned}$$

## Problem using Residue Theorem

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problem:- Evaluate the integral  $\oint_C \frac{ze^z}{(z+1)^3} dz$  where  $C$  is the circle  $|z-1|=3$

sol:- The given function  $\frac{ze^z}{(z+1)^3} = f(z)$  has only one singular point  $z=-1$  which is a pole of order 3.

Also the point  $z=-1$  lies inside the given circle

$$\therefore [\text{Res } f(z)]_{z=-1} = \lim_{z \rightarrow -1} \frac{1}{2!} \frac{d^2}{dz^2} \{ (z+1)^3 \cdot f(z) \}$$

$$= \lim_{z \rightarrow -1} \frac{1}{2} \frac{d^2}{dz^2} \left\{ (z+1)^3 \cdot \frac{ze^z}{(z+1)^3} \right\}$$

$$= \lim_{z \rightarrow -1} \frac{1}{2} \frac{d^2}{dz^2} ze^z$$

$$= \lim_{z \rightarrow -1} e^z = e^{-1}$$

$\therefore$  By Residue Theorem, we get  $\oint_C \frac{ze^z}{(z+1)^3} = 2\pi i [e^{-1}]$

$$\therefore \oint_C \frac{ze^z}{(z+1)^3} = 2\pi i \cdot \frac{1}{e} = \frac{2\pi i}{e}$$

Ans.

Problem:- Evaluate  $\oint_C \frac{4-3z}{z(z-1)(z-2)} dz$  where  $C$  is the circle  $|z| = \frac{3}{2}$  using Residue theorem.

sol:- Let  $f(z) = \frac{4-3z}{z(z-1)(z-2)}$

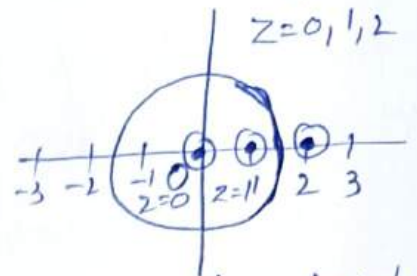
The denominator vanishes at  $z=0$ ,  $z=1$  and  $z=2$ . Hence these are the simple poles of the function  $f(z)$ . Of these, only  $z=0$  and  $z=1$  lie within the given circle.



It is enough if we calculate the residues at  $z=0$  and  $z=1$

$$|z| = \frac{3}{2} = 1.5$$

$$z=0, 1, 2$$



$z=0, 1$  lies inside the circle

$z=2$  outside the circle

$$\text{Now } [\text{Res } f(z)]_{z=0} = \lim_{z \rightarrow 0} (z-0) f(z)$$

$$= \lim_{z \rightarrow 0} z \cdot \frac{4-3z}{z(z-1)(z-2)}$$

$$= \lim_{z \rightarrow 0} \frac{4-3z}{(z-1)(z-2)} = \frac{4-0}{(0-1)(0-2)}$$

$$= \frac{4}{2} = 2$$

$$[\text{Res } f(z)]_{z=1} = \lim_{z \rightarrow 1} (z-1) f(z)$$

$$= \lim_{z \rightarrow 1} (z-1) \cdot \frac{4-3z}{z(z-1)(z-2)}$$

$$= \lim_{z \rightarrow 1} \frac{4-3z}{z(z-2)} = \frac{4-3(1)}{1(1-2)} = \frac{1}{-1} = -1$$

$\therefore$  By Residue Theorem,

$$\oint_C \frac{4-3z}{z(z-1)(z-2)} dz = 2\pi i \times (\text{sum of the residues at } z=0 \text{ and } z=1)$$

$$= 2\pi i \times (2 + (-1))$$

$$= 2\pi i \times 1 = 2\pi i$$

$\rightarrow$  Evaluate  $\int_C \frac{ze^z dz}{z(z-3)}$  where  $C$  is  $|z|=2$  by Residue Theorem.

Sol: The poles of  $f(z) = \frac{ze^z}{z(z-3)}$  are given by

$$z(z-3)=0$$

$$\therefore z=0 \text{ or } z=3$$

Of these poles, only the pole  $z=0$  lies inside the circle  $C$

The residue of  $f(z)$  at  $z=0$  is

$$\lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} \frac{z \cdot 2e^z}{z(z-3)}$$

$$= \lim_{z \rightarrow 0} \frac{2e^z}{z-3}$$

$$= -\frac{2}{3}$$

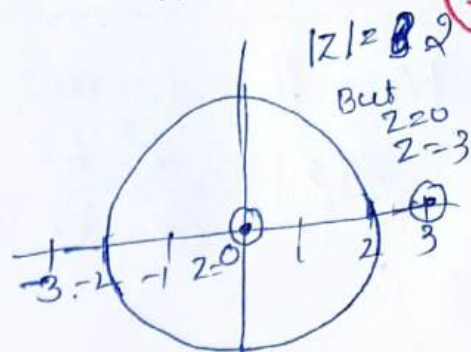
By Residue Theorem,

$$\int_C \frac{2e^z}{z(z-3)} dz = 2\pi i (\text{sum of the residues})$$

$$= 2\pi i \times \left(-\frac{2}{3}\right)$$

$$= -\frac{4\pi i}{3}$$



$z=0$  lies inside the circle  
 $z=3$  lies outside the circle.

7  
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Laurent's Series

Find the Laurent Series expansion of the function

$$f(z) = \frac{z^2 - 6z - 1}{(z-1)(z-3)(z+2)}$$

in the region  $3 < |z+2| < 5$

Sol :-

# Evaluation of Real Definite Integrals By Contour (curve) Integration ①

(1) Integrals of the type  $\int_0^{2\pi} F(\cos\theta, \sin\theta) d\theta$ .

procedure:-

(1) put  $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

let  $e^{i\theta} = z$  then

$$\cos\theta = \frac{z + \frac{1}{z}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

|| by let  $e^{i\theta} = z$

$$\sin\theta = \frac{z - \frac{1}{z}}{2i}$$

Substitute in  $F[\cos\theta, \sin\theta] = f(z)$

(2) Then find the poles for the required  $f(z)$

(3) By Residue theorem,

$$\int_C f(z) dz = 2\pi i \times [\text{sum of the residues of } f(z) \text{ at its poles inside } C]$$

Problem:- Show that  $\int_0^{2\pi} \frac{d\theta}{2 + \cos\theta} = \frac{2\pi}{\sqrt{3}}$

Sol:- let  $I = \int_0^{2\pi} \frac{d\theta}{2 + \cos\theta}$

On the unit circle  $|z|=1$ , we have

$$z = e^{i\theta}$$

$$dz = i e^{i\theta} d\theta$$

$$\Rightarrow d\theta = \frac{dz}{i e^{i\theta}} = \frac{dz}{iz}$$

$$\text{and } \cos\theta = \frac{1}{2} \left[ z + \frac{1}{z} \right] = \frac{z^2 + 1}{2z}$$



By substituting these values, we get

$$I = \int_C \frac{1}{2 + \frac{z^2+1}{2z}} \frac{dz}{iz}$$

$$= \frac{2}{i} \int_C \frac{dz}{z^2+4z+1} \quad \text{--- (1)}$$

where  $C$  is the unit circle  
 $|z|=1$

The integrand  $f(z) = \frac{1}{z^2+4z+1}$  has simple poles given by

$$z = \frac{-4 \pm \sqrt{16-4}}{2(1)} = -2 \pm \sqrt{3}$$

i.e.,  $z = -2 + \sqrt{3}, -2 - \sqrt{3}$   
Out of these  $z = -2 + \sqrt{3}$  lies inside the circle  
 $\therefore$  Residue at  $z = -2 + \sqrt{3}$  is

$$\begin{aligned} [\text{Res } f(z)]_{z=-2+\sqrt{3}} &= \lim_{z \rightarrow -2+\sqrt{3}} \left[ \frac{1}{(z - (-2+\sqrt{3}))} f(z) \right] \\ &= \lim_{z \rightarrow -2+\sqrt{3}} \left[ \frac{1}{(z - (-2+\sqrt{3}))} \cdot \frac{1}{(z - (-2+\sqrt{3}))(z - (-2-\sqrt{3}))} \right] \\ &= \lim_{z \rightarrow -2+\sqrt{3}} \frac{1}{z - (-2-\sqrt{3})} \\ &= \frac{1}{-2+\sqrt{3} + 2 + \sqrt{3}} = \frac{1}{2\sqrt{3}} \quad \text{--- (2)} \end{aligned}$$

$\therefore$  By Residue Theorem,

$$I = 2\pi i \cdot \frac{2}{i} \cdot \frac{1}{2\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

problem:- prove that  $\int_0^{2\pi} \frac{\cos 2\theta}{1-2a\cos\theta+a^2} d\theta = \frac{2\pi a^2}{1-a^2}, (a < 1)$  (2)

Sol:- let  $z = e^{i\theta}$  and  $C$  be the unit  $|z|=1$

$$dz = e^{i\theta} i d\theta$$

$$\Rightarrow d\theta = \frac{dz}{zi}$$

$$\cos\theta = \frac{1}{2} \left[ z + \frac{1}{z} \right] = \frac{z^2+1}{2z}$$

$$\cos 2\theta = \frac{e^{i2\theta} + e^{-i2\theta}}{2} = \frac{z^2 + \frac{1}{z^2}}{2} = \frac{z^4+1}{2z^2}$$

$$\left[ \begin{aligned} e^{i2\theta} &= (e^{i\theta})^2 \\ &= z^2 \end{aligned} \right.$$

Now given, let  $I = \int_0^{2\pi} \frac{\cos 2\theta}{1-2a\cos\theta+a^2} d\theta$

$$= \int_C \frac{(z^4+1)/2z^2}{1-2a(\frac{z^2+1}{2z})+a^2} \frac{dz}{iz}$$

$$= \frac{1}{2i} \int_C \frac{(z^4+1)dz}{z^2(z-az^2-a+az)}$$

$$= \frac{1}{2i} \int_C \frac{(z^4+1)dz}{z^2(z-a)(1-az)}$$

$$= \frac{1}{2i} \int_C f(z) dz \quad \text{--- (1)}$$

Now The poles of  $f(z)$  are.

(i)  $z=0$ , a pole of order 2

(ii)  $z=a$  and (iii)  $z=\frac{1}{a}$

Out of these only  $z=a$  and  $z=0$  are inside the unit circle.

Hence we have to calculate the corresponding residues



$$\begin{aligned}
 \left[ \text{Res } f(z) \right]_{z=a} &= \lim_{z \rightarrow a} (z-a) f(z) \\
 &= \lim_{z \rightarrow a} (z-a) \cdot \frac{z^4+1}{z^2(z-a)(1-az)} \\
 &= \lim_{z \rightarrow a} \frac{z^4+1}{z^2(1-az)} = \frac{a^4+1}{a^2(1-a^2)}
 \end{aligned}$$

$$\begin{aligned}
 \left[ \text{Res } f(z) \right]_{z=0} &= \lim_{z \rightarrow 0} \frac{d}{dz} \left[ (z-0)^2 f(z) \right] \\
 &= \lim_{z \rightarrow 0} \frac{d}{dz} \left[ z^2 \cdot \frac{z^4+1}{z^2(z-a)(1-az)} \right] \\
 &= \lim_{z \rightarrow 0} \frac{d}{dz} \left[ \frac{z^4+1}{(z-a)(1-az)} \right] \\
 &= \lim_{z \rightarrow 0} \frac{d}{dz} \left[ \frac{z^4+1}{z-az^2-a+az^2} \right] \\
 &= \lim_{z \rightarrow 0} \frac{(z-az^2-a+az^2)(4z^3) - (z^4+1)(1-2az+a^2)}{(z-az^2-a+az^2)^2} \\
 &= -\frac{(1+a^4)}{a^2}
 \end{aligned}$$

By Residue Theorem,

$$\int_C f(z) dz = 2\pi i \times \text{Sum of the above two residues}$$

$$= 2\pi i \left[ \frac{a^4+1}{a^2(1-a^2)} - \frac{1+a^4}{a^2} \right]$$

$$= \frac{2\pi i}{a^2} \left[ \frac{a^4+1}{1-a^2} - (1+a^4) \right] = \frac{2\pi i}{a^2(1-a^2)} \left[ (a^4+1) - (1-a^2)(1+a^4) \right]$$

$$= \frac{2\pi i}{a^2(1-a^2)} \cdot [2a^4] = \frac{4a^2\pi i}{1-a^2} \quad \text{--- (2)}$$

Sub (2) in (1)

$$I = \frac{1}{2\pi i} \times \frac{4a^2\pi i}{1-a^2} = \frac{2a^2}{1-a^2}$$

# Integrals of the Type $\int_{-\infty}^{\infty} f(x) dx$

(3)

## Integration around Semi-circles

$$\int_C f(z) dz = \int_{C_R} f(z) dz + \int_{-R}^R f(x) dx$$

$$= 2\pi i \times (\text{Sum of residues at interior poles})$$

Problem:- prove that  $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a+b}$  [ $a>0, b>0, a \neq b$ ]

(Q) Show that  $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2(a+b)}$ .

Sol:- To Evaluate the given integral,

Consider  $\int_C \frac{z^2}{(z^2+a^2)(z^2+b^2)} dz = \int_C f(z) dz$  where  $C$

is the contour consisting of the semi-circle  $C_R$  of radius  $R$  together with the part of the real axis from  $-R$  to  $R$ .

The poles of  $f(z) = \frac{z^2}{(z^2+a^2)(z^2+b^2)}$  are

$$z = \pm ia, z = \pm ib.$$

But  $z = ia$  and  $z = ib$  are the only two poles lie in the upper half of the  $z$ -plane.

$$\therefore [\text{Res } f(z)]_{z=ia} = \lim_{z \rightarrow ia} (z-ia) f(z)$$

$$= \lim_{z \rightarrow ia} \frac{z^2(z-ia)}{(z+ia)(z-ia)(z^2+b^2)}$$



$$= \lim_{z \rightarrow ia} \frac{z^2}{(z+ia)(z^2+b^2)} = \frac{-a^2}{2ia(-a^2+b^2)} = \frac{a}{2i(a^2-b^2)} = R_1$$

$$\begin{aligned} \text{Also } [\text{Res} f(z)]_{z=ib} &= \lim_{z \rightarrow ib} (z-ib) f(z) \\ &= \lim_{z \rightarrow ib} (z-ib) \frac{z^2}{(z+ib)(z^2+a^2)(z-ib)} \\ &= \frac{-b^2}{2ib(-b^2+a^2)} = \frac{-b}{2i(a^2-b^2)} = R_2 \end{aligned}$$

By Residue Theorem, we have.

$$\begin{aligned} \int_C f(z) dz &= 2\pi i \times [R_1 + R_2] \\ &= 2\pi i \times \left[ \frac{a}{2i(a^2-b^2)} - \frac{b}{2i(a^2-b^2)} \right] \\ &= \frac{2\pi i}{2i(a^2-b^2)} [a-b] = \frac{\pi}{a+b} (a-b) \\ &= \frac{\pi}{a+b} \end{aligned}$$

$$\therefore \int_{-R}^R f(x) dx + \int_{CR} f(z) dz = \frac{\pi}{a+b}$$

But  $\int_{CR} f(z) dz \rightarrow 0$  as  $z = Re^{i\theta}$  and  $R \rightarrow \infty$ .

$$\therefore \text{Hence } \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a+b}$$

$$\text{and } \int_{-\infty}^{\infty} f(x) dx = 2 \int_0^{\infty} f(x) dx = \frac{\pi}{a+b}$$

$$\Rightarrow \int_0^{\infty} f(x) dx = \frac{\pi}{2(a+b)}$$

# Integrals of the Type $\int_{-\infty}^{\infty} e^{imx} f(x) dx$ .

(4)

Problem:- P.T  $\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+1} dx = \pi e^{-a}, a \geq 0$

Sol:- we know that  $\cos ax$  is the real part of  $e^{iax}$ .

we consider the function  $f(z) = \frac{e^{iaz}}{z^2+1}$

Now the poles of  $f(z)$  are given by  $z = \pm i$  but  $z = i$  is the only pole lie in the upper half of the  $z$ -plane.

$$\begin{aligned} \left[ \text{Res}[f(z)] \right]_{z=i} &= \lim_{z \rightarrow i} (z-i) f(z) \\ &= \lim_{z \rightarrow i} (z-i) \frac{e^{iaz}}{z^2+1} \\ &= \lim_{z \rightarrow i} (z-i) \frac{e^{iaz}}{(z-i)(z+i)} = \lim_{z \rightarrow i} \frac{e^{iaz}}{z+i} \\ &= \lim_{z \rightarrow i} \frac{e^{iaz}}{z+i} = \frac{e^{-a}}{2i} \end{aligned}$$

$$\text{Thus } \int_C \frac{e^{iaz}}{z^2+1} dz = 2\pi i \times \frac{e^{-a}}{2i} = \pi e^{-a} \quad \text{--- (1)}$$

where  $C$  is the closed curve consisting of the semicircle  $C_R: |z|=R$  together with the real axis from  $-R$  to  $R$ .

On the semi-circle  $C_R$ ,  $z = Re^{i\theta}$ .

$$\int_{C_R} f(z) dz \rightarrow 0 \text{ as } R \rightarrow \infty$$

from (1) we get  $\int_{-\infty}^{\infty} f(x) dx = \pi e^{-a}$ .

Now equating the real part, we get  $\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+1} dx = \pi e^{-a}$



5. Evaluate  $\int_0^{\infty} \frac{x dx}{1+x^6}$  using  $\beta$ - $\Gamma$  functions

Sol:

$$\text{put } x^6 = y \Rightarrow x = y^{1/6}$$

$$\Rightarrow 6x^5 dx = dy$$

$$\Rightarrow 6 \cdot x \cdot x^4 dx = dy$$

$$\Rightarrow x dx = \frac{dy}{6x^4} = \frac{dy}{6 \cdot (y^{1/6})^4} = \frac{dy}{6 \cdot y^{2/3}}$$

Also when  $x \rightarrow 0$  to  $\infty$

$y \rightarrow 0$  to  $\infty$

$$\therefore \int_0^{\infty} \frac{x dx}{1+x^6} = \int_0^{\infty} \frac{dy}{6 y^{2/3} (1+y)} = \frac{1}{6} \int_0^{\infty} \frac{dy}{y^{2/3} (1+y)}$$

$$= \frac{1}{6} \int_0^{\infty} \frac{y^{-2/3}}{(1+y)} dy = \frac{1}{6} \int_0^{\infty} \frac{y^{\frac{1}{3}-1}}{(1+y)^{1+\frac{2}{3}}} dy$$

$$= \frac{1}{6} B\left[\frac{1}{3}, \frac{2}{3}\right] \left[ \because \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = B(m, n) \right]$$

$$= \frac{1}{6} \frac{\Gamma\left(\frac{1}{3}\right) \Gamma\left(\frac{2}{3}\right)}{\Gamma\left[\frac{1}{3} + \frac{2}{3}\right]} = \frac{1}{6} \frac{\Gamma\left(\frac{1}{3}\right) \Gamma\left[1 - \frac{1}{3}\right]}{\Gamma(1)}$$

$$= \frac{1}{6} \Gamma\left[\frac{1}{3}\right] \Gamma\left[1 - \frac{1}{3}\right] = \frac{1}{6} \cdot \frac{\pi}{\sin \frac{\pi}{3}} \quad \left[ \because \Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi} \right]$$

$$= \frac{1}{6} \cdot \frac{\pi}{\sqrt{3}/2} = \frac{1}{3} \frac{\pi}{\sqrt{3}}$$

$$= \frac{\pi}{3\sqrt{3}}$$

(Q) Evaluate

$$\int_0^{\pi/2} [\sqrt{\tan \theta} + \sqrt{\sec \theta}] d\theta.$$

Sol:-

$$\int_0^{\pi/2} [\sqrt{\tan \theta} + \sqrt{\sec \theta}] d\theta = \int_0^{\pi/2} \sqrt{\tan \theta} d\theta + \int_0^{\pi/2} \sqrt{\sec \theta} d\theta.$$

$$= \int_0^{\pi/2} (\tan \theta)^{1/2} d\theta + \int_0^{\pi/2} (\sec \theta)^{1/2} d\theta$$

$\frac{3}{2}$

$$= \int_0^{\pi/2} \left( \frac{\sin \theta}{\cos \theta} \right)^{1/2} d\theta + \int_0^{\pi/2} \left( \frac{1}{\cos \theta} \right)^{1/2} d\theta$$

$$= \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta d\theta + \int_0^{\pi/2} \sin^0 \theta \cos^{-1/2} \theta d\theta$$

$$= \int_0^{\pi/2} \sin^{\frac{1/2+1}{2}} \theta \cos^{\frac{-1/2+1}{2}} \theta d\theta + \int_0^{\pi/2} \sin^{\frac{0+1}{2}} \theta \cos^{\frac{-1/2+1}{2}} \theta d\theta$$

$$= \frac{\Gamma\left[\frac{1/2+1}{2}\right] \Gamma\left[\frac{-1/2+1}{2}\right]}{2 \Gamma\left[\frac{1/2+1}{2} + \frac{-1/2+1}{2} + 1\right]} + \frac{\Gamma\left[\frac{0+1}{2}\right] \Gamma\left[\frac{-1/2+1}{2}\right]}{2 \Gamma\left[\frac{0+1}{2} + \frac{-1/2+1}{2} + 1\right]}$$

$$= \frac{\Gamma\left[\frac{3}{4}\right] \Gamma\left[\frac{1}{4}\right]}{2 \Gamma[1]} + \frac{\Gamma\left[\frac{1}{2}\right] \Gamma\left[\frac{1}{4}\right]}{2 \Gamma\left[\frac{3}{4}\right]}$$

$$= \frac{1}{2} \Gamma\left[\frac{1}{4}\right] \left[ \Gamma\left[\frac{3}{4}\right] + \frac{\sqrt{\pi}}{\Gamma\left[\frac{3}{4}\right]} \right]$$

Ans.

Note:-

$$\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{1}{2} \frac{\Gamma\left[\frac{p+1}{2}\right] \Gamma\left[\frac{q+1}{2}\right]}{\Gamma\left[\frac{p+q+2}{2}\right]}$$



(11) Evaluate  $\int_0^{\pi/2} \sin^{7/2} \theta \cos^{3/2} \theta d\theta$ .

Sol:- We know that  $\int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{1}{2} B(m, n)$

Given  $\int_0^{\pi/2} \sin^{7/2} \theta \cos^{3/2} \theta d\theta$

Let  $2m-1 = 7/2$  and  $2n-1 = 3/2$

$\Rightarrow m = 9/4$   $\Rightarrow n = 5/4$

$$\therefore \int_0^{\pi/2} \sin^{7/2} \theta \cos^{3/2} \theta d\theta = \frac{1}{2} B(9/4, 5/4)$$

$$= \frac{1}{2} \frac{\Gamma(9/4) \Gamma(5/4)}{\Gamma(9/4 + 5/4)} \quad \left[ (n+1)\Gamma(n-1) = \Gamma(n) \right]$$

$$= \frac{1}{2} \frac{[9/4-1] \Gamma[9/4-1] \Gamma[5/4]}{\Gamma[14/4]}$$

$$= \frac{1}{2} \cdot \frac{5}{4} \frac{(\Gamma[5/4])^2}{\Gamma[14/4]}$$

$$= \frac{1}{2} \cdot \frac{5}{4} \frac{\left[ \frac{1}{4} \Gamma[1/4] \right]^2}{\Gamma[7/2]}$$

$$= \frac{1}{2} \cdot \frac{5}{64} \frac{(\Gamma[1/4])^2}{\frac{5}{2} \Gamma[5/2]}$$

$$= \frac{1}{2} \cdot \frac{5}{64} \frac{(\Gamma[1/4])^2}{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma[1/2]}$$

$$= \frac{1}{2} \cdot \frac{5}{64} \cdot \frac{4}{15} \frac{(\Gamma[1/4])^2}{\sqrt{\pi}} = \frac{1}{48\sqrt{\pi}} (\Gamma[1/4])^2$$

(4)

Q(8) Evaluate  $\int_0^{\pi/8} \sin^4 8\theta \cos^6 4\theta d\theta$

(1) sol:- Given integral  $\int_0^{\pi/8} \sin^4 8\theta \cos^6 4\theta d\theta$

S  
$$= \int_0^{\pi/8} (2 \sin 4\theta \cos 4\theta)^4 \cos^6 4\theta d\theta$$

$$\because \sin 8\theta = \sin 2(4\theta)$$

$$\sin 2A = 2 \sin A \cos A$$

$$= 16 \int_0^{\pi/8} \sin^4 4\theta \cos^{10} 4\theta d\theta$$

$$\text{put } 4\theta = t$$

$$d\theta = \frac{dt}{4}$$

$$\theta \rightarrow 0 \text{ to } \pi/8$$

$$t \rightarrow 0 \text{ to } \pi/2$$

$$\therefore \int_0^{\pi/8} \sin^4 8\theta \cos^6 4\theta d\theta = 16 \int_0^{\pi/2} \sin^4 t \cos^{10} t \frac{dt}{4}$$

$$= 4 \int_0^{\pi/2} \sin^4 t \cos^{10} t dt$$

$$= 4 \cdot \frac{\pi}{2} \left( \frac{4+1}{2} \right) \frac{\pi}{2} \left( \frac{10+1}{2} \right)$$

$$\frac{\pi [4+10+2]}{2}$$

$$= 2 \cdot \frac{\pi [5/2] \pi [11/2]}{\pi(8)}$$

$$= \frac{2}{7!} \left[ \frac{3}{2} \times \frac{1}{2} \times \pi(1/2) \right] \left[ \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \pi(1/2) \right]$$

$$= \frac{2}{7!} \left[ \frac{9 \times 7 \times 5 \times 3 \times 3}{16 \times 8} \right] \pi$$

$$= \frac{9}{1024} \pi$$



→

(1) Define Harmonic function ? and verify  $u = 2 \log(x^2 + y^2)$  is harmonic and find its harmonic conjugate.

sol:- Given  $u = 2 \log(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = \frac{2}{x^2 + y^2} \cdot 2x = \frac{4x}{x^2 + y^2}, \quad \frac{\partial^2 u}{\partial x^2} = \frac{4(y^2 - x^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{4y}{x^2 + y^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{4(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\text{Now } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{4(y^2 - x^2) + 4(x^2 - y^2)}{(x^2 + y^2)^2} = 0$$

Hence  $u$  is a harmonic function.

Let  $v(x, y)$  be the harmonic conjugate of  $u(x, y)$

$$\text{Then } dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

$$= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

$$= -\frac{4y}{x^2 + y^2} dx + \frac{4x}{x^2 + y^2} dy$$

$$\Rightarrow dv = \frac{4(x dy - y dx)}{x^2 + y^2}$$

Integrating on b.s,

$$\text{we get } v = 4 \tan^{-1} \frac{y}{x} + c$$

(P1) Show that the function  $f(z) = z^n$  is analytic every where and also find its derivative. (8)

Sol:-  $f(z) = z^n$

$$= (x+iy)^n = (re^{i\theta})^n$$

$$= r^n e^{in\theta}$$

$$= r^n [\cos n\theta + i \sin n\theta]$$

$$= r^n \cos n\theta + i r^n \sin n\theta$$

$$u(x,y) = r^n \cos n\theta, \quad v(x,y) = r^n \sin n\theta$$

$$\frac{\partial u}{\partial r} = n r^{n-1} \cos n\theta, \quad \frac{\partial u}{\partial \theta} = -n r^n \sin n\theta$$

$$\frac{\partial v}{\partial r} = n r^{n-1} \sin n\theta, \quad \frac{\partial v}{\partial \theta} = n r^n \cos n\theta$$

$$\therefore \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\Rightarrow \frac{\partial v}{\partial \theta} = r \frac{\partial u}{\partial r}$$

$u, v$  satisfies C-R Equations

Since  $r^n, \cos n\theta, \sin n\theta$  are continuous product of two continuous functions again continuous.

$\therefore \frac{\partial u}{\partial r}, \frac{\partial u}{\partial \theta}, \frac{\partial v}{\partial r}, \frac{\partial v}{\partial \theta}$  are continuous

Therefore  $f(z)$  is analytic

$$f'(z) = \left[ \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right] e^{-i\theta}$$

$$= n r^{n-1} (\cos n\theta + i \sin n\theta) e^{-i\theta}$$

$$= n r^{n-1} [e^{in\theta}] e^{-i\theta}$$

$$\Rightarrow f'(z) = n r^{n-1} (e^{i\theta})^{n-1} = n (re^{i\theta})^{n-1}$$

$$= n (z^{n-1})$$

$\therefore$  formula in polar coordinates

$$\left\{ \begin{array}{l} f(z) = f(re^{i\theta}) \\ f'(z) = f'(re^{i\theta}) e^{i\theta} \\ \Rightarrow f'(re^{i\theta}) = \frac{f'(z)}{e^{i\theta}} \end{array} \right.$$